

# ARIA: Score- and Memory-Conditioned Retrieval and Compression for Latent-Compression Retrieval-Augmented Generation

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## Abstract

We study retrieval–compression coupling in latent-compression retrieval-augmented generation (RAG). ARIA coordinates three forward-pass operations: Compression-Feedback Reranking Scheme (CFRS), which uses a teacher-forced next-token squared-probability proxy for auxiliary training and final passage ordering; Adaptive Compression Rate (ACR), which maps relative retrieval scores to per-passage memory gates; and Memory-Token Feedback Retrieval Loop (MTFRL), which performs one memory-conditioned second retrieval. Phase II updates the query, compressor, and generator adapters together with the feedback projection; the BGE encoder and index remain frozen, and hard selections are discrete. Across NQ, HotpotQA, MuSiQue, and 2WikiMultiHopQA, the complete ARIA configuration records a five-seed mean Avg F1 of 44.50% at nominal 16 $\times$ ; our CLaRa reimplementation, matched on reader, corpus, Phase II data, optimization schedule, nominal ratio, and seeds, records 38.94%. In a separate retraining study that matches the mean evidence-token budget and retains a two-round retrieval topology, the matched full control records 44.46% Avg F1, compared with 43.01% when all three operations are disabled.

## 1 Introduction

When used without external updates, large language models (LLMs) rely on parameters fixed at training time and can produce unsupported or outdated factual outputs (Gao et al., 2023). Retrieval-Augmented Generation (RAG) (Lewis et al., 2020) conditions generation on retrieved evidence for knowledge-intensive tasks. ARIA (Adaptive Retrieval–Compression Integrated Architecture) adds three operations to latent-compression RAG: CFRS incorporates a per-document teacher-forced next-token squared-probability proxy into

final passage ordering; ACR maps relative relevance scores to per-passage memory gates; and MTFRL projects first-pass memory tokens into one additional retrieval query (Figure 1).

Retrieval scores therefore condition compression, while compressed memory conditions one further retrieval. During Phase II, the teacher-forced proxy loss and retrieval-side auxiliary losses update the continuous query and memory mappings; the BGE index and hard top- $k$  document identities form a discrete selection boundary. MTFRL executes exactly one additional retrieval step. Experiments instantiate the design with the five-stage retrieval backbone—Query Complexity Assessment (QCA), Adaptive Hybrid Retrieval (AHR), Iterative Gap-Filling Retrieval (IGFR), Multi-Axis Document Scoring (MADS), and Cross-Component Evidence Fusion (CCEF)—detailed in Appendix Figure A1.

**Contribution and evaluation scope.** The methodological proposal is the CFRS–ACR–MTFRL forward-pass layer; QCA–CCEF is its retrieval implementation in our experiments. We evaluate the complete ARIA and CLaRa configurations using the same reader, corpus, Phase II mixture, optimization schedule, nominal ratios, and seed set. Reported ARIA–CLaRa differences are complete-system endpoints; published systems provide broader contextual reference results.

## 2 Related Work

**Retrieval-Augmented Generation and adaptive retrieval.** RAG (Lewis et al., 2020) conditions generation on non-parametric knowledge; advances include dense retrieval (Karpukhin et al., 2020), contrastive pre-training (Izacard et al., 2022), fusion-in-decoder (Izacard and Grave, 2021), in-context retrieval augmentation (Ram et al., 2023), and joint training at scale (Izacard et al., 2023). Long contexts exhibit positional

bias (Liu et al., 2024), while irrelevant evidence can distract RAG models (Yoran et al., 2024); both motivate selective compression. Adaptive-retrieval methods (Asai et al., 2024; Jiang et al., 2023c; Trivedi et al., 2023; Yan et al., 2024; Shi et al., 2024) adapt retrieval timing, routing, or evidence selection, whereas ARIA uses latent compressed memory for one additional retrieval. AdaptiveRAG (Jeong et al., 2024) uses a trained query-complexity classifier to select among fixed strategies; ARIA instead couples latent compression with the retrieval path. For multi-hop QA (Yang et al., 2018; Trivedi et al., 2022), prior work spans interleaved retrieval and reasoning (Trivedi et al., 2023), generate-then-ground (Shi et al., 2024), DRO (Shi et al., 2026), and KG retrieval (Gutiérrez et al., 2024). ARIA combines teacher-forced next-token-proxy reranking, score-conditioned per-passage memory gating, and memory-conditioned second retrieval in one forward path.

**Context Compression.** Techniques range from prompt compression (Jiang et al., 2023b; Pan et al., 2024; Jiang et al., 2024) and selective augmentation (Xu et al., 2024) to gist tokens (Mu et al., 2023), in-context autoencoders (Ge et al., 2024), Autocompressor (Chevalier et al., 2023), xRAG (Cheng et al., 2024), and PISCO (Louis et al., 2025). Unlike Fusion-in-Decoder (Izacard and Grave, 2021), latent compression replaces retrieved passages with a small number of memory tokens *before* the decoder. PISCO jointly distills its compressor and decoder while keeping its retriever/reranker fixed. CLaRa (He et al., 2026) retrieves over continuous compressed representations and jointly trains its query reasoner and generator; its reported design uses uniform memory budgets. ARIA adds score-conditioned per-passage budgets and memory-conditioned second retrieval. Appendix Table A1 compares the three ARIA operations explicitly.

### 3 Method

Figure 1 presents CFRS, ACR, and MTFRL as parallel conceptual views. Execution is staged: first-pass retrieval and ACR produce memory; MTFRL retrieves and re-scores one additional candidate set; ACR is recomputed on the final survivor set; and CFRS orders the passages before generation. Phase II uses soft ACR gates, whereas inference retains thresholded memory states. We de-

note the nominal pre-ACR compression ratio by  $r$  and the base memory length for passage  $i$  by  $K_i^{(0)}(r) = \max\{1, \lfloor L_i/r \rfloor\}$ . Because ACR sparsifies this base allocation, equal nominal ratios correspond to different realized evidence-memory budgets. Appendix A.3 gives the supporting-backbone details, pool sizes, and realized-ratio definition.

#### 3.1 Problem Formulation and Training Losses

Given a query  $q$  and corpus  $\mathcal{C}$ , let  $\mathcal{G}_\theta(q, \mathcal{C})$  denote ARIA’s coupled evidence–memory operator. It executes the staged forward pass above and returns an *ordered* list of final passage–memory pairs, so both MTFRL evidence replacement and CFRS reranking remain explicit. ARIA produces an answer by

$$\hat{a} = \text{Decode}_\eta[p_\theta(\cdot \mid q, \mathcal{G}_\theta(q, \mathcal{C}))], \quad (1)$$

where  $\eta$  is greedy one-beam decoding without sampling, stopped at EOS or 64 generated tokens, and is held fixed for ARIA and the protocol-controlled CLaRa reimplementation. Inside  $\mathcal{G}_\theta$ , document identities and ordering are evaluated by hard selection on detached scores. Continuous query and memory representations are trained through the explicit objectives below; gradients do not pass through `argsort`, dense-index lookup, or hard top- $k$ . AHR and MADS consume the shared query representation  $\mathbf{q}_{\text{rep}} = \text{QR}(q) \in \mathbb{R}^{d_h}$ , implemented as the final-token hidden state of a Query Reasoner (QR). Its frozen language-model base carries a rank-16 LoRA adapter in Phase II. The adapter is initialized from the corresponding Phase I compressor adapter. QCA instead uses surface rules; IGFR and CCEF consume the intermediate signals specified above. A pre-fitted and then fixed projection  $W_{\text{BGE}}$  maps  $\mathbf{q}_{\text{rep}}$  into the frozen BGE embedding space (Appendix A.3).  $\mathcal{L}_{\text{QR}}$  aligns the normalized projected QR state with the frozen BGE embedding of an annotated supporting passage. The support construction and both retrieval-space auxiliary losses are defined below; projection calibration details are in Appendix A.4.

Phase I pre-trains the compressor with a conditional generation objective

$$\mathcal{L}_{\text{pre}} = -\mathbb{E}_{(d, I, y) \sim \mathcal{D}_{\text{pre}}} \log p_\theta(y \mid I, \mathcal{F}(d)), \quad (2)$$

where  $I$  is the task instruction and  $y$  is the target for the simple QA, multi-fact/reasoning,

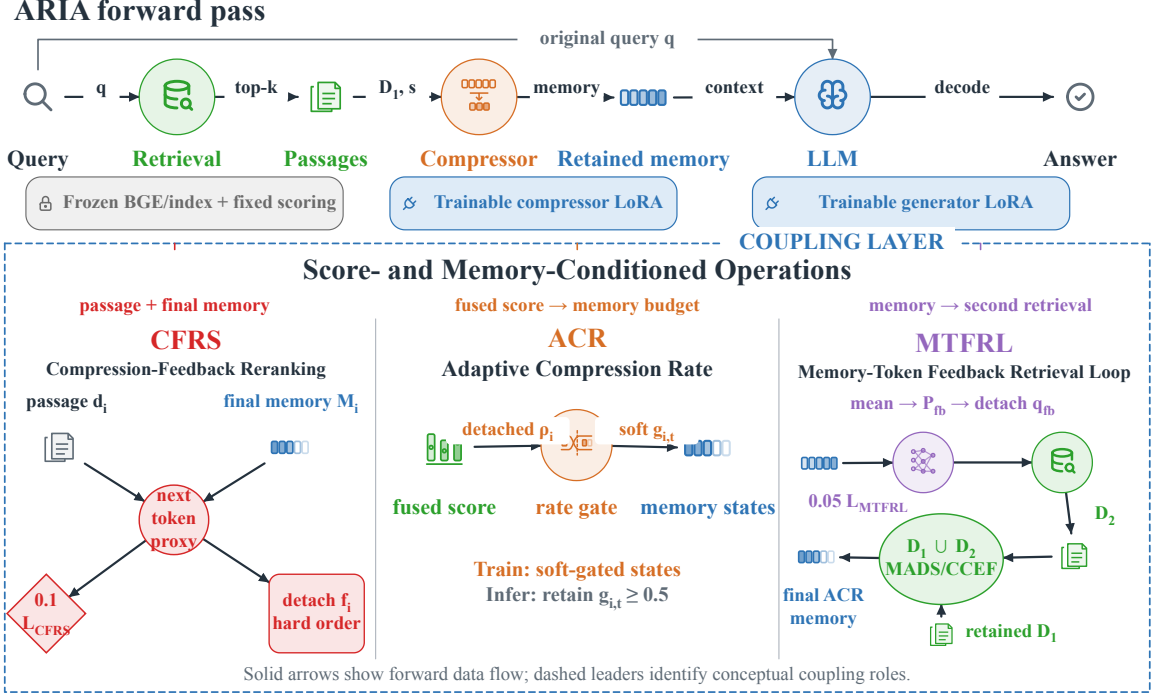


Figure 1: **ARIA comprises CFRS, ACR, and MTFRL.** The parallel panels show teacher-forced next-token-proxy ordering, score-conditioned memory gating, and one memory-conditioned second retrieval. Hard retrieval selections remain discrete; Appendix Figure A1 shows the supporting backbone.

entity-augmented, or paraphrase example (Appendix A.11). Phase II minimizes

$$\mathcal{L} = \mathcal{L}_{QA} + \lambda \mathcal{L}_{MSE} + \mu \mathcal{L}_{CFRS} + \nu \mathcal{L}_{QR} + \xi \mathcal{L}_{MTFRL}, \quad (3)$$

Here  $\lambda$ ,  $\mu$ ,  $\nu$ , and  $\xi$  respectively scale the MSE, CFRS, QR positive-pair alignment, and MTFRL contrastive terms. The default full-ARIA values are (0.10, 0.10, 0.05, 0.05); reported ablations state any variation. These coefficients balance losses with different numerical scales. The answer-generation term is

$$\mathcal{L}_{QA} = -\mathbb{E}_{(q, a^*, \mathcal{C}) \sim \mathcal{D}_{tr}} [\log p_{\theta}(a^* | q, \mathcal{G}_{\theta}(q, \mathcal{C}))], \quad (4)$$

$$\mathcal{L}_{MSE} = \mathbb{E}_{x \sim \mathcal{D}_{tr}} \left[ \frac{1}{d_h} \|\bar{\mathbf{h}}_{\mathcal{M}} - \bar{\mathbf{h}}_{\mathcal{U}}\|_2^2 \right], \quad (5)$$

where  $x$  denotes a Phase II training example and  $d_h$  is the hidden dimension. In the final decoder layer of the teacher-forced QA forward pass after CFRS ordering,  $\mathcal{M}$  indexes valid memory-token positions and  $\mathcal{U}$  indexes valid query and gold-answer positions; padding and structural delimiters are excluded. Overlines denote sequence-level means over the corresponding hidden states,

and  $\mathcal{D}_{tr}$  is the Phase II training mixture (Appendix A.16). The factor  $1/d_h$  makes the loss a coordinate-wise mean squared error. Gradients reach both means:  $\mathcal{L}_{MSE}$  is a symmetric representation-alignment regularizer, not alignment to a fixed anchor, and it constrains sequence means rather than individual token states. It is distinct from CFRS’s document-level teacher-forced proxy objective. Its presence and  $\lambda$  sensitivity are reported in Appendix A.12.

### 3.2 Forward-Pass Operations

Figure A2 details the three mechanisms. They exchange information during Phase II and inference. The retrieval encoder and document index remain frozen, the score formulas and rules remain fixed, and candidate identities and hard ranking/top- $k$  operations remain discrete. The QR, CFRS, and MTFRL auxiliary losses train their continuous query- and memory-side modules without differentiating through those operations.

**CFRS: compression→final passage ordering.** Let  $x_i = (x_{i1}, \dots, x_{iL_i})$  be the tokenized candidate passage after the same truncation used by the compressor, and let  $\mathcal{T}_i$  contain its non-padding, non-delimiter target positions; preprocessing re-

tains at least one such position. Let  $\mathbf{m}_i^{\text{ctx}}$  be the final ACR-processed memory produced by the current compressor  $\mathcal{F}_{\theta_c}$  for  $d_i$  and actually supplied to the generator in the current forward pass: soft-gated during Phase II and hard-retained at inference. CFRS evaluates this same memory. With the Phase II generator LoRA disabled, the frozen base decoder  $\phi_0$  uses the fixed textual prefix  $u_{\text{rec}} = \text{“Reconstruct the original passage from the memory tokens. Output only the reconstructed passage.”}$  Under teacher forcing it produces

$$p_{it}^{\text{rec}}(v) = p_{\phi_0}(v \mid u_{\text{rec}}, \mathbf{m}_i^{\text{ctx}}, x_{i,<t})$$

over vocabulary  $\mathcal{V}$ . CFRS defines a per-document token-mean teacher-forced squared-probability error

$$e_i \equiv \hat{\mathcal{E}}_{\text{rec}}(d_i) = \frac{1}{|\mathcal{T}_i|} \sum_{t \in \mathcal{T}_i} \ell_{it}^{\text{rec}},$$

$$\ell_{it}^{\text{rec}} = \sum_{v \in \mathcal{V}} (p_{it}^{\text{rec}}(v) - \mathbf{1}[v = x_{it}])^2.$$

(6)

Unlike  $\mathcal{L}_{\text{MSE}}$ , which aligns hidden-state means in the QA pass,  $e_i$  is a teacher-forced next-token squared-probability proxy conditioned on both the compressed memory and the gold prefix  $x_{i,<t}$ . This definition is used for both the auxiliary compressor objective and the detached ordering score. Its undetached training value supplies  $\mathcal{L}_{\text{CFRS}} = \mathbb{E}_x[|D|^{-1} \sum_{d_i \in D} e_i]$ . The decoder parameters remain frozen in this pass, while the gradient with respect to its memory input updates the compressor. For ordering, CFRS uses the detached copy  $\bar{e}_i = \text{sg}[e_i]$  and reverse-min-max normalization over the fixed final CCEF survivor set  $D$ . With  $\bar{e}_{\min} = \min_{d_j \in D} \bar{e}_j$ ,  $\bar{e}_{\max} = \max_{d_j \in D} \bar{e}_j$ ,  $\Delta \bar{e} = \bar{e}_{\max} - \bar{e}_{\min}$ , and  $\epsilon = 10^{-6}$ ,

$$f_i = \begin{cases} (\bar{e}_{\max} - \bar{e}_i) / \Delta \bar{e}, & \Delta \bar{e} > \epsilon, \\ 0.5, & \text{otherwise.} \end{cases} \quad (7)$$

Thus  $f_i \in [0, 1]$ , with higher values denoting lower teacher-forced proxy error. The tied-score fallback uses stable sorting to preserve the CCEF order. The hard reranking operation is discrete; compressor supervision comes from Eq. 18. The final forward-pass ranking interpolates the CCEF score and the normalized proxy score:

$$s_{\text{final}}(d_i) = (1 - w_{\text{cfrs}}) s_{\text{fused}}(d_i) + w_{\text{cfrs}} f_i. \quad (8)$$

Here  $w_{\text{cfrs}} = 0.3$ . CFRS reuses the frozen base decoder for the teacher-forced proxy pass. The undetached error trains the compressor, while its detached copy orders passages in the Phase II forward pass and at inference.

**ACR: retrieval→compression.** ACR applies a sparsifying mask that is differentiable with respect to the selected memory states. The per-document scaling *rate*  $\rho_i$  is computed from retrieval scores, treated as a detached constant, and distinguished from the nominal global *ratio*  $r$ . For passage  $d_i$  with truncated source length  $L_i$ , define its base pre-ACR memory length as  $K_i^{(0)}(r) = \max\{1, \lfloor L_i/r \rfloor\}$  and let  $\widetilde{\mathbf{m}}_i^{(t)} = [\mathcal{F}_{\theta_c}(d_i)]_t$  for  $t \in \{1, \dots, K_i^{(0)}(r)\}$ . ACR sparsifies the base allocation without redistributing removed tokens to higher-ranked passages (realized totals in Appendix A.35):

$$g_{it} = \sigma\left(\beta(\rho_i K_i^{(0)}(r) - t)\right),$$

$$\mathbf{m}_{i,\text{train}}^{(t)} = g_{it} \widetilde{\mathbf{m}}_i^{(t)}. \quad (9)$$

Here  $\sigma(\cdot) \in (0, 1)$  is a scalar gate broadcast over the  $d_h$ -dimensional hidden state at position  $t$ ,  $\beta = 10$ , and  $\rho_i \in [\rho_{\min}, \rho_{\max}] = [0.25, 1.0]$  is computed from  $s_i \equiv s_{\text{fused}}(d_i)$ . For either current CCEF survivor set  $S \in \{D_1, D\}$ , let  $N = |S|$ ,  $s_{\min} = \min_{d_j \in S} s_j$ ,  $s_{\max} = \max_{d_j \in S} s_j$ ,  $R_s = s_{\max} - s_{\min}$ , and  $\Delta \rho = \rho_{\max} - \rho_{\min}$ . When  $R_s > \epsilon$ , define  $z_i = (s_i - s_{\min}) / R_s$ :

$$\rho_i = \begin{cases} \rho_{\max}, & N = 1, \\ (\rho_{\min} + \rho_{\max})/2, & N > 1 \text{ and } R_s \leq \epsilon, \\ \rho_{\min} + \Delta \rho z_i, & \text{otherwise.} \end{cases} \quad (10)$$

Here  $\epsilon = 10^{-6}$ . Thus a singleton receives  $\rho_{\max}$ , multiple tied survivors receive the midpoint, and otherwise higher-scored documents receive larger scale factors  $\rho_i$ ; retained token counts also depend on source length  $L_i$  and integer rounding. The rates  $\rho_i$  are detached constants: gradients through gated memory states update the compressor while retrieval scores and selected documents remain fixed. Equation 9 is the soft training-time mask; at inference, let  $\mathcal{I}_i = \{t : g_{it} \geq 0.5\}$ . If this set is empty, it is replaced by  $\{\arg \max_t g_{it}\}$ ; only the corresponding raw states  $\widetilde{\mathbf{m}}_i^{(t)}$  are retained for MTFRL, CFRS, and generation. This realizes an integer memory length with a one-token minimum. A complexity-adaptive floor for multi-hop queries at 128× is analyzed in Appendix A.28.

**MTFRL: compression→retrieval.** MTFRL executes exactly one memory-conditioned second-round retrieval. Let  $T_i = |\mathcal{I}_i| = \max\{1, \lfloor \rho_i K_i^{(0)}(r) \rfloor\}$  under the monotone gate and empty-set fallback above. To make the soft-hard distinction explicit, MTFRL pools all soft-gated positions during Phase II and only retained positions at inference:

$$\begin{aligned}\bar{\mathbf{m}}_{\text{train}} &= \frac{1}{N} \sum_{i=1}^N \frac{\sum_{t=1}^{K_i^{(0)}(r)} \mathbf{m}_{i,\text{train}}^{(t)}}{\max\left\{\sum_{t=1}^{K_i^{(0)}(r)} g_{it}, \epsilon\right\}}, \\ \bar{\mathbf{m}}_{\text{inf}} &= \frac{1}{N} \sum_{i=1}^N \frac{1}{T_i} \sum_{t \in \mathcal{I}_i} \tilde{\mathbf{m}}_i^{(t)}, \\ \mathbf{q}_{\text{fb}} &= \text{Normalize}_{2,\epsilon}(P_{\text{fb}}(\bar{\mathbf{m}}_{\text{stage}})).\end{aligned}\quad (11)$$

Here  $\bar{\mathbf{m}}_{\text{stage}}$  selects the training or inference expression. Gate-mass normalization converts each document’s gated sum to a weighted mean, limiting scale variation due solely to total gate mass. We use  $\text{Normalize}_{2,\epsilon}(\mathbf{z}) = \mathbf{z} / \max(\|\mathbf{z}\|_2, \epsilon)$ ; the pool spans the realized  $N = |D_1| \in [1, 5]$  first-pass survivors.

For minibatch  $\mathcal{B}$ , let  $\mathcal{B}_{\text{sup}}$  contain examples with annotated supports  $\mathcal{P}(x)$ . A run-seeded sampler draws the shared positive  $d^+(x)$  uniformly from supports absent from  $D_1(x)$  when possible, otherwise from  $\mathcal{P}(x)$ . After masking all annotated supports,  $\mathcal{N}(x)$  page-ID-deduplicates the union of non-supporting  $D_1(x)$  passages and other minibatch positives. Let  $\mathcal{B}_{\text{mt}} = \{x \in \mathcal{B}_{\text{sup}} : \mathcal{N}(x) \neq \emptyset\}$  and  $\mathcal{A}(x) = \{d^+(x)\} \cup \mathcal{N}(x)$ . With normalized frozen BGE document embedding  $e_d$ , normalized projected query  $\mathbf{z}_q$ , and  $s_x(d) = \mathbf{q}_{\text{fb}}^\top e_d / 0.05$ , the objectives are

$$\begin{aligned}\ell_{\text{QR}}(x) &= 1 - \mathbf{z}_q^\top e_{d^+(x)}, \\ \ell_{\text{MTFRL}}(x) &= -s_x(d^+(x)) + \text{LSE}_{d \in \mathcal{A}(x)} s_x(d).\end{aligned}\quad (12)$$

Here  $\mathbf{z}_q = \text{Normalize}_{2,\epsilon}(W_{\text{BGE}} \mathbf{q}_{\text{rep}})$  and LSE is log-sum-exp.  $\mathcal{L}_{\text{QR}}$  and  $\mathcal{L}_{\text{MTFRL}}$  average their pointwise losses over  $\mathcal{B}_{\text{sup}}$  and  $\mathcal{B}_{\text{mt}}$ , respectively, and are zero when that set is empty. Hence a supported example with no valid negative contributes only to  $\mathcal{L}_{\text{QR}}$ . MTFRL trains  $P_{\text{fb}}$  and the soft memory path; detached  $\mathbf{q}_{\text{fb}}$  retrieves  $|D_2| = 200$ , and  $D_1 \cup D_2$  is re-scored by MADS+CCEF. Appendix A.4 gives exact projection and sampling details.

### 3.3 Supporting Retrieval Backbone

The QCA–AHR–IGFR–MADS–CCEF backbone supplies ARIA’s first-pass survivor set and fused scores. Its rules, equations, thresholds, and pool sizes are in Appendix A.3 and Figure A1.

### 3.4 Training Procedure

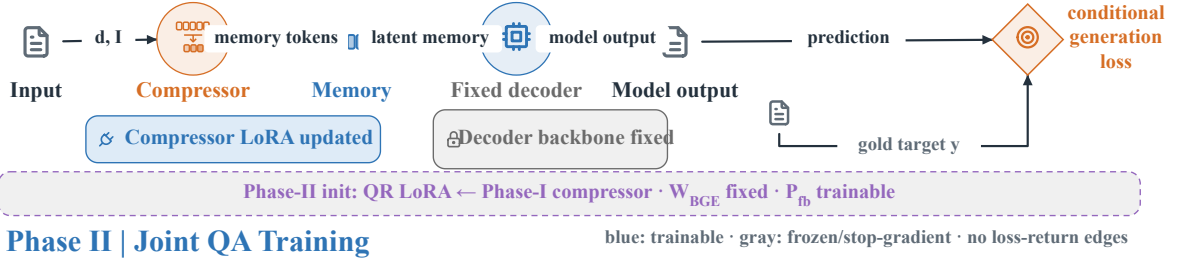
We train a separate model for each nominal ratio  $r \in \{4, 16, 32, 64, 128\}$  because ACR depends on  $K_i^{(0)}(r)$ . Phase I trains the compressor adapter on 7,808,465 examples for three epochs. For each ratio, Phase II initializes the compressor from the corresponding Phase-I checkpoint, copies that adapter into the QR branch, initializes a separate generator LoRA, and Xavier-initializes  $P_{\text{fb}}$  with zero biases. Phase II jointly optimizes Eq. 3 with rank-16 LoRA adapters (Hu et al., 2022) on  $\mathbf{q}_{\text{proj}}$  in the QR, compressor, and generator ( $\alpha_{\text{LoRA}} = 32$ ; dropout 0.10) and with all parameters of  $P_{\text{fb}}$ . Language-model base weights, the pre-fitted  $W_{\text{BGE}}$ , the BGE encoder and document index, and fixed retrieval rules remain frozen. All candidate selection and reranking operations use detached hard decisions; the continuous modules receive their training signals through the explicit losses in Eq. 3. Projection fitting and auxiliary-objective details are in Appendix A.4. At 32× under Normal retrieval, full Phase-II joint training has 1.47 pp higher mean Avg CEM than generator-only tuning (Appendix A.13). Optimization settings and the two-phase diagram are in Appendix A.39 and Figure 2; Algorithm 1 gives the full inference procedure.

## 4 Experiments

### 4.1 Experimental Setup

**Setup and metrics.** The primary backbone is Mistral-7B-Instruct-v0.2 (Jiang et al., 2023a); focused checks use Llama-3-8B-Instruct (Llama Team, 2024) and Qwen-2.5-14B-Instruct (Qwen Team, 2024). We evaluate on NQ (Kwiatkowski et al., 2019), HotpotQA (Yang et al., 2018), MuSiQue (Trivedi et al., 2022), and 2WikiMulti-HopQA (Ho et al., 2020). The primary metric is normalized whitespace-token F1, maximized over benchmark-provided gold aliases. CEM, EM, and gold-document Recall@ $k$  follow the definitions in Appendix A.38. Avg F1 and Avg CEM are unweighted means over the four benchmark-level scores, computed within each seed and then averaged over five seeds.

## Phase I | Conditional Generation Pre-Training



## Phase II | Joint QA Training

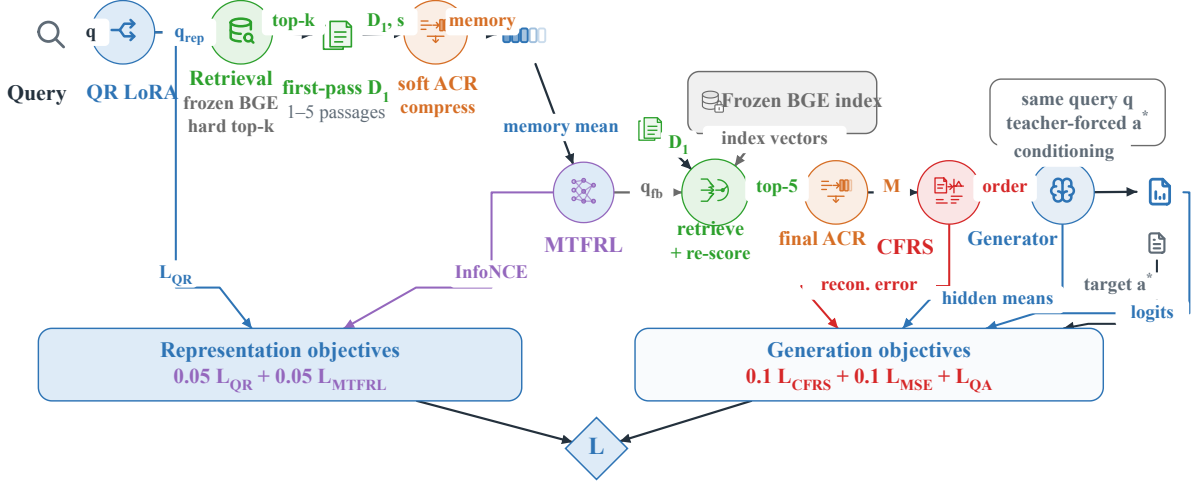


Figure 2: **ARIA’s two-phase training procedure.** Phase I trains the compressor with conditional generation targets;  $W_{BGE}$  is pre-fitted once and then frozen. Phase II updates the QR, compressor, and generator LoRA adapters together with  $P_{fb}$  under Eq. 3. The BGE encoder and index remain frozen, and hard selection remains discrete.

**Retrieval and comparison.** Normal retrieval searches the KILT Wikipedia corpus (Petroni et al., 2021) with frozen BGE-large-en-v1.5 and BM25 infrastructure. The protocol-controlled Oracle comparison reranks a page-ID-deduplicated 100-candidate pool constructed to contain the annotated positives. We retrain CLaRa (He et al., 2026) with the same reader, corpus, Phase II data, optimization schedule, nominal ratios, and five seeds as ARIA, while retaining CLaRa’s Phase-II next-token-prediction objective, straight-through top- $k$  selection over uniformly sized compressed document representations, all-linear LoRA placement, and method-specific initialization detailed in Appendix A.25. ACR changes ARIA’s realized memory budget. Published systems with other readers, training procedures, or evaluation settings are listed as contextual references only in Appendix A.9.

**Training.** Phase II uses 38,400 benchmark-balanced examples per epoch for five epochs. ARIA and CLaRa use the same augmented

MuSiQue partition and page-URL deduplication against evaluation splits. Full data composition, optimization settings, and the data-equalized MuSiQue control are in Appendices A.16, A.39, and A.22.

**Statistics and selection.** We report means over seeds  $\{42, 123, 456, 789, 2024\}$ ; Figure 3 additionally shows seed SD for Avg CEM. For the primary Mistral ARIA-CLaRa family (5 ratios  $\times$  (4 benchmarks + Avg)  $\times$  2 metrics), we use two-sided paired query bootstrap with 10,000 resamples after averaging the five seed-pair differences (Berg-Kirkpatrick et al., 2012) and control the 50 comparisons at family-wise  $\alpha = 0.01$  with the Bonferroni threshold  $p < 2.0 \times 10^{-4}$ . Both Avg-metric gains at  $4 \times -32 \times$  meet this threshold. Appendix A.23 retains pointwise thresholds for cell-level resolution; supplemental control studies report their stated within-study tests. Hyperparameters are selected on training-derived development partitions disjoint from final evaluation sets (Appendices A.23 and A.39).

## 4.2 End-to-End QA Performance

**Headline result at 16×**. Under Normal retrieval, ARIA records 44.50% Avg F1 at nominal 16×, compared with 38.94% for our five-seed CLaRa reimplementation, a 5.56-point difference. Under Oracle retrieval at 4×, the corresponding means are 67.31% and 63.22% (Appendix A.9).

**Across compression ratios.** ARIA has a higher five-seed mean Avg F1 than the reimplementation at all five tested ratios (Table 1). At 128×, however, it trails CLaRa on 2Wiki F1 and on MuSiQue/2Wiki CEM; Appendix A.28 analyzes this high-compression regime.

**Cross-backbone check at 16×**. The ARIA-CLaRa mean difference is positive in both Avg CEM and Avg F1 on each evaluated backbone (Table 2). The Avg-F1 differences are 5.56 points on Mistral-7B, 6.14 on Llama-3-8B, and 4.41 on Qwen-2.5-14B. Qwen is evaluated at this ratio only; the Llama experiment covers all five ratios, with full per-benchmark results in Appendices A.17 and A.30.

## 4.3 First-Pass Retrieval Results

At 16× Normal retrieval, ARIA’s macro-averaged first-pass Recall@5 is 4.2 points above CLaRa’s, with differences of +3.1, +5.8, +4.1, and +3.8 points on NQ, HotpotQA, MuSiQue, and 2Wiki, respectively (Table 3). For ARIA, this diagnostic is measured after AHR-IGFR-MADS-CCEF; CLaRa is measured at its corresponding first-pass top-5 output. Both measurements precede compression and ARIA’s MTFRL and CFRS operations. Full results are in Appendices A.10 and A.32.

Holding the frozen encoder, index, corpus, candidate construction, and evaluation queries fixed, Phase II QR adaptation raises ARIA’s macro-averaged first-pass Recall@5 from 58.5% at the pre-Phase-II operating point to 61.3% after Phase II (+2.8 points). The four benchmark differences range from +2.5 to +3.4 points and their paired-bootstrap 95% intervals exclude zero (Appendix A.32).

## 4.4 Sampled Error Categorization

We categorize 200 ARIA errors per benchmark at 16× Normal retrieval. Answer-bearing text is present in the retrieved source passages for 52.5% of the sampled NQ errors, categorized as evidence-present answer failures. Missing source evidence

is the largest sampled MuSiQue category at 56.5%. On the pooled 800-item taxonomy sample, a second annotator independently labels the same items with 89.9% raw agreement and Cohen’s  $\kappa = 0.83$  (95% CI [0.80, 0.86]); disagreements are arbitrated only after agreement is computed. Table 4 reports all category counts; Appendix A.37 gives one observed case per category, and Appendix A.7 reports related controls.

## 4.5 Coupling Intervention and Sensitivity Analyses

**Budget- and topology-matched retraining.** Table 5 compares separately retrained controls with the same mean evidence-token budget ( $\approx 108$  per query), training data, initialization protocol, schedule, seed set, first-round candidate-pool construction and size, and final document count. Because each variant is retrained independently, the retrieved document identities can differ. When MTFRL is disabled, a static second retrieval preserves the two-round topology and candidate count. The independently retrained matched-full endpoint is 44.46% Avg F1; disabling all three operations records 43.01% (−1.45 points). In the complementary fixed-checkpoint intervention, disabling the three forward-path operations changes Avg F1 from 44.50% to 42.00% (−2.50 points); that intervention keeps learned parameters fixed but changes the mean evidence budget from 108 to 184 tokens. Candidate provenance and statistical results for both protocols are reported in Appendix A.18.

## 4.6 Realized Evidence Allocation

At 16×, ARIA retains a measured mean of 108 evidence-memory tokens per query, versus 184 for CLaRa. Across 4×–128×, the corresponding ARIA means are 432/108/54/27/13 tokens, while the CLaRa means are 738/184/92/46/23. Reporting these realized totals alongside the nominal ratios makes ACR’s rank-conditioned sparsification explicit; Appendix A.35 gives the allocation calculation and per-rank measurements.

## 5 Analysis

**Compression regime.** The ARIA-CLaRa Avg-CEM difference narrows from 2.41 points at 16× to 0.16 at 128× (Figure 3). At 128×, ARIA’s 2Wiki F1 is 6.95 points below its 64× value, versus a 2.26-point difference for CLaRa. The

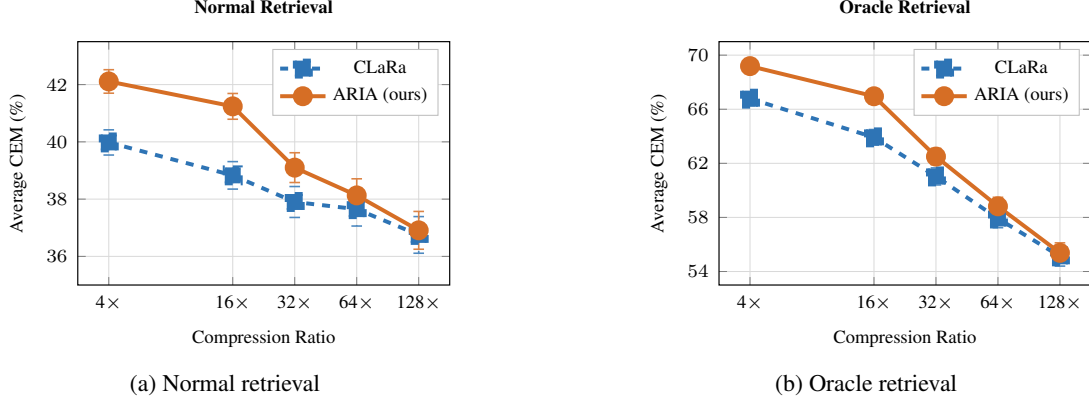


Figure 3: Average CEM across compression ratios (five-seed mean; whiskers show  $\pm 1$  SD). Across the five tested ratios, the ARIA–CLaRa mean difference is largest at 16 $\times$  in both settings (+2.41 points under Normal and +3.04 under Oracle) and is +0.16 and +0.26 points at 128 $\times$ , respectively.

Table 1: End-to-end Normal-retrieval results (five-seed means). Each cell reports ARIA/CLaRa. CR is nominal; ACR gives ARIA a different realized evidence-memory budget. Figure 3 shows Avg-CEM seed SD; Appendix A.9 reports full F1, CEM, EM, Oracle, and contextual-baseline results.

| CR           | NQ F1       | Hotpot F1   | MuSiQue F1  | 2Wiki F1    | Avg F1      | Avg CEM     |
|--------------|-------------|-------------|-------------|-------------|-------------|-------------|
| 4 $\times$   | 56.20/48.34 | 54.10/46.07 | 22.30/17.55 | 51.10/47.32 | 45.93/39.82 | 42.11/39.98 |
| 16 $\times$  | 55.12/47.95 | 52.64/45.84 | 21.07/17.68 | 49.19/44.30 | 44.50/38.94 | 41.24/38.83 |
| 32 $\times$  | 54.80/47.28 | 49.47/44.28 | 19.04/16.72 | 47.38/42.82 | 42.67/37.77 | 39.10/37.90 |
| 64 $\times$  | 52.90/46.41 | 49.82/44.65 | 19.12/16.08 | 44.80/40.57 | 41.66/36.93 | 38.13/37.65 |
| 128 $\times$ | 51.74/45.54 | 47.88/42.29 | 17.94/14.50 | 37.85/38.31 | 38.85/35.16 | 36.91/36.75 |

Table 2: Normal-retrieval cross-backbone results at nominal 16 $\times$  (five-seed means). Differences are ARIA minus the protocol-controlled CLaRa reimplementation; realized evidence-memory budgets differ.

| Backbone     | CLaRa |       | ARIA  |       | Difference |       |
|--------------|-------|-------|-------|-------|------------|-------|
|              | CEM   | F1    | CEM   | F1    | CEM        | F1    |
| Mistral-7B   | 38.83 | 38.94 | 41.24 | 44.50 | +2.41      | +5.56 |
| Llama-3-8B   | 41.43 | 41.44 | 44.28 | 47.58 | +2.85      | +6.14 |
| Qwen-2.5-14B | 40.49 | 41.10 | 42.23 | 45.51 | +1.74      | +4.41 |

two lowest-ranked passages’ mean soft cutoffs approach the one-token boundary. In the reported adaptive-floor grid, the  $\rho_{\min}^{\text{MH}} = 0.45$  row records 0.44 points higher Avg CEM and 1.77 points higher 2Wiki F1 than the default row (Appendices A.35 and A.28). Independently trained hard-gate checkpoints remain within 0.24 points in Avg CEM and 0.32 points in Avg F1 of the soft-gate checkpoints at all five ratios (Appendix A.19). With the mean evidence budget held at 13 tokens at 128 $\times$ , the adaptive-floor allocation records 39.32% Avg F1 and 39.28% 2Wiki F1, compared with 38.85% and 37.85% for the default allocation (Appendix A.29).

**Distribution across tasks and ratios.** At 16 $\times$ , ARIA’s F1 differences over CLaRa are +7.17, +6.80, +3.39, and +4.89 points on NQ, HotpotQA, MuSiQue, and 2Wiki. Across the five ratios, the Avg-F1 difference remains positive, contracting from +6.11 points at 4 $\times$  to +3.69 at 128 $\times$  (Table 1).

**Cross-backbone pattern.** At 16 $\times$ , Avg-CEM and Avg-F1 differences are positive on all three backbones (Table 2). At 128 $\times$ , the 2Wiki F1 difference reverses to  $-0.46$  on Mistral-7B and  $-1.42$  on Llama-3-8B, while Llama’s Avg-CEM difference remains +1.05 points (Appendix A.17).

**Error distribution and compression profile.** Answer-bearing text is present in the retrieved passages for 52.5% of the sampled NQ errors, categorized as evidence-present answer failures. Missing source evidence is the largest sampled MuSiQue category (56.5%). Across ARIA’s corresponding endpoint rows, Avg F1 changes from 45.93 to 38.85 as the nominal ratio changes from 4 $\times$  to 128 $\times$ , and retained evidence changes from 432 to 13 tokens.

Table 3: First-pass Recall@5 (%) at nominal 16 $\times$  under Normal retrieval, measured after each system’s complete Phase-II training and before compression and ARIA’s MTFRL/CFRS operations.

| Model    | NQ   | Hotpot | MuSiQue | 2Wiki | Avg  |
|----------|------|--------|---------|-------|------|
| CLaRa    | 71.2 | 58.4   | 34.1    | 64.7  | 57.1 |
| ARIA     | 74.3 | 64.2   | 38.2    | 68.5  | 61.3 |
| $\Delta$ | +3.1 | +5.8   | +4.1    | +3.8  | +4.2 |

Table 4: Sampled ARIA errors at 16 $\times$  Normal retrieval (200 per benchmark). M: missing source evidence; A: evidence-present answer failure; J: partial evidence and answer failure. Parentheses give within-sample percentages;  $\kappa$  is human–model label agreement.

| Benchmark | M          | A          | J         | $\kappa$ |
|-----------|------------|------------|-----------|----------|
| NQ        | 71 (35.5)  | 105 (52.5) | 24 (12.0) | 0.83     |
| HotpotQA  | 94 (47.0)  | 77 (38.5)  | 29 (14.5) | 0.82     |
| MuSiQue   | 113 (56.5) | 64 (32.0)  | 23 (11.5) | 0.81     |
| 2Wiki     | 89 (44.5)  | 84 (42.0)  | 27 (13.5) | 0.80     |

Table 5: Separately retrained coupling controls (Normal, 16 $\times$ ; five-seed means). Each row retains a rounded mean of 108 evidence-memory tokens per query; Matched full is the within-study retraining baseline. Final R@5 is measured on the final five-document set after the second retrieval and final ordering; all reported metrics are percentages. Appendix A.18 gives per-benchmark F1, candidate provenance, and statistical tests.

| Variant                  | Avg CEM      | Avg EM       | Avg F1       | Final R@5   |
|--------------------------|--------------|--------------|--------------|-------------|
| <b>Matched full</b>      | <b>41.20</b> | <b>35.49</b> | <b>44.46</b> | <b>65.3</b> |
| –CFRS, retrained         | 40.73        | 34.96        | 43.94        | 64.8        |
| Uniform ACR, retrained   | 40.64        | 34.82        | 43.80        | 64.6        |
| Static second retrieval  | 40.76        | 34.99        | 43.98        | 63.7        |
| All three off, retrained | 39.88        | 34.09        | 43.01        | 62.8        |

## 6 Conclusion

ARIA couples teacher-forced next-token-proxy ordering, score-conditioned memory gating, and one memory-conditioned second retrieval. In the Mistral-7B comparison matched on reader, corpus, Phase-II data, optimization schedule, nominal ratios, and five seeds, the complete ARIA configuration records higher mean Avg F1 than the CLaRa reimplement at all five ratios (+5.56 points at 16 $\times$ ). In the separately retrained, budget- and retrieval-topology-matched study, the all-off control records 43.01% Avg F1 versus 44.46% for its matched-full control.

## Limitations

**Empirical scope.** The experiments cover three 7–14B backbones, four English Wikipedia-based QA benchmarks, and one retrieval backbone. The

reported error proportions use four 200-error samples.

**Budgets, data, and annotation.** Compression ratios are nominal, and ARIA and CLaRa retain different evidence-memory totals at the same ratio. Phase-I data are reported by functional category and aggregate count. The overlap audits use fixed exact, n-gram, entity, and semantic thresholds. Two in-house annotators independently label the QCA and error samples; these secondary analyses do not replace full-benchmark QA evaluation. Per-operator counts for the MuSiQue augmentation were not retained.

**Attribution and statistics.** The matched retraining study equalizes the mean token budget and preserves a two-round retrieval topology, but disabling memory conditioning still changes which documents the second round introduces; it therefore does not hold final candidate identities fixed. CFRS is a teacher-forced next-token proxy. The primary 50-comparison ARIA–CLaRa family uses the stated Bonferroni correction; secondary control tables retain pointwise within-study bootstrap thresholds. All bootstrap summaries are conditional on the five trained seed pairs. Fixed-checkpoint budget-allocation and adaptive-floor results are sensitivity analyses.

**Comparability and reproducibility.** CLaRa retains its next-token-prediction objective, all-linear adapter placement, straight-through selection path, checkpoint initialization, and uniform memory allocation; published systems serve as contextual references. ARIA-NOCOMP is an inference-only diagnostic. The release package will provide executable QCA matchers, full BM25/index configurations, training and evaluation code, all reported checkpoints, per-query result files, and audit sampling/configuration manifests at [repository URL to be inserted].

## Ethical Considerations

Phase II uses publicly released QA benchmarks (NQ, HotpotQA, MuSiQue, and 2WikiMultiHopQA) and a public Wikipedia retrieval corpus. Mistral-7B-Instruct-v0.2 and Qwen-2.5-14B-Instruct use Apache 2.0; Llama-3-8B-Instruct uses the Meta Llama 3 Community License and acceptable-use policy; BGE-large-en-v1.5 uses the MIT license. We collected no sensitive participant data; two in-house annotators independently supplied the secondary-analysis labels described below. NQ contains anonymized, aggregated search queries, and the retrieval corpus contains public Wikipedia text. *Generative models used in the study*. “GPT-5.5” denotes the gpt-5.5-2026-04-23 snapshot accessed through the OpenAI Chat Completions API. It was used as a consistency check for in-house diagnostic annotations and to generate question paraphrases for the MuSiQue entity-variant augmentation. Exact roles and the sampled model-substitution check are documented in Appendix A.2. Human–human and human–model  $\kappa$  values are identified separately. *Train–test overlap*. Phase II uses the publicly released training splits of NQ, HotpotQA, MuSiQue, and 2WikiMultiHopQA together with the KILT Wikipedia retrieval corpus (Petroni et al., 2021). We deduplicated Phase II training passages against evaluation splits at the page-URL level and removed matches before training. Appendix A.2 reports the subsequent normalized 13-gram, entity-tuple, and BERTScore audits and a removal sensitivity check. The MuSiQue training augmentation (Appendix A.36) combines deterministic decomposition operators with GPT-5.5 entity-variant paraphrases over the original MuSiQue training partition; no new crowd-sourced data were collected, and the augmented partition contains no page-URL matches to test splits. The overlap audit in Appendix A.2 reports semantic similarity for the Phase-II mixture and a stratified Phase-I sample. The manuscript reports the augmentation procedure; redistribution is governed by the source MuSiQue dataset’s CC BY 4.0 terms.

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## A Supporting Methods, Results, and Reproducibility

The appendix presents additional results, implementation details, and sensitivity analyses. All experiments follow §4 unless otherwise noted.

### A.1 Method-Axis Comparison

Table A1: Latent-compression methods organized by the three operations evaluated in ARIA. “Next-token proxy” denotes teacher-forced proxy reranking.

| Method             | Next-token proxy rerank | Adaptive budget | Feedback retrieval |
|--------------------|-------------------------|-----------------|--------------------|
| Autocompressor     | ×                       | ×               | ×                  |
| ICAE               | ×                       | ×               | ×                  |
| PISCO              | ×                       | ×               | ×                  |
| CLaRa              | ×                       | ×               | ×                  |
| <b>ARIA (ours)</b> | ✓                       | ✓               | ✓                  |

### A.2 Responsible-Research Audits

**Train–test overlap audit.** We audit all 485,006 Phase-II training items remaining after exact page-URL/page-ID deduplication against every evaluation split. Exact and normalized 13-gram checks operate on supporting-passage text; the latter uses the normalization in Appendix A.38 followed by contiguous whitespace-token 13-grams. Entity-tuple matching extracts the PERSON, ORG, GPE, LOC, FAC, EVENT, PRODUCT, and WORK\_OF\_ART types used by QCA with spaCy en\_core\_web\_sm and flags an item when an unordered pair of distinct normalized entity surface forms also occurs in an evaluation passage. BERTScore F1 (Zhang et al., 2020) uses the standard English roberta-large checkpoint with IDF weighting and baseline rescaling disabled; each training item is assigned its maximum passage-text score over evaluation passages. Candidate rate is the fraction flagged by the stated automatic check. For the 13-gram, entity-tuple, and BERTScore- $\geq 0.90$  checks, every flagged item is manually reviewed; confirmed risky denotes a shared answer-bearing statement or reasoning chain that would reveal an evaluation answer, rather than topical or entity overlap alone. Both rates therefore use the full 485,006-item denominator. The 0.80 BERTScore threshold is reported as a screening rate and was not manually adjudicated. Table A2 reports the results.

For Phase I, we draw 50,000 examples from each functional category with the fixed seed sched-

Table A2: Phase-II train–test overlap audit after exact page deduplication. Rates are percentages of all 485,006 Phase-II training items, rounded to two decimals; “—” denotes a screening threshold without manual adjudication.

| Check                    | Candidate (%) | Confirmed risky (%) |
|--------------------------|---------------|---------------------|
| Exact page URL/ID        | 0.00          | 0.00                |
| Normalized 13-gram       | 0.14          | 0.02                |
| Entity tuple             | 0.76          | 0.03                |
| BERTScore F1 $\geq 0.90$ | 0.32          | 0.05                |
| BERTScore F1 $\geq 0.80$ | 2.03          | —                   |

ule and apply the same passage-level BERTScore- $\geq 0.90$  and manual-review protocol (Table A3). Across 200,000 examples, the BERTScore- $\geq 0.90$  and confirmed-risky rates round to 0.20% and 0.04%. Removing the confirmed risky items from both training stages and retraining changes Avg CEM from 41.24% to 41.19% and Avg F1 from 44.50% to 44.46% in the Dedup+ sensitivity checkpoint.

Table A3: Stratified Phase-I semantic-overlap audit (50,000 examples per category).

| Category         | BERTScore $\geq .90$ (%) | Risky (%) |
|------------------|--------------------------|-----------|
| SimpleQA         | 0.18                     | 0.03      |
| ComplexQA        | 0.25                     | 0.05      |
| Paraphrase       | 0.14                     | 0.03      |
| Entity-Augmented | 0.22                     | 0.05      |
| Overall          | 0.20                     | 0.04      |

**Independent human-annotation reliability.** Annotator 2 independently labels the same 1,000 QCA items and 800 error-taxonomy items. Agreement is computed before arbitration. The 95% intervals for Cohen’s  $\kappa$  are percentile intervals from 10,000 nonparametric item-level bootstrap resamples of the paired pre-arbitration labels. The corresponding raw agreement/Cohen’s  $\kappa$  values are 91.1%/0.85 (95% CI [0.82, 0.88]) for QCA and 89.9%/0.83 (95% CI [0.80, 0.86]) for the error taxonomy; arbitration rates are 8.9% and 10.1%. Table A4 gives the integer confusion matrices.

**Consistency-check model substitution.** On a random 200-query subset, replacing the gpt-5.5-2026-04-23 consistency check with gpt-5.4-2026-03-05 records human–GPT-5.4  $\kappa = 0.79$  (vs. 0.82 with GPT-5.5). Category-wise labels disagree on 4.5% of items. These

Table A4: Human–human confusion matrices before arbitration. Left: QCA labels ( $n = 1,000$ ). Right: error taxonomy ( $n = 800$ ; M/A/J follow Table 4). Rows are Annotator 1 and columns are Annotator 2. For QCA, S/A/H denote Simple/Multi-Aspect/Multi-Hop.

| QCA   |     |     |     |       | Error taxonomy |     |     |     |       |
|-------|-----|-----|-----|-------|----------------|-----|-----|-----|-------|
|       | S   | A   | H   | Total |                | M   | A   | J   | Total |
| S     | 252 | 18  | 11  | 281   | M              | 337 | 18  | 12  | 367   |
| A     | 15  | 124 | 15  | 154   | A              | 19  | 296 | 15  | 330   |
| H     | 8   | 22  | 535 | 565   | J              | 9   | 8   | 86  | 103   |
| Total | 275 | 164 | 561 | 1000  | Total          | 365 | 322 | 113 | 800   |

consistency-check labels are diagnostic and do not enter the deterministic router.

### A.3 Supporting Retrieval Backbone and Coupling Details

For query  $q$ , let  $D_q(r)$  be the final CCEF survivor set,  $L_{qi}$  passage  $i$ ’s source length, and  $T_{qi}(r)$  its retained memory-token count. The aggregate realized evidence-memory ratio is

$$r_{\text{eff}}(r) = \frac{\sum_q \sum_{i \in D_q(r)} L_{qi}}{\sum_q \sum_{i \in D_q(r)} T_{qi}(r)}. \quad (13)$$

CFRS changes order but not set membership.

#### A.3.1 Stage 1: Query Complexity Assessment

QCA classifies each query  $q$  into Simple, Multi-Aspect, or Multi-Hop via 38 surface-level rules (14 Multi-Hop, 12 Multi-Aspect, 12 Simple; hop indicators, aspect indicators, simplicity indicators) and named-entity counts from spaCy (Honnibal et al., 2020). The declarative rule inventory, weights  $w_\ell$ , and the 500-query dev-set development procedure are in Appendix A.6. The overall rule-activation score  $c(q) = \sum_\ell w_\ell I_\ell(q) / \sum_\ell w_\ell$ , where  $I_\ell(q) \in \{0, 1\}$  indicates that rule  $\ell$  fires, is logged only as a heuristic summary; it is neither a calibrated probability nor an input to routing. QCA evaluates the rule-specific conditions in Tables A56–A57, with the listed explicit phrase rules evaluated before count-only rules. If any Multi-Hop rule fires, the type is Multi-Hop; otherwise it is Multi-Aspect if any Multi-Aspect rule fires and Simple otherwise. Hop rules therefore take precedence over aspect rules, and the deterministic type selects the AHR defaults. Multi-Hop queries also produce deterministic subquery templates for Stage 3; IGFR fills later-hop bridge slots online, with no LLM generation. On the annotated routing sample, consensus-reference routing records 41.74% Avg CEM and 44.91% Avg F1, versus 41.14% and 44.38% for rule-based QCA;

on the full evaluation, a zero-shot LLM router records 44.69% Avg F1, versus 44.50% for rule-based QCA (Appendices A.31 and A.33).

#### A.3.2 Retrieval Pipeline (Stages 2–5)

**Stage 2 – Adaptive Hybrid Retrieval (AHR).** AHR interpolates BM25 and dense scores from BGE-large-en-v1.5 (Xiao et al., 2024) via the fixed  $W_{\text{BGE}}$  projection. The weight pair  $(w_{\text{bm25}}, w_{\text{dense}})$  is type-conditioned with defaults  $(0.75, 0.25)/(0.30, 0.70)/(0.25, 0.75)$  for Simple/Multi-Aspect/Multi-Hop respectively (Appendix A.15). We retrieve the top 4,000 from each channel, form their union, and min–max normalize each channel over that union; missing-channel entries take that channel’s minimum and an all-tied channel takes value 0.5. AHR ranks  $w_{\text{bm25}} \tilde{s}_{\text{bm25}} + w_{\text{dense}} \tilde{s}_{\text{dense}}$  and retains the top 4,000.

**Stage 3 – Iterative Gap-Filling Retrieval (IGFR).** For Multi-Hop queries, IGFR computes entity coverage  $\text{cov}(q, D) = |\mathcal{E}_q \cap \mathcal{E}_D| / |\mathcal{E}_q|$ ; the branch is entered only when  $|\mathcal{E}_q| > 0$ , so the denominator is defined. Starting from  $D^{(0)} = D_{\text{AHR}}$ , for  $k = 1, \dots, K_{\text{igfr}}$  the procedure stops early if  $\text{cov}(q, D^{(k-1)}) \geq \gamma$  ( $\gamma = 0.5$ ), or if no unused template remains. Otherwise, it instantiates the next QCA template using bridge entities extracted from  $D^{(k-1)}$ , retrieves at most 200 new candidates for that iteration, and sets  $D^{(k)} = D^{(k-1)} \cup R_k$  after duplicate removal. This is a surface-heuristic decomposition, not the model-generated chain-of-thought–retrieval interleaving used by IRCOT (Trivedi et al., 2023).  $K_{\text{igfr}} = 0$  returns  $D^{(0)}$  unchanged; the resulting union is passed to MADS.

**Stage 4 – Multi-Axis Document Scoring (MADS).** Three scoring modules—Lexical  $\mathcal{S}_{\text{lex}}$  (word-unigram TF–IDF cosine, fitted to the query and its current candidate pool), Semantic  $\mathcal{S}_{\text{sem}}$

(cosine between the L2-normalized projected query and BGE-large-en-v1.5 document embedding), and Entity  $\mathcal{S}_{\text{ent}}$  (named-entity coverage)—operate on each candidate. Entity score is  $|\mathcal{E}_q \cap \mathcal{E}_d|/|\mathcal{E}_q|$ , set to 0 when  $\mathcal{E}_q$  is empty, using spaCy en\_core\_web\_sm. Each module is min-max normalized per query. For module  $j$ , let  $R_j = s_j^{\text{max}} - s_j^{\text{min}}$ ; then

$$\tilde{s}_{ij} = \begin{cases} (s_{ij} - s_j^{\text{min}})/R_j, & R_j > \epsilon, \\ 0.5, & R_j \leq \epsilon. \end{cases}$$

Scores are combined with equal weights  $w_j = \frac{1}{3}$ , selected on the dev partition. In the evaluation grid in Appendix A.27, learned per-module weights record 0.20 pp lower Avg F1. Inter-module dispersion feeds CCEF’s agreement signal. MADS uses a stable descending sort, breaks exact score ties by corpus document ID, and re-ranks the current candidate pool to top-100.

**Stage 5 – Cross-Component Evidence Fusion (CCEF).** CCEF fuses the three normalized module scores with an agreement multiplier:

$$s_{\text{fused}}(d_i) = (\sum_j w_j \tilde{s}_{ij})[\alpha + (1-\alpha)\text{agr}_i], \quad (14)$$

with  $\alpha = 0.5$  and  $\text{agr}_i = \text{clip}(1 - \sigma_i/(\mu_i + \epsilon), 0, 1)$ , where  $\mu_i = \text{mean}_j(\tilde{s}_{ij})$  and  $\sigma_i$  is the population standard deviation across the three modules. The bounded heuristic multiplier reaches 1 at perfect agreement and falls toward  $\alpha$  otherwise. Documents below  $\tau = 0.30$  are discarded, after which at most the top-5 survivors are retained. Precisely, CCEF stably sorts  $\{d_i : s_{\text{fused}}(d_i) \geq \tau\}$  by descending fused score, uses incoming MADS order only to break exact fused-score ties, and retains the first five documents. If this set is empty, it retains the first document attaining  $\max_i s_{\text{fused}}(d_i)$  under incoming MADS order. Downstream equations therefore use the realized set size  $1 \leq N \leq 5$ .

**Inference algorithm.** Algorithm 1 gives ARIA’s end-to-end inference pipeline. The QR forward pass is listed first because AHR and MADS use  $q_{\text{rep}}$ ; QCA operates on surface-level rules. IGFR consumes QCA’s subquery templates and coverage signal, while CCEF fuses MADS outputs.

**CFRS training and inference branches.** For each selected candidate, CFRS computes the teacher-forced squared-probability proxy in Eq. 6 from the final ACR-processed memory supplied

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#### Algorithm 1 ARIA inference pipeline.

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**Require:** Query  $q$ ; corpus  $\mathcal{C}$ ; frozen BGE index, pre-fitted fixed  $W_{\text{BGE}}$ , and fixed rules; trained QR, compressor, generator, and  $P_{\text{fb}}$ .

- 1:  $q_{\text{rep}} \leftarrow \text{QR}(q)$  ▷ query representation
- 2: type,  $c(q)$ ,  $\{u_k\} \leftarrow \text{QCA}(q)$  ▷  $c(q)$  logged only
- 3:  $D_{\text{AHR}} \leftarrow \text{AHR}(q_{\text{rep}}, \text{type}; \mathcal{C})$ ,  $|D_{\text{AHR}}| = 4,000$  ▷ Stage 2
- 4:  $D_{\text{pool}} \leftarrow D_{\text{AHR}}$
- 5: **if** type = Multi-Hop and  $|\mathcal{E}_q| > 0$  and  $\text{cov}(q, D_{\text{pool}}) < \gamma$  **then**
- 6:  $D_{\text{pool}} \leftarrow \text{IGFR}(\{u_k\}, D_{\text{pool}}; K_{\text{igfr}}, \gamma)$  ▷ Stage 3
- 7: **end if**
- 8:  $D_{\text{MADS}}^{(1)} \leftarrow \text{MADS}(q_{\text{rep}}, D_{\text{pool}})$ ,  $|D_{\text{MADS}}^{(1)}| = 100$  ▷ Stage 4
- 9:  $D_1 \leftarrow \text{CCEF}(D_{\text{MADS}}^{(1)}; \tau, \alpha)$ ,  $N \leftarrow |D_1| \in [1, 5]$  ▷ Stage 5
- 10:  $\{\rho_i\} \leftarrow \text{ACR-scale}(s_{\text{fused}}(D_1))$  ▷ Eq. 10
- 11:  $\{\mathbf{m}_i^{(1)}\} \leftarrow \text{ACR-Hard}(\text{Compressor}(D_1), \{\rho_i\})$  ▷ Eq. 9
- 12:  $\overline{\mathbf{m}}_{\text{inf}} \leftarrow \text{PoolInf}(\{\mathbf{m}_i^{(1)}\})$  ▷ Eq. 11
- 13:  $q_{\text{fb}} \leftarrow \text{Normalize}_{2, \epsilon}(P_{\text{fb}}(\overline{\mathbf{m}}_{\text{inf}}))$
- 14:  $D_2 \leftarrow \text{DenseSearch}(\text{sg}[q_{\text{fb}}], \mathcal{C})$ ,  $|D_2| = 200$  ▷ 2nd round
- 15:  $D' \leftarrow \text{MADS}(q_{\text{rep}}, D_1 \cup D_2)$ ,  $|D'| = 100$
- 16:  $D \leftarrow \text{CCEF}(D'; \tau, \alpha)$ ,  $|D| \in [1, 5]$
- 17:  $\{\rho_i^{\text{final}}\} \leftarrow \text{ACR-scale}(s_{\text{fused}}(D))$  ▷ recompute on final  $D$
- 18:  $\{\mathbf{m}_i^{\text{final}}\} \leftarrow \text{ACR-Hard}(\text{Compressor}(D), \{\rho_i^{\text{final}}\})$
- 19:  $\{e_i\}_{d_i \in D} \leftarrow \text{CFRSReconError}(\{(d_i, \mathbf{m}_i^{\text{final}})\}_{d_i \in D})$  ▷ Eq. 6
- 20:  $\{f_i^{\text{final}}\} \leftarrow \text{CFRSNorm}(\{\text{sg}[e_i]\}_{d_i \in D})$  ▷ Eq. 7
- 21:  $s_{\text{final}}(d_i) \leftarrow (1 - w_{\text{cfrs}})s_{\text{fused}}(d_i) + w_{\text{cfrs}}f_i^{\text{final}}$  ▷ Eq. 8
- 22:  $\pi \leftarrow \text{stable-argsort}_{i \in D} s_{\text{final}}(d_i)$  ▷ descending; CCEF tie order
- 23:  $\hat{a} \leftarrow \text{Generator}(q, [\mathbf{m}_{\pi(1)}^{\text{final}}, \dots, \mathbf{m}_{\pi(|D|)}^{\text{final}}])$
- 24: **return**  $\hat{a}$  ▷ realizes Eq. 1

---

to the generator. During training, the undetached value enters  $\mathcal{L}_{\text{CFRS}}$  and backpropagates through the frozen decoder to the compressor input. For passage ordering, CFRS normalizes the stopped-gradient copy  $\bar{e}_i$  into  $f_i$  and applies Eq. 8. Thus the proxy loss and hard reranking share the same measured quantity but have separate gradient and selection branches.

#### A.4 Retrieval-Space Maps and Auxiliary Objectives

**Supporting-passage rule.** For a Phase II example  $x$ , let  $\mathcal{P}(x)$  be its possibly empty set of annotated supporting corpus passages, matched by corpus page ID. Let  $\text{pid}(d)$  denote passage  $d$ ’s page ID and, for any passage set  $S$ , let  $\text{pid}(S) = \{\text{pid}(d) : d \in S\}$ . Define  $\mathcal{D}_{\text{sup}} = \{x : \mathcal{P}(x) \neq \emptyset\}$ . For each seeded training occurrence  $x \in \mathcal{D}_{\text{sup}}$ , after its current first-pass set  $D_1(x)$  has been computed, define

$$\mathcal{P}_{\text{miss}}(x) = \{d \in \mathcal{P}(x) : \text{pid}(d) \notin \text{pid}(D_1(x))\}.$$

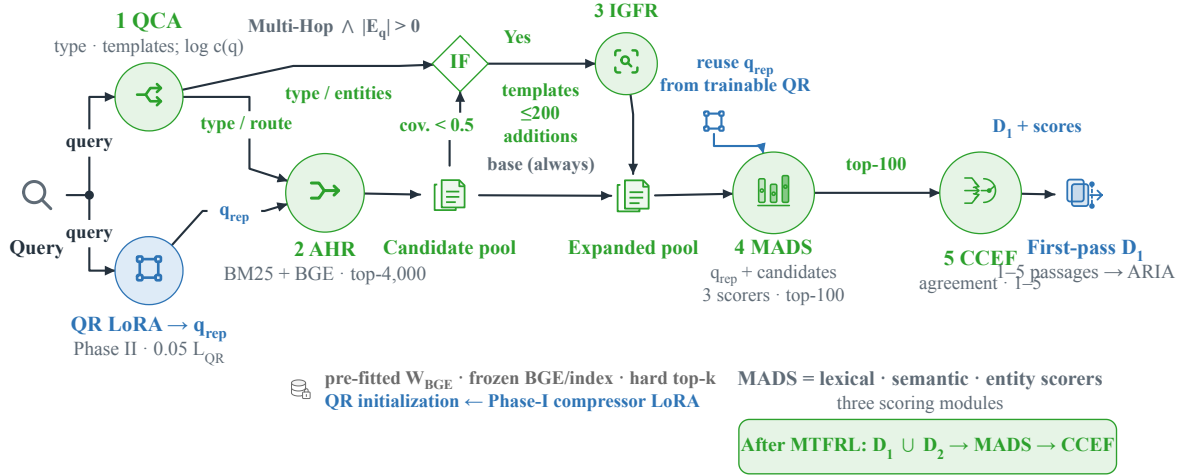


Figure A1: **Retrieval backbone used in the ARIA experiments.** QCA conditions AHR; IGFR conditionally expands multi-hop candidates; MADS and CCEF produce the agreement-weighted evidence set supplied to the coupling layer. The local  $q_{rep}$  glyph beside MADS reuses the same Phase-II-trained QR output shown on the left. After MTFRL’s second-round retrieval,  $D_1 \cup D_2$  is re-scored by MADS and then filtered by CCEF.

The run-seeded sampler draws the shared positive  $d^+(x)$  uniformly from  $\mathcal{P}_{miss}(x)$  when that set is nonempty and otherwise from  $\mathcal{P}(x)$ . Thus an annotated support absent from the current first-pass survivor set is selected whenever available. Both  $\mathcal{L}_{QR}$  and  $\mathcal{L}_{MTFRL}$  use this same  $d^+(x)$ . For a minibatch  $\mathcal{B}$ , define the in-batch pool

$$\begin{aligned} \mathcal{I}_{\mathcal{B}}(x) &= \{d^+(x') : x' \in (\mathcal{B} \cap \mathcal{D}_{sup}) \setminus \{x\}\}, \\ \mathcal{U}(x) &= \{d \in D_1(x) : \text{pid}(d) \notin \text{pid}(\mathcal{P}(x))\} \\ &\quad \cup \mathcal{I}_{\mathcal{B}}(x), \\ \tilde{\mathcal{U}}(x) &= \text{Unique}_{\text{pid}}(\mathcal{U}(x)), \\ \mathcal{N}(x) &= \{d \in \tilde{\mathcal{U}}(x) : \text{pid}(d) \notin \text{pid}(\mathcal{P}(x))\}. \end{aligned}$$

The MTFRL negatives for  $x$  are  $\mathcal{N}(x)$ , which combines all non-supporting first-pass survivors with in-batch positives, deduplicates by page ID, and masks annotated false negatives. Define  $\mathcal{B}_{sup} = \mathcal{B} \cap \mathcal{D}_{sup}$  and  $\mathcal{B}_{mt} = \{x \in \mathcal{B}_{sup} : \mathcal{N}(x) \neq \emptyset\}$ . Examples without document-level support annotations still contribute to  $\mathcal{L}_{QA}$ ,  $\mathcal{L}_{MSE}$ , and  $\mathcal{L}_{CFRS}$ .

**Pre-fitted query-space map.** Before Phase I/II training, the linear map  $W_{BGE} : \mathbb{R}^{d_h} \rightarrow \mathbb{R}^{1024}$  is fitted once per backbone from its base QR representation on 50,000 question-supporting-passage pairs from the training corpus. Its normalized output is aligned by cosine loss to the frozen BGE document embedding of the supporting passage. This fitting uses AdamW, learning rate  $5 \times 10^{-4}$ , batch size 128, and two epochs.  $W_{BGE}$  is there-

after shared across compression ratios for that backbone and excluded from downstream optimization. After Phase I, each ratio-specific QR LoRA adapter is initialized from its corresponding compressor adapter. In Phase II, Eq. 15 updates that adapter against the frozen BGE embedding of the shared positive defined above while retaining the fixed map and document index. This defines a positive-pair cosine-alignment regularizer. With  $e_{d^+} = \text{Normalize}_{2,\epsilon}(E_{BGE}^d(d^+))$  and  $z_q = \text{Normalize}_{2,\epsilon}(W_{BGE}q_{rep})$ , the objective is

$$\begin{aligned} \ell_{QR}(x) &= 1 - z_q^\top e_{d^+(x)}, \\ \mathcal{L}_{QR} &= \begin{cases} \frac{1}{|\mathcal{B}_{sup}|} \sum_{x \in \mathcal{B}_{sup}} \ell_{QR}(x), & |\mathcal{B}_{sup}| > 0, \\ 0, & |\mathcal{B}_{sup}| = 0. \end{cases} \end{aligned} \quad (15)$$

**Trainable feedback projection.** The two-layer map

$$P_{fb}(x) = W_2 \text{GELU}(W_1 x + b_1) + b_2 \quad (16)$$

uses  $4096 \rightarrow 2048 \rightarrow 1024$  for Mistral-7B/Llama-3-8B and  $5120 \rightarrow 2560 \rightarrow 1024$  for Qwen-2.5-14B. Its weights use Xavier-uniform initialization and its biases are zero. Unlike  $W_{BGE}$ , all parameters of  $P_{fb}$  are updated in Phase II. Equation 17 uses the positive and negative sets defined above. The temperature is  $\tau_{fb} = 0.05$ . Writing

$s_x(d) = \mathbf{q}_{\text{fb}}^\top \mathbf{e}_d / \tau_{\text{fb}}$  and  $\mathcal{A}(x) = \{d^+(x)\} \cup \mathcal{N}(x)$ ,

$$\ell_{\text{mt}}(x) = -s_x(d^+(x)) + \text{LSE}_{d \in \mathcal{A}(x)} s_x(d),$$

$$\mathcal{L}_{\text{MTFRL}} = \begin{cases} \frac{1}{|\mathcal{B}_{\text{mt}}|} \sum_{x \in \mathcal{B}_{\text{mt}}} \ell_{\text{mt}}(x), & |\mathcal{B}_{\text{mt}}| > 0, \\ 0, & |\mathcal{B}_{\text{mt}}| = 0. \end{cases} \quad (17)$$

Here LSE is the log-sum-exp operator; the implementation uses the equivalent log-softmax form for numerical stability.

**Loss paths and discrete boundaries.** The CFRS auxiliary term averages the undetached document errors from Eq. 6:

$$\mathcal{L}_{\text{CFRS}} = \mathbb{E}_x \left[ \frac{1}{|D|} \sum_{d_i \in D} e_i \right]. \quad (18)$$

Equation 3 uses the default coefficients  $(\lambda, \mu, \nu, \xi) = (0.10, 0.10, 0.05, 0.05)$ ; reported variants state any change explicitly. The QA and hidden-state MSE terms update the compressor and generator adapters;  $\mathcal{L}_{\text{CFRS}}$  additionally updates the compressor through the frozen decoder used for proxy evaluation;  $\mathcal{L}_{\text{QR}}$  updates the QR adapter through fixed  $W_{\text{BGE}}$ ; and  $\mathcal{L}_{\text{MTFRL}}$  updates  $P_{\text{fb}}$  and the soft memory path. The score copy used by CFRS ordering and the feedback-query copy sent to the BGE index are detached. Accordingly, gradients stop at hard sorting, index lookup, and top- $k$  selection, while the continuous modules are optimized by their stated objectives.

**One Phase II training step.** Algorithm 2 makes the two loss/search branches explicit. FIRSTPASS denotes QCA-AHR-IGFR-MADS-CCEF, and FINALSELECT denotes the second MADS-CCEF pass. Both return detached document identities and scores.

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#### Algorithm 2 One ARIA Phase II training step.

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**Require:** Minibatch  $\mathcal{B}$ ; corpus  $\mathcal{C}$ ; current QR, compressor, and generator LoRA adapters and  $P_{\text{fb}}$ ; frozen base models,  $W_{\text{BGE}}$ , BGE index, and fixed rules.

- 1: **for all**  $x = (q, a^*, \mathcal{P}(x)) \in \mathcal{B}$  **do**
- 2:  $\mathbf{q}_{\text{rep}}(x) \leftarrow \text{QR}(q); \quad D_1(x) \leftarrow \text{sg}[\text{FIRSTPASS}(q, \mathbf{q}_{\text{rep}}(x); \mathcal{C})]$
- 3:  $\mathbf{M}_1(x) \leftarrow \text{ACR-Soft}(\text{Compressor}(D_1(x)))$
- 4:  $\mathbf{q}_{\text{fb}}^{\text{loss}}(x) \leftarrow \text{Normalize}_{2,\epsilon}(P_{\text{fb}}(\text{PoolTrain}(\mathbf{M}_1(x))))$
- 5:  $D_2(x) \leftarrow \text{DenseSearch}(\text{sg}[\mathbf{q}_{\text{fb}}^{\text{loss}}(x)], \mathcal{C})$
- 6:  $D(x) \leftarrow \text{sg}[\text{FINALSELECT}(q, \mathbf{q}_{\text{rep}}(x), D_1(x) \cup D_2(x))]$
- 7:  $\mathbf{M}(x) \leftarrow \text{ACR-Soft}(\text{Compressor}(D(x)))$
- 8:  $e_i^{\text{loss}}(x) \leftarrow \text{CFRSReconError}_{\phi_0}(d_i, \mathbf{M}_i(x))$  for  $d_i \in D(x)$   $\triangleright$  generator LoRA disabled
- 9:  $e_i^{\text{rank}}(x) \leftarrow \text{sg}[e_i^{\text{loss}}(x)]; \quad \pi(x) \leftarrow \text{sg}[\text{CFRSOrder}(D(x), \{e_i^{\text{rank}}(x)\})]$
- 10:  $\mathcal{L}_{\text{QA}}^x, \mathcal{L}_{\text{MSE}}^x \leftarrow \text{TeacherForcedQA}(q, a^*, \mathbf{M}_{\pi(x)}(x))$
- 11: **end for**
- 12: Using the stored current  $D_1(x)$ , select  $d^+(x)$ ,  $\mathcal{N}(x)$ ,  $\mathcal{B}_{\text{sup}}$ , and  $\mathcal{B}_{\text{mt}}$  by the supporting-passage rule above.
- 13: Compute  $\mathcal{L}_{\text{CFRS}}$  from  $e^{\text{loss}}(x)$ ,  $\mathcal{L}_{\text{QR}}$  from  $\mathbf{q}_{\text{rep}}(x)$ , and  $\mathcal{L}_{\text{MTFRL}}$  from  $\mathbf{q}_{\text{fb}}^{\text{loss}}(x)$ .
- 14: Form Eq. 3 and update only the three LoRA adapters and  $P_{\text{fb}}$ .

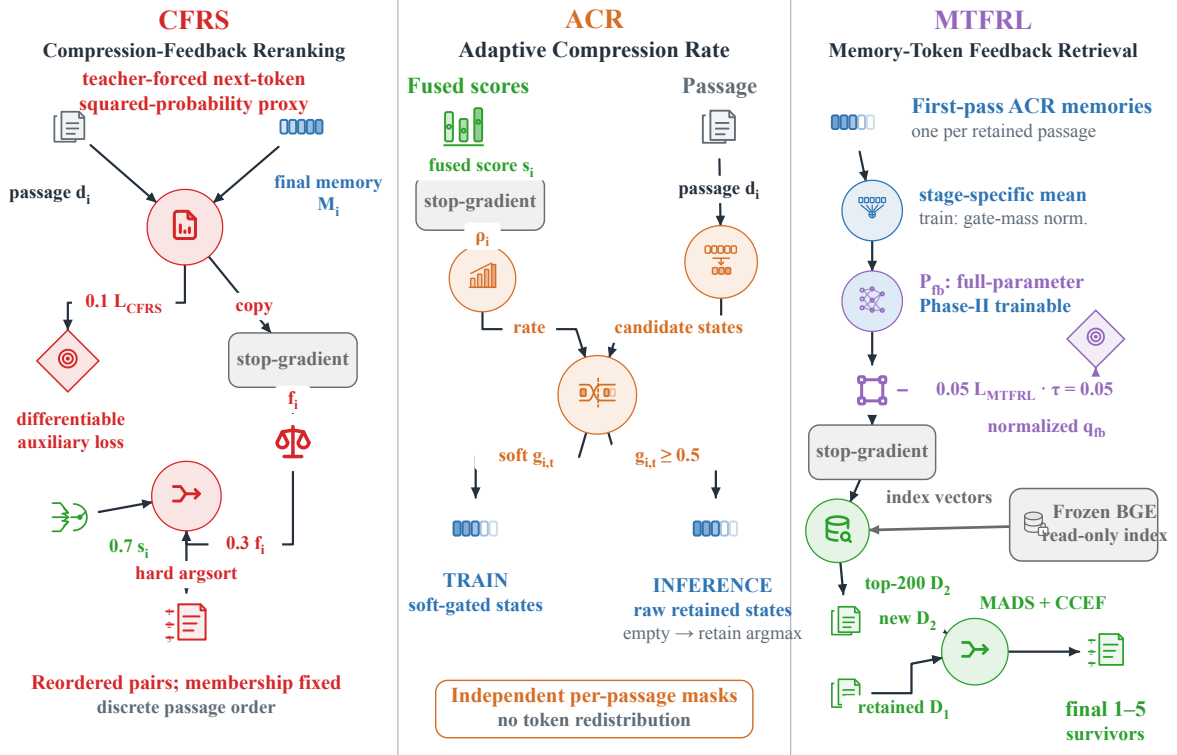
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### A.5 Phase II Training Dynamics

Figure A3 plots Phase II diagnostics on the four-benchmark validation set under Normal retrieval at nominal CR=16 $\times$  for the seed-42 run. At the first and last displayed samples,  $\mathcal{L}_{\text{QA}}$  is 2.4224 and 1.3721,  $\lambda \cdot \mathcal{L}_{\text{MSE}}$  is 0.4753 and 0.1593, and validation Avg CEM is 35.93% and 40.15%, respectively. Final evaluation-set values are reported separately.

### A.6 QCA Rules and Routing Diagnostics

**QCA declarative rule inventory.** Tables A56 and A57 list the 38 surface-level rule families used by QCA, organized by category. Each rule  $\ell$  contributes indicator  $I_\ell(q) \in \{0, 1\}$  and weight  $w_\ell$ ; the diagnostic rule-activation score is  $c(q) = \sum_\ell w_\ell I_\ell(q) / \sum_\ell w_\ell$ . This score is logged for analysis only and does not select the route. All string matching is case-insensitive; NER is performed with spaCy (Honnibal et al., 2020) en\_core\_web\_sm. All reported QCA results use this model. The tables summarize the declarative rule families. Explicit phrase rules are evaluated before count-only rules. A query is classified as Multi-Hop if any Multi-Hop rule fires, as Multi-Aspect if no hop rule but at least one aspect rule fires, and as Simple otherwise. Thus hop rules take precedence over aspect rules; Simple rules provide matched-rule diagnostics within the Sim-



Hard argsort and top-k are discrete selection boundaries outside autograd.

Figure A2: **Detailed retrieval-compression coupling mechanisms.** CFRS uses the same teacher-forced squared-probability proxy as an auxiliary compressor loss and, through a detached copy, as a hard ordering score; ACR gates latent positions independently for each passage; MTFRL trains  $P_{fb}$  and the soft memory path with a supporting-passage contrastive objective, while a detached feedback query drives the second dense search over the frozen index. Solid arrows denote forward computation; explicit stop-gradient boxes mark detached copies, and the gray index arrow denotes a frozen resource input. Hard argsort and top- $k$  are discrete selection boundaries.

ple route and S12 is the fallback. Rules within each group are ordered by descending weight  $w_\ell$  for diagnostic reporting only.

**Sub-question generation for IGFR.** For queries classified as Multi-Hop, QCA additionally produces deterministic sub-questions through templates for bridge connectives, possessive nesting, and “who-action” patterns (rules H02–H12 and H14 in Table A56). Each detected bridge splits the query into a prefix sub-question (targeting the bridge entity, e.g., “Where is Microsoft HQ?”) and a suffix sub-question instantiated with the resolved bridge (e.g., “tallest mountain Washington state”); the resolution happens online during IGFR, using entities extracted from retrieved passages. For count-driven H01/H13 cases, a deterministic fallback takes the first two named-entity spans in surface order: the first template queries the main

predicate around the first span, and the second retains the second span and predicate while replacing the first with a {BRIDGE} slot. If that parse fails, the sole fallback template is the original query. IGFR stops when the template list is exhausted. This stage uses deterministic templates and no LLM generation.

**Reference-label sample.** The routing diagnostics use 1,000 manually annotated queries stratified across the four benchmarks: NQ = 225, HotpotQA = 257, MuSiQue = 84, and 2Wiki = 434. The operational taxonomy labels a query Simple when it requests one fact, Multi-Aspect when it requests multiple attributes or a comparison without a bridge dependency, and Multi-Hop when it requires linked evidence across at least two facts. Two in-house annotators independently assigned labels before arbitration, with 91.1% raw agree-

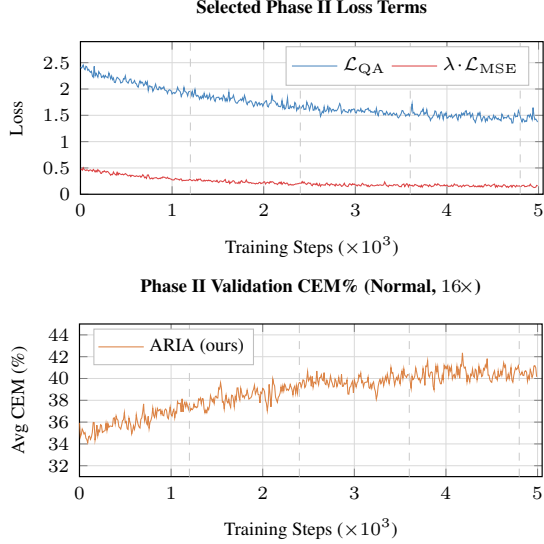


Figure A3: Selected Phase II training diagnostics for seed = 42 at CR = 16 $\times$ . The curves are unsmoothed and sampled every 10 optimizer steps. **Upper:**  $\mathcal{L}_{QA}$  and  $\lambda \cdot \mathcal{L}_{MSE}$ ; the three auxiliary terms in Eq. 3 are not plotted. **Lower:** validation Avg CEM% (four benchmarks, Normal, 16 $\times$ ). Dashed vertical lines mark the 1,200-step epoch boundaries. The plot displays optimizer steps 0–5,000.

ment and Cohen’s  $\kappa = 0.85$  (95% CI [0.82, 0.88]). GPT-5.5 served as a separate consistency check against Annotator 1 ( $\kappa = 0.82$  human–model agreement). Appendix A.31 reports the corresponding reference-label routing intervention.

Table A5 reports the observed with-QCA versus without-QCA endpoints by question-complexity stratum. For reporting, both variants are stratified by the Full-ARIA QCA route recorded before the intervention; the without-QCA path does not consume those route labels. The without-QCA rows are fixed-checkpoint routing interventions on the same ARIA-trained checkpoints. QCA is bypassed, and routing-dependent stages receive the implementation’s fixed fallback inputs rather than the recorded route labels. The largest differences occur on Multi-Hop questions, specifically +3.00 pp on NQ, +2.80 pp on HotpotQA, +2.00 pp on MuSiQue, and +3.50 pp on 2Wiki. The observed NQ differences are +0.57 pp for Simple and +1.82 pp for Multi-Aspect questions.

#### A.7 Sampled Error Categorization and Additional Controlled Experiments

Table 4 in the main text reports the category counts, within-sample proportions, and human–model agreement values. Appendix A.37 gives

one observed case from each category.

The main operational sensitivity analyses (§4.5) report performance differences when individual pipeline stages and coupling mechanisms are disabled in trained Full-ARIA checkpoints. Two further evaluations report: (a) an end-to-end direct-context configuration without memory-token compression, and (b) a matched MuSiQue training-partition control.

**ARIA retrieval without compression (Appendix A.21).** ARIA-NOCOMP—ARIA’s five-stage retrieval feeding the top-5 documents directly into Mistral-7B without memory-token compression—records 45.43% Avg F1 (5 seeds), 0.93 points above the compressed ARIA 16 $\times$  configuration. The largest per-benchmark difference is on MuSiQue, where ARIA-NOCOMP is 6.34 points higher; the compressed configuration is 0.81, 0.80, and 1.05 points higher on NQ, HotpotQA, and 2Wiki, respectively.

**MuSiQue training-partition control (Appendix A.22).** Under a controlled re-training of both ARIA and CLaRa on the standard MuSiQue partition (19,938 examples) rather than the augmented split (168,745), the ARIA–CLaRa Avg-F1 gap narrows from +5.56 pp to +5.50 pp, a measured change of 0.06 pp in the reported aggregate comparison. The larger per-benchmark change occurs on MuSiQue. Full per-benchmark numbers are in Table A34.

#### A.8 Complete Compressor Results: Normal and Oracle Retrieval

The largest measured mean CEM difference between ARIA and CLaRa is +2.41 pp at moderate compression (16 $\times$ ) and narrows to +0.16 pp at 128 $\times$  ( $p \geq 0.05$  in the pointwise bootstrap summary); under Oracle retrieval the largest measured difference is +3.04 pp at 16 $\times$  (Figure 3). All 128 $\times$  results use  $K_{igfr} = 1$  (selected per Appendix A.24). ARIA trails CLaRa at 128 $\times$  on MuSiQue and 2Wiki in the displayed 128 $\times$  results.

#### A.9 Complete End-to-End QA Results

In the protocol-controlled comparison, ARIA records higher mean Avg F1 than the protocol-controlled CLaRa reimplementation at every ratio from 4 $\times$  to 128 $\times$ . DRO’s published 41.96% single-run Avg F1 and ARIA’s 4 $\times$ –32 $\times$  values come from different settings and serve as source-reported contextual rows. At 128 $\times$ , ARIA trails CLaRa on

Table A5: CEM% by the Full-ARIA QCA route recorded before intervention, with and without QCA under Normal retrieval at CR = 16×. Both variants use this reporting partition; the without-QCA path does not consume the route labels.

| Variant | CR  | Type         | NQ           | HotpotQA     | MuSiQue      | 2Wiki        |
|---------|-----|--------------|--------------|--------------|--------------|--------------|
| w/ QCA  | 16× | Simple       | <b>62.21</b> | <b>51.15</b> | <b>14.83</b> | <b>55.64</b> |
| w/ QCA  | 16× | Multi-hop    | <b>54.93</b> | <b>41.82</b> | <b>10.47</b> | <b>43.87</b> |
| w/ QCA  | 16× | Multi-aspect | <b>58.93</b> | <b>45.74</b> | <b>12.36</b> | <b>49.23</b> |
| w/o QCA | 16× | Simple       | 61.64        | 50.87        | 14.53        | 54.84        |
| w/o QCA | 16× | Multi-hop    | 51.93        | 39.02        | 8.47         | 40.37        |
| w/o QCA | 16× | Multi-aspect | 57.11        | 44.58        | 11.06        | 47.23        |

Table A6: Complete compressor CEM% under Normal and Oracle retrieval. ARIA and CLaRa: 5-seed means; other baselines: single-run. Values for contextual baselines other than Self-RAG/CRAG use their source-reported metrics and normalization. Source-reported averages are retained rather than recomputed from independently rounded cells. \*CLaRa: 5-seed means. <sup>◊</sup>All 128× ARIA rows use  $K_{\text{igfr}} = 1$  (selected on dev partition; Appendix A.24).

| Model                   | CR   | Ret. | NQ           | HotpotQA     | MuSiQue      | 2Wiki        | Avg          |
|-------------------------|------|------|--------------|--------------|--------------|--------------|--------------|
| <i>Normal Retrieval</i> |      |      |              |              |              |              |              |
| Autocompressor          | 1×   | N    | 17.24        | 14.61        | 3.81         | 19.89        | 13.89        |
| Mistral-7B              | 1×   | N    | 54.58        | 42.94        | 8.94         | 44.24        | 37.67        |
| LLMLingua-2             | 4×   | N    | 47.53        | 37.05        | 9.02         | 44.35        | 34.49        |
| PISCO                   | 16×  | N    | 54.39        | 41.94        | 10.09        | 44.88        | 37.83        |
| Self-RAG                | 1×   | N    | 56.21        | 44.83        | 10.81        | 46.12        | 39.49        |
| CRAG                    | 1×   | N    | 55.84        | 45.18        | 11.23        | 46.52        | 39.69        |
| CLaRa                   | 4×   | N    | 57.18        | 45.24        | 10.40        | 47.08        | 39.98*       |
| CLaRa                   | 16×  | N    | 55.72        | 43.86        | 10.22        | 45.52        | 38.83*       |
| CLaRa                   | 32×  | N    | 54.79        | 42.55        | 9.72         | 44.53        | 37.90*       |
| CLaRa                   | 64×  | N    | 53.91        | <b>42.72</b> | 9.88         | 44.10        | 37.65*       |
| CLaRa                   | 128× | N    | 53.06        | 41.14        | <b>9.61</b>  | <b>43.19</b> | 36.75*       |
| ARIA                    | 4×   | N    | <b>59.82</b> | <b>47.31</b> | <b>12.15</b> | <b>49.17</b> | <b>42.11</b> |
| ARIA                    | 16×  | N    | <b>58.64</b> | <b>46.10</b> | <b>11.68</b> | <b>48.53</b> | <b>41.24</b> |
| ARIA                    | 32×  | N    | <b>56.81</b> | <b>44.02</b> | <b>9.98</b>  | <b>45.60</b> | <b>39.10</b> |
| ARIA                    | 64×  | N    | <b>55.30</b> | 42.50        | <b>10.05</b> | <b>44.65</b> | <b>38.13</b> |
| ARIA                    | 128× | N    | <b>54.05</b> | <b>41.65</b> | 9.55         | 42.40        | <b>36.91</b> |
| <i>Oracle Retrieval</i> |      |      |              |              |              |              |              |
| Mistral-7B              | 1×   | O    | 71.64        | 70.77        | 45.72        | 68.83        | 64.24        |
| LLMLingua-2             | 4×   | O    | 63.99        | 52.42        | 27.47        | 53.92        | 49.45        |
| PISCO                   | 16×  | O    | 73.44        | 66.53        | 33.80        | 60.45        | 58.56        |
| CLaRa                   | 4×   | O    | 76.38        | 73.96        | 46.14        | 70.64        | 66.78*       |
| CLaRa                   | 16×  | O    | 75.37        | 70.93        | 43.06        | 66.31        | 63.92*       |
| CLaRa                   | 32×  | O    | 73.06        | 68.35        | 38.54        | 64.17        | 61.03*       |
| CLaRa                   | 64×  | O    | 71.42        | 64.25        | 35.12        | <b>60.89</b> | 57.92*       |
| CLaRa                   | 128× | O    | 68.84        | 61.93        | 31.12        | <b>58.59</b> | 55.12*       |
| ARIA                    | 4×   | O    | <b>79.10</b> | <b>76.95</b> | <b>47.41</b> | <b>73.30</b> | <b>69.19</b> |
| ARIA                    | 16×  | O    | <b>78.10</b> | <b>74.08</b> | <b>45.92</b> | <b>69.74</b> | <b>66.96</b> |
| ARIA                    | 32×  | O    | <b>75.20</b> | <b>69.50</b> | <b>40.10</b> | <b>65.20</b> | <b>62.50</b> |
| ARIA                    | 64×  | O    | <b>72.40</b> | <b>65.80</b> | <b>36.74</b> | 60.35        | <b>58.82</b> |
| ARIA                    | 128× | O    | <b>69.40</b> | <b>62.10</b> | <b>31.80</b> | 58.20        | <b>55.38</b> |

2Wiki F1/EM in the displayed 128 $\times$  results (per-benchmark MuSiQue/2Wiki CEM reversals are documented in Appendix A.8). Under Oracle retrieval, the measured 4 $\times$  difference is +4.09 pp.

### A.10 Oracle Top-100 Reranking Recall Results

Table A8 reports Recall@ $k$  after system-specific reranking of an Oracle top-100 candidate pool containing annotated positives, across all four benchmarks and compression ratios.

*Matched ARIA–CLaRa recall differences at 4 $\times$ .* Within the protocol-controlled ARIA–CLaRa comparison at 4 $\times$ , ARIA records higher Recall@ $k$  at every displayed  $k$  in Table A8. The 1 $\times$  rows are source-reported contextual references under CLaRa’s Oracle evaluation and are not part of the protocol-controlled comparison. The R@5 differences between ARIA and the CLaRa reimplementation are: +3.12 pp on NQ, +3.39 pp on HotpotQA, +2.94 pp on MuSiQue, and +2.97 pp on 2Wiki. These aggregate measurements summarize the recorded Oracle reranking output.

*Oracle Recall endpoints from 4 $\times$  to 128 $\times$ .* From 4 $\times$  to 128 $\times$ , ARIA’s R@1 on NQ drops by 3.61 pp from 33.42 to 29.81, while CLaRa drops by 6.82 pp from 32.62 to 25.80, so ARIA’s measured drop is 3.21 pp smaller. On MuSiQue R@1, ARIA drops by 11.70 pp from 31.50 to 19.80, versus CLaRa’s 9.71 pp from 30.45 to 20.74, resulting in ARIA trailing CLaRa at 128 $\times$ . Four adjacent-CR reversals occur: (CLaRa MuSiQue R@3: 32 $\times$  = 42.58, 64 $\times$  = 43.12; CLaRa 2Wiki R@5: 32 $\times$  = 77.33, 64 $\times$  = 77.42; ARIA MuSiQue R@3: 32 $\times$  = 49.34, 64 $\times$  = 49.82; ARIA 2Wiki R@5: 32 $\times$  = 80.65, 64 $\times$  = 81.05). Their absolute changes are at most 0.54 pp.

At 128 $\times$ , CLaRa records higher R@1 on MuSiQue and 2Wiki, while ARIA records higher displayed R@3 and R@5 on both benchmarks.

### A.11 Pretraining Data Composition

Table A9 reports four cumulative pretraining-mixture configurations and a standalone paraphrase-only control (CEM%, Oracle, nominal CR=32 $\times$ ). Adding Entity-Augmented data to the SQ+CQ+Para configuration is associated with a +1.99 pp Avg-CEM difference (MuSiQue +2.95 pp; 2Wiki +1.25 pp). The No-Pretrain-to-SimpleQA and SQ-to-SQ+ComplexQA transitions record +3.43 pp and +1.66 pp differences, respectively. Paraphrase-only training reaches

59.79% avg, and the full four-category mixture has the highest measured Avg CEM at 62.50%.

### A.12 MSE Alignment Loss: Presence and $\lambda$ Sensitivity

Table A10 reports the with-loss versus without-loss difference for  $\mathcal{L}_{\text{MSE}}$  (Eq. 5) by comparing ARIA trained with and without the loss under Oracle retrieval. At 32 $\times$ , the with-loss condition records a +0.77 pp Avg-CEM difference from the without-loss condition. The largest per-benchmark difference is on HotpotQA (+1.38 pp), followed by MuSiQue (+0.70 pp). At 128 $\times$ , the corresponding difference is +0.88 pp (HotpotQA +1.05 pp), computed as the mean of the four displayed per-benchmark differences. The with-loss condition records higher Avg CEM in both measured compression regimes.

*Sensitivity to  $\lambda$ .* Table A11 sweeps  $\lambda \in \{0, 0.01, 0.05, 0.1, 0.2, 0.5\}$  at 32 $\times$  Oracle. The  $\lambda = 0.1$  row records the highest measured aggregate value (default; 62.50% Avg CEM);  $\lambda = 0.01$  records +0.24 pp over the  $\lambda = 0$  baseline, and  $\lambda = 0.5$  records lower Avg CEM in this sweep (61.92%). The [0.05, 0.2] range is within 0.18 pp of that measured value at  $\lambda = 0.1$ .

### A.13 Generator-Only vs. Full Phase-II Training

Table A12 compares generator-only fine-tuning against full Phase-II joint training of the QR, compressor, and generator LoRA adapters plus  $P_{\text{fb}}$ . Full joint training records higher means in all tested settings: +1.47 pp Avg CEM at 32 $\times$  Normal, +2.22 pp at 32 $\times$  Oracle, and +1.25 pp at 128 $\times$  Normal. At 32 $\times$  Oracle, the largest per-benchmark difference is a three-way tie across NQ, HotpotQA, and 2Wiki; under both Normal settings, NQ records the largest difference. The measured joint-tuning mean difference narrows at 128 $\times$ .

### A.14 MADS Scorer Combination Study

Table A13 examines MADS scoring-module combinations under Oracle retrieval at 32 $\times$ ; the corresponding study under Normal retrieval at 16 $\times$  (Appendix A.27) records the same scorer ordering. The three-scorer combination (Lex+Sem+Ent) records the highest measured values among the tested subsets; adding entity-aware scoring corresponds to the largest measured margin (+0.90 pp

Table A7: Complete end-to-end QA (F1%/EM%) across all compression ratios and retrieval conditions. **Bold:** higher value between the two compressed systems at the same nominal CR. Source-reported uncompressed rows are shown in regular type. At Normal 128 $\times$ , ARIA trails CLaRa on 2Wiki F1/EM (§5); at Oracle 32 $\times$ /128 $\times$ , ARIA trails CLaRa on HotpotQA F1/EM. The displayed adjacent-CR F1 non-monotonicities under Normal retrieval are +0.13 pp (CLaRa MuSiQue, 4 $\times$   $\rightarrow$  16 $\times$ ), +0.37 pp (CLaRa HotpotQA, 32 $\times$   $\rightarrow$  64 $\times$ ), +0.35 pp (ARIA HotpotQA, 32 $\times$   $\rightarrow$  64 $\times$ ), and +0.08 pp (ARIA MuSiQue, 32 $\times$   $\rightarrow$  64 $\times$ ). \*RECOMP: variable compression. \*CLaRa: 5-seed means.  $\diamond$ All 128 $\times$  ARIA rows use  $K_{\text{igfr}} = 1$  (selected on dev partition; Appendix A.24).

| Model            | CR    | Ret. | NQ           |              | HotpotQA     |              | MuSiQue      |              | 2Wiki        |              | Avg F1       | Avg EM       |
|------------------|-------|------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                  |       |      | F1           | EM           | F1           | EM           | F1           | EM           | F1           | EM           |              |              |
| Normal Retrieval |       |      |              |              |              |              |              |              |              |              |              |              |
| In-ctx RAG       | 1×    | N    | 44.69        | 38.07        | 40.27        | 37.14        | 20.11        | 16.78        | 41.02        | 38.51        | 36.52        | 32.62        |
| RECOMP           | var.* | N    | 42.67        | 37.47        | 42.72        | 38.72        | 24.96        | 17.34        | 38.26        | 32.17        | 37.15        | 31.43        |
| GenGround        | 1×    | N    | 42.31        | 40.60        | 44.71        | 41.27        | 24.36        | 20.77        | 42.58        | 39.61        | 38.49        | 35.56        |
| RetRobust        | 1×    | N    | 43.82        | 37.03        | 40.54        | 35.59        | 18.16        | 18.11        | 39.11        | 38.65        | 35.41        | 32.34        |
| DRO              | 1×    | N    | 51.01        | 42.41        | 47.87        | 40.37        | 25.32        | 21.36        | 43.65        | 42.12        | 41.96        | 36.56        |
| Self-RAG         | 1×    | N    | 49.52        | 41.18        | 48.18        | 40.84        | 23.82        | 19.51        | 44.10        | 40.28        | 41.41        | 35.45        |
| CRAG             | 1×    | N    | 48.73        | 40.38        | 47.05        | 39.47        | 22.42        | 18.13        | 46.91        | 42.96        | 41.28        | 35.23        |
| CLaRa            | 4×    | N    | 48.34        | 38.16        | 46.07        | 35.12        | 17.55        | 8.11         | 47.32        | 43.11        | 39.82*       | 31.12*       |
| CLaRa            | 16×   | N    | 47.95        | 37.84        | 45.84        | 34.95        | 17.68        | 8.20         | 44.30        | 40.08        | 38.94*       | 30.27*       |
| CLaRa            | 32×   | N    | 47.28        | 37.22        | 44.28        | 33.95        | 16.72        | 7.85         | 42.82        | 38.52        | 37.77*       | 29.38*       |
| CLaRa            | 64×   | N    | 46.41        | 36.51        | 44.65        | 34.18        | 16.08        | 7.52         | 40.57        | 36.34        | 36.93*       | 28.64*       |
| CLaRa            | 128×  | N    | 45.54        | 35.63        | 42.29        | 31.92        | 14.50        | 6.68         | <b>38.31</b> | <b>34.11</b> | 35.16*       | 27.08*       |
| LLMLingua-2      | 4×    | N    | 40.22        | 31.84        | 37.84        | 28.93        | 15.27        | 7.09         | 44.63        | 40.75        | 34.49        | 27.15        |
| PISCO            | 16×   | N    | 46.71        | 36.95        | 43.77        | 33.48        | 17.48        | 8.13         | 43.34        | 39.35        | 37.83        | 29.48        |
| ARIA             | 4×    | N    | <b>56.20</b> | <b>46.10</b> | <b>54.10</b> | <b>42.50</b> | <b>22.30</b> | <b>12.10</b> | <b>51.10</b> | <b>47.50</b> | <b>45.93</b> | <b>37.05</b> |
| ARIA             | 16×   | N    | <b>55.12</b> | <b>44.83</b> | <b>52.64</b> | <b>41.17</b> | <b>21.07</b> | <b>11.34</b> | <b>49.19</b> | <b>44.73</b> | <b>44.50</b> | <b>35.52</b> |
| ARIA             | 32×   | N    | <b>54.80</b> | <b>44.20</b> | <b>49.47</b> | <b>39.10</b> | <b>19.04</b> | <b>9.72</b>  | <b>47.38</b> | <b>42.80</b> | <b>42.67</b> | <b>33.96</b> |
| ARIA             | 64×   | N    | <b>52.90</b> | <b>42.50</b> | <b>49.82</b> | <b>39.31</b> | <b>19.12</b> | <b>9.98</b>  | <b>44.80</b> | <b>40.60</b> | <b>41.66</b> | <b>33.10</b> |
| ARIA             | 128×  | N    | <b>51.74</b> | <b>41.83</b> | <b>47.88</b> | <b>37.62</b> | <b>17.94</b> | <b>9.03</b>  | 37.85        | 33.45        | <b>38.85</b> | <b>30.48</b> |
| Oracle Retrieval |       |      |              |              |              |              |              |              |              |              |              |              |
| CLaRa            | 4×    | O    | 75.78        | 67.64        | 69.81        | 56.92        | 33.28        | 22.42        | 74.02        | 69.74        | 63.22*       | 54.18*       |
| CLaRa            | 16×   | O    | 71.68        | 63.29        | 68.57        | 55.38        | 30.86        | 20.56        | 67.98        | 62.64        | 59.77*       | 50.47*       |
| CLaRa            | 32×   | O    | 69.88        | 65.17        | <b>66.08</b> | <b>53.41</b> | 28.94        | 18.45        | 64.52        | 59.32        | 57.36*       | 49.09*       |
| CLaRa            | 64×   | O    | 67.57        | 62.08        | 63.61        | 51.43        | 27.16        | 16.73        | 61.61        | 56.28        | 54.99*       | 46.63*       |
| CLaRa            | 128×  | O    | 65.00        | 58.94        | <b>61.36</b> | <b>49.63</b> | 23.65        | 14.97        | 58.43        | 53.17        | 52.11*       | 44.18*       |
| ARIA             | 4×    | O    | <b>79.84</b> | <b>72.31</b> | <b>75.42</b> | <b>62.18</b> | <b>36.74</b> | <b>25.17</b> | <b>77.23</b> | <b>73.08</b> | <b>67.31</b> | <b>58.19</b> |
| ARIA             | 16×   | O    | <b>76.51</b> | <b>68.44</b> | <b>73.18</b> | <b>59.73</b> | <b>33.91</b> | <b>23.04</b> | <b>73.15</b> | <b>68.47</b> | <b>64.19</b> | <b>54.92</b> |
| ARIA             | 32×   | O    | <b>73.80</b> | <b>66.10</b> | 65.80        | 53.20        | <b>33.10</b> | <b>22.40</b> | <b>69.40</b> | <b>64.80</b> | <b>60.53</b> | <b>51.62</b> |
| ARIA             | 64×   | O    | <b>71.50</b> | <b>63.80</b> | <b>65.12</b> | <b>52.68</b> | <b>28.50</b> | <b>18.60</b> | <b>68.10</b> | <b>63.10</b> | <b>58.31</b> | <b>49.55</b> |
| ARIA             | 128×  | O    | <b>69.84</b> | <b>61.37</b> | 61.35        | 49.12        | <b>27.13</b> | <b>17.84</b> | <b>63.91</b> | <b>58.86</b> | <b>55.56</b> | <b>46.80</b> |

on MuSiQue). All rows are inference-time scoring interventions on the same trained checkpoints.

#### A.15 CCEF Threshold, IGFR Iteration, CFRS Weight, and AHR Weight Sensitivity

Tables A14–A18 examine hyperparameter sensitivity. Defaults were selected on the 500-query held-out dev partition (disjoint from the 1,000-query annotated sample described in Appendix A.31) or, for the Multi-Aspect AHR weights, on the separate 260-query Multi-Aspect validation subset specified below. Unless a caption identifies that validation subset, the tables below report evaluation sensitivity around dev-selected defaults.

*Other hyperparameters fixed by dev-set selection.* The CCEF agreement floor  $\alpha = 0.5$ , ACR sigmoid sharpness  $\beta = 10$ , and ACR scale bounds  $(\rho_{\min}, \rho_{\max}) = (0.25, 1.0)$  were selected on the same 500-query dev partition by coarse grid search ( $\alpha \in \{0.3, 0.5, 0.7\}$ ,  $\beta \in \{5, 10, 20\}$ ,  $\rho_{\min} \in \{0.1, 0.25, 0.4\}$ ); the chosen values were no more than 0.30 pp Avg CEM below the highest measured dev-set row and held fixed throughout all main and ablation experiments. The complexity-adaptive floor at 128 $\times$  (Appendix A.28) is the only exception, applied at inference time only.

*CCEF threshold.* The selected  $\tau = 0.30$  row has the highest measured aggregate value; relative to this row, the  $\tau = 0.20$

Table A8: **Oracle** Recall@ $k$  (%) after system-specific reranking of a top-100 candidate pool constructed to contain the annotated positives. The Normal first-pass Recall@5 diagnostic in Appendix A.32 uses a different candidate-pool protocol, so the two sets of values are reported separately. The 1 $\times$  rows are source-reported contextual values from CLaRa’s Oracle evaluation and were not recomputed under our page-ID-deduplicated top-100 protocol; only the ARIA–CLaRa reimplementation rows form the shared top-100 comparison. Bold marks the higher value within each same-CR ARIA–CLaRa pair. BGE-Reranker uses the released BGE reranking model from the resource family of Xiao et al. (2024); Setwise follows Zhuang et al. (2024); Rank-R1 follows Zhuang et al. (2026); and RankZephyr-7B follows Pradeep et al. (2023).

| Model         | CR           | NQ           |              |              | HotpotQA     |              |              | MuSiQue      |              |              | 2Wiki        |              |              |
|---------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|               |              | R@1          | R@3          | R@5          | R@1          | R@3          | R@5          | R@1          | R@3          | R@5          | R@1          | R@3          | R@5          |
| BM25          | 1 $\times$   | 6.57         | 21.89        | 35.99        | 25.12        | 48.86        | 62.09        | 13.00        | 29.12        | 39.13        | 15.23        | 37.06        | 51.40        |
| BGE-Reranker  | 1 $\times$   | 11.18        | 33.56        | 47.78        | 28.53        | 67.91        | 85.93        | 18.32        | 42.45        | 54.13        | 22.21        | 54.95        | 68.32        |
| Setwise       | 1 $\times$   | 12.60        | 38.20        | 52.10        | 30.40        | 60.30        | 74.60        | 18.50        | 40.30        | 51.60        | 26.20        | 57.99        | 71.61        |
| Rank-R1       | 1 $\times$   | 8.95         | 30.60        | 46.72        | 24.87        | 56.56        | 73.06        | 16.77        | 38.49        | 51.68        | 24.27        | 59.02        | 77.02        |
| RankZephyr-7B | 1 $\times$   | 16.00        | 40.20        | 52.40        | 38.00        | 63.70        | 75.90        | 21.04        | 40.61        | 50.65        | 33.40        | 61.55        | 74.69        |
| CLaRa         | 4 $\times$   | 32.62        | 63.71        | 76.38        | 45.07        | 81.90        | 88.21        | 30.45        | 59.99        | 72.46        | 34.37        | 68.08        | 79.13        |
| CLaRa         | 16 $\times$  | 28.45        | 58.97        | 71.88        | 42.01        | 77.80        | 83.40        | 22.57        | 44.34        | 58.83        | 32.04        | 66.47        | 78.34        |
| CLaRa         | 32 $\times$  | 27.70        | 57.66        | 70.53        | 41.44        | 76.52        | 81.95        | 21.85        | 42.58        | 57.55        | 31.46        | 65.12        | 77.33        |
| CLaRa         | 64 $\times$  | 26.95        | 56.81        | 69.18        | 40.87        | 75.24        | 80.50        | 21.13        | 43.12        | 56.27        | 30.87        | 64.05        | 77.42        |
| CLaRa         | 128 $\times$ | 25.80        | 54.91        | 67.74        | 40.70        | 73.60        | 79.10        | <b>20.74</b> | 41.29        | 54.86        | <b>30.14</b> | 61.83        | 74.97        |
| ARIA          | 4 $\times$   | <b>33.42</b> | <b>65.10</b> | <b>79.50</b> | <b>48.12</b> | <b>84.40</b> | <b>91.60</b> | <b>31.50</b> | <b>62.10</b> | <b>75.40</b> | <b>36.15</b> | <b>70.50</b> | <b>82.10</b> |
| ARIA          | 16 $\times$  | <b>32.80</b> | <b>63.10</b> | <b>77.03</b> | <b>45.30</b> | <b>80.15</b> | <b>86.10</b> | <b>27.80</b> | <b>51.40</b> | <b>65.90</b> | <b>35.79</b> | <b>70.44</b> | <b>81.30</b> |
| ARIA          | 32 $\times$  | <b>31.80</b> | <b>62.02</b> | <b>75.42</b> | <b>44.05</b> | <b>78.68</b> | <b>84.25</b> | <b>25.98</b> | <b>49.34</b> | <b>64.40</b> | <b>34.90</b> | <b>68.62</b> | <b>80.65</b> |
| ARIA          | 64 $\times$  | <b>30.80</b> | <b>60.20</b> | <b>73.80</b> | <b>42.80</b> | <b>77.20</b> | <b>82.40</b> | <b>24.15</b> | <b>49.82</b> | <b>62.89</b> | <b>34.01</b> | <b>66.80</b> | <b>81.05</b> |
| ARIA          | 128 $\times$ | <b>29.81</b> | <b>58.54</b> | <b>72.62</b> | <b>41.90</b> | <b>75.39</b> | <b>80.65</b> | 19.80        | <b>45.87</b> | <b>59.74</b> | 29.50        | <b>65.17</b> | <b>79.63</b> |

Table A9: Pretraining-mixture configurations (CEM%, Oracle, nominal CR=32 $\times$ ). The No-Pretrain, SQ, SQ+CQ, SQ+CQ+Para, and Full rows form the cumulative sequence; Paraphrase only is a standalone control. SQ = SimpleQA, CQ = ComplexQA, Para = Paraphrase, Ent = Entity-Augmented.

| Data                         | NQ           | HotpotQA     | MuSiQue      | 2Wiki        | Avg          |
|------------------------------|--------------|--------------|--------------|--------------|--------------|
| No Pretrain                  | 70.94        | 62.38        | 29.41        | 58.17        | 55.23        |
| +SimpleQA                    | 72.48        | 66.15        | 34.12        | 61.88        | 58.66        |
| SQ+ComplexQA                 | 74.02        | 68.32        | 35.80        | 63.14        | 60.32        |
| SQ+CQ+Para                   | 73.55        | 67.40        | 37.15        | 63.95        | 60.51        |
| <b>Full (+Ent)</b>           | <b>75.20</b> | <b>69.50</b> | <b>40.10</b> | <b>65.20</b> | <b>62.50</b> |
| Paraphrase only (standalone) | 73.10        | 67.50        | 36.04        | 62.50        | 59.79        |

Table A10:  $\mathcal{L}_{\text{MSE}}$  presence ablation (CEM%, Oracle retrieval, 5-seed mean). All other objective coefficients and training settings use their defaults. Avg cells are computed before independent two-decimal display rounding.

| Model                          | CR           | NQ           | HotpotQA     | MuSiQue      | 2Wiki        | Avg          |
|--------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|
| w/o $\mathcal{L}_{\text{MSE}}$ | 32 $\times$  | 74.85        | 68.12        | 39.40        | 64.55        | 61.73        |
| w/ $\mathcal{L}_{\text{MSE}}$  | 32 $\times$  | <b>75.20</b> | <b>69.50</b> | <b>40.10</b> | <b>65.20</b> | <b>62.50</b> |
| w/o $\mathcal{L}_{\text{MSE}}$ | 128 $\times$ | 68.65        | 61.05        | 30.82        | 57.45        | 54.49        |
| w/ $\mathcal{L}_{\text{MSE}}$  | 128 $\times$ | <b>69.40</b> | <b>62.10</b> | <b>31.80</b> | <b>58.20</b> | <b>55.38</b> |

NQ/HotpotQA/MuSiQue/2Wiki values differ by  $-0.14/ -0.20/ -0.28/ -0.43$  pp, and the  $\tau = 0.40$  values differ by  $-1.24/ -0.25/ -0.43/ -0.73$  pp.

*IGFR iterations.*  $K = 2$  records the highest Avg CEM in the displayed grid (41.24%); the  $K = 3$  and  $K = 4$  rows are lower in this evaluation.

Table A11: MSE loss weight  $\lambda$  sensitivity (CEM%, Oracle, CR=32 $\times$ , 5-seed mean).  $\lambda = 0$  is the w/o  $\mathcal{L}_{\text{MSE}}$  baseline from Table A10. All other objective coefficients and training settings use their defaults. **Bold**: selected default.

| $\lambda$ | NQ           | HotpotQA     | MuSiQue      | 2Wiki        | Avg CEM      | $\Delta$ |
|-----------|--------------|--------------|--------------|--------------|--------------|----------|
| 0.00      | 74.85        | 68.12        | 39.40        | 64.55        | 61.73        | -0.77    |
| 0.01      | 75.04        | 68.45        | 39.62        | 64.78        | 61.97        | -0.53    |
| 0.05      | 75.12        | 69.21        | 39.88        | 65.07        | 62.32        | -0.18    |
| 0.10      | <b>75.20</b> | <b>69.50</b> | <b>40.10</b> | <b>65.20</b> | <b>62.50</b> | 0.00     |
| 0.20      | 75.03        | 69.32        | 39.83        | 65.31        | 62.37        | -0.13    |
| 0.50      | 74.25        | 68.87        | 38.94        | 65.62        | 61.92        | -0.58    |

*CFRS blending weight.* Table A16 sweeps  $w_{\text{cfrs}}$ .  $w_{\text{cfrs}}=0.30$  records the highest aggregate value in this grid; the  $w_{\text{cfrs}}=0$  row is 1.22 pp lower on NQ CEM.

*AHR type-specific default weights.* Tables A17 and A18 report sensitivity around the  $(w_{\text{bm25}}, w_{\text{dense}})$  defaults for each QCA category

Table A12: Fine-tuning scope (CEM%, mean over 5 seeds). Gen-only = only the generator LoRA is updated; Full FT = QR/compressor/generator LoRA and  $P_{fb}$  are jointly updated (default).

| Scope          | CR   | Ret. | NQ           | HotpotQA     | MuSiQue      | 2Wiki        | Avg          |
|----------------|------|------|--------------|--------------|--------------|--------------|--------------|
| Gen-only       | 32×  | N    | 54.50        | 42.80        | 9.63         | 43.60        | 37.63        |
| <b>Full FT</b> | 32×  | N    | <b>56.81</b> | <b>44.02</b> | <b>9.98</b>  | <b>45.60</b> | <b>39.10</b> |
| Gen-only       | 128× | N    | 52.14        | 40.10        | 8.90         | 41.50        | 35.66        |
| <b>Full FT</b> | 128× | N    | <b>54.05</b> | <b>41.65</b> | <b>9.55</b>  | <b>42.40</b> | <b>36.91</b> |
| Gen-only       | 32×  | O    | 72.80        | 67.10        | 38.40        | 62.80        | 60.28        |
| <b>Full FT</b> | 32×  | O    | <b>75.20</b> | <b>69.50</b> | <b>40.10</b> | <b>65.20</b> | <b>62.50</b> |

Table A13: MADS scorer combination study (CEM%, Oracle, CR = 32×).

| Config             | Lex | Sem | Ent | NQ           | HotpotQA     | MuSiQue      | 2Wiki        |
|--------------------|-----|-----|-----|--------------|--------------|--------------|--------------|
| Lex only           | ✓   | ×   | ×   | 73.80        | 68.10        | 39.20        | 63.50        |
| Sem only           | ×   | ✓   | ×   | 74.90        | 68.90        | 39.50        | 64.60        |
| Lex+Sem            | ✓   | ✓   | ×   | 74.80        | 69.20        | 39.20        | 64.80        |
| <b>Lex+Sem+Ent</b> | ✓   | ✓   | ✓   | <b>75.20</b> | <b>69.50</b> | <b>40.10</b> | <b>65.20</b> |

Table A14: CCEF threshold  $\tau$  sensitivity (CEM%, Normal, CR = 16×).

| $\tau$      | NQ           | Hotpot       | MuSiQue      | 2Wiki        |
|-------------|--------------|--------------|--------------|--------------|
| 0.10        | 56.80        | 44.10        | 10.90        | 47.20        |
| 0.20        | 58.50        | 45.90        | 11.40        | 48.10        |
| <b>0.30</b> | <b>58.64</b> | <b>46.10</b> | <b>11.68</b> | <b>48.53</b> |
| 0.40        | 57.40        | 45.85        | 11.25        | 47.80        |
| 0.50        | 58.05        | 44.60        | 10.95        | 47.50        |

Table A15: IGFR iteration count  $K$  sensitivity (CEM%, Normal, CR = 16×).

| $K$      | NQ           | Hotpot       | MuSiQue      | 2Wiki        | Avg          |
|----------|--------------|--------------|--------------|--------------|--------------|
| 0        | 56.83        | 44.52        | 10.73        | 47.14        | 39.81        |
| 1        | 58.12        | 45.80        | 11.30        | 48.20        | 40.86        |
| <b>2</b> | <b>58.64</b> | 46.10        | <b>11.68</b> | <b>48.53</b> | <b>41.24</b> |
| 3        | 58.40        | 45.95        | 11.40        | 48.10        | 40.96        |
| 4        | 57.90        | <b>46.20</b> | 11.25        | 47.95        | 40.83        |

Table A16: CFRS blending weight  $w_{cfrs}$  sensitivity (CEM%, Normal, CR = 16×).  $w_{cfrs} = 0$  corresponds to next-token-proxy reranking disabled.

| $w_{cfrs}$  | NQ           | Hotpot       | MuSiQue      | 2Wiki        | Avg          |
|-------------|--------------|--------------|--------------|--------------|--------------|
| 0.00        | 57.42        | 45.60        | 11.38        | 48.00        | 40.60        |
| 0.10        | 58.15        | 45.85        | 11.52        | 48.28        | 40.95        |
| 0.20        | 58.40        | 46.00        | 11.60        | 48.45        | 41.11        |
| <b>0.30</b> | <b>58.64</b> | <b>46.10</b> | <b>11.68</b> | <b>48.53</b> | <b>41.24</b> |
| 0.40        | 58.30        | 45.70        | 11.45        | 48.10        | 40.89        |
| 0.50        | 57.95        | 45.20        | 11.20        | 47.65        | 40.50        |

(Avg CEM at 16× Normal, 5 seeds). The Simple setting spans 0.21 pp in the reported sweep. Simple-routed queries are concentrated on NQ, where the BM25-heavy settings record similar values. The Multi-Hop setting is more sensitive: (0.25, 0.75) records 0.68 pp higher mean Avg CEM than (0.50, 0.50) and 0.31 pp higher than a fully dense (0.10, 0.90) setting; the highest measured row among these settings uses dense-heavy weights with a nonzero lexical component. The Multi-Aspect default (0.30, 0.70) is chosen via Table A18 on a 260-query Multi-Aspect sub-sample, giving an observed +0.25 pp on the sub-sample over balanced weights.

## A.16 Dataset and Pretraining Statistics

Tables A19 and A20 summarize the pretraining corpus and QA evaluation benchmark statistics.

*Phase II training subsample protocol.* The four benchmarks together contain 485,006 training examples; Phase II uses a *benchmark-balanced* subsample of 38,400 examples per epoch (9,600 from each benchmark), drawn uniformly without replacement and reshuffled at every epoch with a fixed seed schedule. Across 5 epochs at effective batch size 32 this gives 192,000 total example views and 6,000 optimizer steps. The balanced schedule gives each benchmark equal sampling weight rather than weighting by its raw split size. On the held-out dev trace, the dis-

Table A17: AHR default weight sensitivity (Avg CEM%, Normal, CR = 16 $\times$ , 5 seeds). Each row varies the weight pair for one QCA category while fixing the other two at their defaults. **Bold**: selected row per category. Multi-Aspect is fixed at its chosen default (0.30, 0.70) (see Table A18 and §3). All “Avg CEM” values are 4-benchmark averages over NQ, HotpotQA, MuSiQue, and 2Wiki; for the Simple category the displayed differences are concentrated on NQ (see text).

| Category  | $(w_{\text{bm25}}, w_{\text{dense}})$ | Avg CEM      | $\Delta$ |
|-----------|---------------------------------------|--------------|----------|
| Simple    | (0.90, 0.10)                          | 41.18        | −0.06    |
|           | <b>(0.75, 0.25)</b>                   | <b>41.24</b> | 0.00     |
|           | (0.50, 0.50)                          | 41.03        | −0.21    |
| Multi-Hop | (0.10, 0.90)                          | 40.93        | −0.31    |
|           | <b>(0.25, 0.75)</b>                   | <b>41.24</b> | 0.00     |
|           | (0.50, 0.50)                          | 40.56        | −0.68    |

played lower-panel curve covers the first 5,000 of 6,000 optimizer steps (Figure A3) as a training-progress diagnostic. The augmented MuSiQue split contributes 9,600 examples per epoch from its 168,745-example pool. The draw schedule gives each MuSiQue example 0.06 selections per epoch on average (0.28 across the 5-epoch run) and each NQ example from the 58K pool 0.16 per epoch (0.82 across the run), with all four frequencies rounded.

*Aggregate Phase-I mixture records.* The Phase-I mixture contains 7,808,465 samples grouped into four functional pipeline categories. SimpleQA and ComplexQA each contribute 2M samples with different output length profiles: SimpleQA averages 158.18 output tokens (single-answer paraphrase targets), while ComplexQA averages 253.90 tokens (multi-step reasoning chains). The Entity-Augmented corpus (1,842,174 samples) has the highest pairs-per-document ratio after SimpleQA (5.31 vs. 7.80) and is used for entity-centric supervision. Paraphrase data contributes 1,966,291 samples with the shortest output length (108.67 tokens), targeting surface-level reformulation skill. Here SimpleQA, ComplexQA, Paraphrase, and Entity-Augmented are internal functional mixture labels. Table A19 reports their aggregate composition.

*Evaluation benchmarks.* The four benchmarks span single-hop (NQ), two-hop (HotpotQA, 2Wiki), and 2–4 hop (MuSiQue) reasoning, covering retrieval–compression behavior from one to

Table A18: Multi-Aspect AHR weight sensitivity analysis (CEM% on Multi-Aspect query sub-sample, Normal, CR = 16 $\times$ , 5 seeds). Sub-sample = 260 queries drawn from the four benchmark training splits and routed by QCA as Multi-Aspect; this sub-sample is held out from model training and disjoint from the 500-query dev partition and the 1,000-query annotated sample. Four candidate weight pairs evaluated; (0.30, 0.70) is the chosen default (bold weight pair), recording 0.25 pp higher mean Avg CEM than the balanced (0.50, 0.50) alternative on the sub-sample. Bold metric cells mark column maxima;  $\Delta$  Avg is computed from full-precision per-benchmark averages.

| $(w_{\text{bm25}}, w_{\text{dense}})$ | NQ           | Hot.         | Mq.          | 2Wiki        | $\Delta$ Avg |
|---------------------------------------|--------------|--------------|--------------|--------------|--------------|
| (0.50, 0.50)                          | 58.93        | 45.74        | 12.36        | 49.23        | 0.00         |
| (0.40, 0.60)                          | 59.01        | 46.02        | 12.41        | 49.48        | +0.17        |
| <b>(0.30, 0.70)</b>                   | 59.01        | <b>46.17</b> | 12.47        | <b>49.61</b> | <b>+0.25</b> |
| (0.20, 0.80)                          | <b>59.04</b> | 46.10        | <b>12.48</b> | 49.52        | +0.22        |

four hops. MuSiQue has the smallest evaluation set (2,417 examples) and the deepest reasoning chains. 2Wiki has the largest evaluation set (12,576 examples).

*MuSiQue training set size.* The 168,745 MuSiQue training samples result from sub-question decomposition, partial-chain generation, and entity-variant augmentation applied to the answerable training split: 8.5 $\times$  the 19,938 original examples after rounding. Evaluation uses the standard 2,417-example split without augmentation.

Table A19: Aggregate Phase-I mixture statistics under internal functional pipeline labels.

|              | SimpleQA  | ComplexQA | Para.     | Ent-Aug.  |
|--------------|-----------|-----------|-----------|-----------|
| Samples      | 2,000,000 | 2,000,000 | 1,966,291 | 1,842,174 |
| Pairs/Doc    | 7.80      | 4.62      | 1.00      | 5.31      |
| Input (tok)  | 95.56     | 95.56     | 95.56     | 97.14     |
| Output (tok) | 158.18    | 253.90    | 108.67    | 184.32    |

## A.17 Cross-Backbone Results: Llama-3-8B

We repeat the core experiments on Llama-3-8B-Instruct (Llama Team, 2024) under the same stated two-phase protocol used for Mistral-7B: Phase I runs for 3 epochs, and Phase II uses learning rate  $2 \times 10^{-4}$ , a 500-step warmup, 5 epochs, effective batch 32, and LoRA rank  $r_{\text{LoRA}} = 16$ . The MTFRL projection head uses the same dimensions as for Mistral-7B (4096  $\rightarrow$  2048  $\rightarrow$  1024); training data is unchanged from the Mistral-7B configuration and uses the same augmented MuSiQue training partition (Appendix A.16); 5 seeds {42, 123, 456, 789, 2024}. Tables A21 and

Table A20: QA evaluation benchmark statistics. <sup>†</sup>HotpotQA: 2-document questions. <sup>‡</sup>MuSiQue: 2–4 hop chains. <sup>§</sup>MuSiQue Train of 168,745 is our *augmented* training partition (Appendix A.36); the standard answerable training split contains 19,938 examples. NQ, HotpotQA, and 2WikiMultiHopQA columns report the counts used after benchmark-specific preprocessing; training counts additionally reflect the stated page-URL deduplication.

| Dataset  | Train                | Eval   | Hops             |
|----------|----------------------|--------|------------------|
| NQ       | 58,622               | 6,489  | 1                |
| HotpotQA | 90,185               | 7,384  | 2 <sup>†</sup>   |
| MuSiQue  | 168,745 <sup>§</sup> | 2,417  | 2–4 <sup>‡</sup> |
| 2Wiki    | 167,454              | 12,576 | 2                |

Table A22 report Normal retrieval; Table A24 gives pointwise paired-bootstrap  $p$ -value thresholds from 10,000 query resamples after averaging the five seed-pair differences.

ARIA records higher mean CEM than CLaRa across all tested CRs, with the largest measured mean difference at +2.85 pp at 16 $\times$  (vs. +2.41 pp on Mistral-7B). At the same 16 $\times$  point, the F1 difference is +6.14 pp (47.58% vs. 41.44%; vs. +5.56 pp on Mistral-7B). The gap narrows at 64 $\times$  (+2.39 pp CEM) and 128 $\times$  (+1.05 pp). On Llama-3-8B, as on Mistral-7B, the 128 $\times$  2Wiki differences are negative in CEM (45.13 vs. 45.60) and F1 (40.38 vs. 41.80). Under Oracle retrieval (Table A23), ARIA’s 16 $\times$  Avg-CEM difference is +3.18 pp, 0.14 pp larger than Mistral-7B’s +3.04 pp.

All pointwise query-bootstrap summaries for the reported CEM/F1 deltas at 4 $\times$ –64 $\times$  have  $p < 0.001$  (Table A24); at 128 $\times$ , MuSiQue is  $p < 0.05$  and 2Wiki has  $p \geq 0.05$  for both metrics. Oracle results (Table A23) record the corresponding second-backbone differences (+3.18 pp at 16 $\times$ ; +0.69 pp at 128 $\times$ ).

### A.18 Coupling-Mechanism Sensitivity Across Compression Ratios

**Matched retraining at 16 $\times$ .** The matched study retrains each variant with the main Phase-II data, initialization protocol, optimization schedule, and five seeds. All rows use the same first-round candidate-pool construction, pool size, retrieval protocol, and final document count, and retain a rounded mean of 108 evidence-memory tokens per query. Candidate identities remain variant-specific because every row is trained and evaluated separately. The no-MTFRL and all-off variants replace

memory-conditioned retrieval with a static second retrieval so that every row retains two retrieval rounds and the same candidate count. Table A25 reports per-benchmark F1; Table A26 traces the final document sources.

Relative to matched full, the Avg-F1 differences are  $-0.52$  for  $-CFRS$  ( $p < .01$ ),  $-0.66$  for uniform ACR ( $p < .01$ ),  $-0.48$  for the static second retrieval ( $p < .05$ ), and  $-1.45$  when all three operations are disabled ( $p < .001$ ), using the paired-bootstrap procedure in Appendix A.39. The corresponding final Recall@5 values are 65.3, 64.8, 64.6, 63.7, and 62.8%.

**Fixed-checkpoint sensitivity.** The following tables retain the original main Full-ARIA checkpoint and apply inference-time interventions. They complement the retrained controls by measuring changes at a common learned parameter state.

Table A28 extends the coupling-mechanism sensitivity analysis to all compression ratios. Removing CFRS or MTFRL lowers Avg CEM at every ratio. At 128 $\times$ , the uniform-rate control records higher HotpotQA, MuSiQue, 2Wiki, and Avg CEM than the default ACR setting.

### A.19 Independent Soft- and Hard-Gate Checkpoints

We train soft- and hard-gate variants independently from the same ratio-specific initialization, Phase-II data, schedule, and five seeds. Table A29 shows that the two training choices track closely across compression ratios. The largest absolute difference is 0.24 points in Avg CEM and 0.32 points in Avg F1; the hard-gate checkpoint is slightly higher at 64 $\times$  in F1 and at 128 $\times$  in CEM.

### A.20 Configured-System Endpoint Comparisons

This appendix reports performance for configured systems spanning the CLaRa and Full-ARIA endpoints. Table A30 summarizes their retrieval, compression, and forward-path configurations.

Table A31 reports four configured-system endpoints; their Avg F1 values are 38.94, 39.58, 43.86, and 44.50.

*CFRS-off and CFRS-on configurations.* The CFRS-off configuration, ARIA-COMP (forward-path off; CLaRa-trained), uses the full five-stage ARIA retrieval pipeline *without* the CFRS re-ranking step, followed by CLaRa uniform com-

Table A21: Cross-backbone result: compressor CEM% with Llama-3-8B-Instruct under Normal retrieval averaged over 5 seeds. At 128×, the 2Wiki values are 45.13 for ARIA and 45.60 for CLaRa, the same directional reversal observed on Mistral-7B ( $p \geq 0.05$  in the pointwise query bootstrap; Table A24).

| Model       | CR   | NQ           | Hotpot       | MuSiQue      | 2Wiki        | Avg          |
|-------------|------|--------------|--------------|--------------|--------------|--------------|
| CLaRa       | 4×   | 59.84        | 47.62        | 13.05        | 49.53        | 42.51        |
| CLaRa       | 16×  | 58.21        | 46.58        | 12.83        | 48.10        | 41.43        |
| CLaRa       | 32×  | 57.30        | 45.12        | 11.95        | 46.81        | 40.30        |
| CLaRa       | 64×  | 56.40        | 45.38        | 11.24        | 45.60        | 39.66        |
| CLaRa       | 128× | 55.50        | 43.20        | 10.40        | <b>45.60</b> | 38.68        |
| <b>ARIA</b> | 4×   | <b>62.99</b> | <b>50.24</b> | <b>14.89</b> | <b>52.33</b> | <b>45.11</b> |
| <b>ARIA</b> | 16×  | <b>61.92</b> | <b>49.09</b> | <b>14.38</b> | <b>51.74</b> | <b>44.28</b> |
| <b>ARIA</b> | 32×  | <b>60.56</b> | <b>47.86</b> | <b>13.29</b> | <b>50.23</b> | <b>42.99</b> |
| <b>ARIA</b> | 64×  | <b>59.33</b> | <b>46.53</b> | <b>13.46</b> | <b>48.88</b> | <b>42.05</b> |
| <b>ARIA</b> | 128× | <b>57.85</b> | <b>44.72</b> | <b>11.22</b> | 45.13        | <b>39.73</b> |

Table A22: Cross-backbone result: end-to-end F1% with Llama-3-8B-Instruct under Normal retrieval averaged over 5 seeds. At 128×, the 2Wiki F1 values are 40.38 for ARIA and 41.80 for CLaRa, the same directional reversal observed on Mistral-7B. The 32× → 64× HotpotQA F1 reversals are +0.28 pp for CLaRa and +0.42 pp for ARIA.

| Model       | CR   | NQ           | Hotpot       | MuSiQue      | 2Wiki        | Avg          |
|-------------|------|--------------|--------------|--------------|--------------|--------------|
| CLaRa       | 4×   | 51.20        | 49.15        | 20.82        | 48.31        | 42.37        |
| CLaRa       | 16×  | 50.43        | 48.35        | 20.14        | 46.82        | 41.44        |
| CLaRa       | 32×  | 49.51        | 46.84        | 19.35        | 45.08        | 40.20        |
| CLaRa       | 64×  | 48.60        | 47.12        | 18.72        | 43.40        | 39.46        |
| CLaRa       | 128× | 47.70        | 44.70        | 17.80        | <b>41.80</b> | 38.00        |
| <b>ARIA</b> | 4×   | <b>59.89</b> | <b>57.54</b> | <b>25.32</b> | <b>53.72</b> | <b>49.12</b> |
| <b>ARIA</b> | 16×  | <b>58.53</b> | <b>55.74</b> | <b>23.84</b> | <b>52.19</b> | <b>47.58</b> |
| <b>ARIA</b> | 32×  | <b>57.62</b> | <b>53.38</b> | <b>22.34</b> | <b>50.92</b> | <b>46.07</b> |
| <b>ARIA</b> | 64×  | <b>56.04</b> | <b>53.80</b> | <b>21.07</b> | <b>49.64</b> | <b>45.14</b> |
| <b>ARIA</b> | 128× | <b>54.38</b> | <b>49.73</b> | <b>19.22</b> | 40.38        | <b>40.93</b> |

Table A23: Cross-backbone result: compressor CEM% with Llama-3-8B-Instruct under Oracle retrieval averaged over 5 seeds. ARIA’s +3.18 pp difference at 16× is 0.14 pp larger than the Mistral-7B difference of +3.04 pp.

| Model       | CR   | NQ           | Hotpot       | MuSiQue      | 2Wiki        | Avg          |
|-------------|------|--------------|--------------|--------------|--------------|--------------|
| CLaRa       | 16×  | 77.82        | 73.15        | 45.50        | 69.40        | 66.47        |
| <b>ARIA</b> | 16×  | <b>80.55</b> | <b>76.75</b> | <b>48.55</b> | <b>72.74</b> | <b>69.65</b> |
| CLaRa       | 128× | 71.20        | 63.80        | 33.50        | 59.75        | 57.06        |
| <b>ARIA</b> | 128× | <b>71.96</b> | <b>64.39</b> | <b>34.03</b> | <b>60.63</b> | <b>57.75</b> |

Table A24: Pointwise paired-bootstrap  $p$ -value thresholds with 10,000 query resamples using five-seed outputs for ARIA vs. CLaRa on Llama-3-8B-Instruct, Normal retrieval. Markers denote pointwise thresholds:  $p < 0.001$  (\*\*\*),  $p < 0.01$  (\*\*),  $p < 0.05$  (\*), or  $p \geq 0.05$ . Markers are pointwise summaries. Dagger (†) indicates ARIA trails CLaRa.

| CR             | NQ  | Hotpot | MuSiQue | 2Wiki              | Avg |
|----------------|-----|--------|---------|--------------------|-----|
| <i>CEM (%)</i> |     |        |         |                    |     |
| 4×             | *** | ***    | ***     | ***                | *** |
| 16×            | *** | ***    | ***     | ***                | *** |
| 32×            | *** | ***    | ***     | ***                | *** |
| 64×            | *** | ***    | ***     | ***                | *** |
| 128×           | *** | ***    | *       | $\geq .05^\dagger$ | *** |
| <i>F1 (%)</i>  |     |        |         |                    |     |
| 4×             | *** | ***    | ***     | ***                | *** |
| 16×            | *** | ***    | ***     | ***                | *** |
| 32×            | *** | ***    | ***     | ***                | *** |
| 64×            | *** | ***    | ***     | ***                | *** |
| 128×           | *** | ***    | *       | $\geq .05^\dagger$ | *** |

Table A25: Per-benchmark F1% for the separately retrained, budget- and retrieval-topology-matched controls (Normal, 16 $\times$ , five-seed means).

| Variant                  | NQ           | HotpotQA     | MuSiQue      | 2Wiki        | Avg          |
|--------------------------|--------------|--------------|--------------|--------------|--------------|
| <b>Matched full</b>      | <b>55.08</b> | <b>52.60</b> | <b>21.04</b> | <b>49.12</b> | <b>44.46</b> |
| –CFRS, retrained         | 54.47        | 52.15        | 20.65        | 48.49        | 43.94        |
| Uniform ACR, retrained   | 54.63        | 51.91        | 20.33        | 48.33        | 43.80        |
| Static second retrieval  | 54.78        | 51.84        | 20.58        | 48.72        | 43.98        |
| All three off, retrained | 54.00        | 50.88        | 19.76        | 47.40        | 43.01        |

Table A26: Candidate provenance in the matched retraining study. D1 is the first-round survivor set; “new” denotes final documents first introduced by round 2.

| Variant                 | Final docs from D1 (%) | Round-2 new docs (%) | Queries with new gold page (%) |
|-------------------------|------------------------|----------------------|--------------------------------|
| Matched full            | 74.8                   | 25.2                 | 8.6                            |
| –CFRS                   | 75.1                   | 24.9                 | 8.2                            |
| Uniform ACR             | 75.7                   | 24.3                 | 7.9                            |
| Static second retrieval | 81.5                   | 18.5                 | 5.4                            |
| All three off           | 82.0                   | 18.0                 | 5.1                            |

Table A27: Full per-benchmark coupling sensitivity analysis (CEM% / F1%, Normal, 16 $\times$ , 5-seed mean). “–X” removes only operation X while retaining the five-stage retrieval backbone; Forward-path-off disables CFRS, ACR, and MTFRL together, with ACR replaced by uniform  $\rho_i = 1$  (mean 184 evidence-memory tokens at nominal 16 $\times$ , versus 108 for Full ARIA).

| Variant                | CEM%         |              |              |              |              | F1%          |              |              |              |              |
|------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                        | NQ           | HotpotQA     | MuSiQue      | 2Wiki        | Avg          | NQ           | HotpotQA     | MuSiQue      | 2Wiki        | Avg          |
| <b>Full</b>            | <b>58.64</b> | <b>46.10</b> | <b>11.68</b> | <b>48.53</b> | <b>41.24</b> | <b>55.12</b> | <b>52.64</b> | <b>21.07</b> | <b>49.19</b> | <b>44.50</b> |
| –CFRS                  | 57.42        | 45.60        | 11.38        | 48.00        | 40.60        | 54.00        | 52.04        | 20.52        | 48.67        | 43.81        |
| Uniform $\rho = 0.625$ | 58.10        | 45.40        | 10.90        | 47.75        | 40.54        | 54.63        | 51.81        | 19.67        | 48.42        | 43.63        |
| –MTFRL                 | 58.15        | 44.95        | 11.20        | 48.15        | 40.61        | 54.68        | 51.29        | 20.20        | 48.82        | 43.75        |
| Forward-path-off       | 55.90        | 43.80        | 10.30        | 46.08        | 39.02        | 52.72        | 49.98        | 18.58        | 46.72        | 42.00        |
| CLaRa*                 | 55.72        | 43.86        | 10.22        | 45.52        | 38.83        | 47.95        | 45.84        | 17.68        | 44.30        | 38.94        |

pression.

Table A32 reports this experiment (5 seeds, Normal, CR = 16 $\times$ ). The CLaRa-trained forward-path-off configuration records 42.02 % Avg F1, and the corresponding CFRS-on configuration records 43.86 %.

### A.21 ARIA Retrieval Pipeline Without Compression

ARIA-NoCOMP runs ARIA’s full five-stage retrieval pipeline (QCA, AHR, IGFR, MADS, CCEF) with the top-5 documents concatenated directly into the Mistral-7B decoder prompt without memory-token compression. No additional fine-tuning is performed; decoder adapter weights are frozen at the Phase II checkpoint. Results use 5 seeds. ARIA-NoCOMP is the fixed-checkpoint direct-context diagnostic used in this study. The measured sample means are 590 tokens per document and 2,950 context tokens per query after rounding.

ARIA-NoCOMP records 45.43% Avg F1, 0.93 points above ARIA 16 $\times$ ; see Table A33. DRO is included as a source-reported contextual row.

The end-to-end F1 difference is largest on MuSiQue (+6.34 pp for ARIA-NoCOMP vs. ARIA 16 $\times$ ), whereas the compressed configuration is higher on NQ, HotpotQA, and 2Wiki by 0.81, 0.80, and 1.05 pp, respectively. At the CEM level, ARIA-NoCOMP scores 42.40%, +3.57 pp over CLaRa and +1.16 pp over ARIA 16 $\times$ .

### A.22 MuSiQue Training-Set Augmentation Control

ARIA’s Phase II training uses the augmented MuSiQue partition (168,745 examples; Appendix A.16), which is 8.5 $\times$  the standard answerable subset of 19,938 examples after rounding. To measure how the ARIA–CLaRa MuSiQue difference changes when the augmentation is removed for both systems, we re-train both ARIA and CLaRa using *only the standard MuSiQue train-*

Table A28: Coupling-mechanism interventions across compression ratios (CEM%, Normal retrieval, 5-seed mean). All 128× rows use  $K_{\text{igfr}} = 1$ , matching the main-table configuration.

| CR   | Variant                | NQ    | HotpotQA | MuSiQue | 2Wiki | Avg   |
|------|------------------------|-------|----------|---------|-------|-------|
| 4×   | Full                   | 59.82 | 47.31    | 12.15   | 49.17 | 42.11 |
|      | −CFRS                  | 58.95 | 46.70    | 11.85   | 48.60 | 41.53 |
|      | Uniform $\rho = 0.625$ | 59.40 | 46.90    | 11.50   | 48.40 | 41.55 |
|      | −MTFRL                 | 59.50 | 46.38    | 11.74   | 48.75 | 41.59 |
|      | Forward-path-off       | 57.32 | 45.15    | 10.82   | 47.12 | 40.10 |
| 16×  | Full                   | 58.64 | 46.10    | 11.68   | 48.53 | 41.24 |
|      | −CFRS                  | 57.42 | 45.60    | 11.38   | 48.00 | 40.60 |
|      | Uniform $\rho = 0.625$ | 58.10 | 45.40    | 10.90   | 47.75 | 40.54 |
|      | −MTFRL                 | 58.15 | 44.95    | 11.20   | 48.15 | 40.61 |
|      | Forward-path-off       | 55.90 | 43.80    | 10.30   | 46.08 | 39.02 |
| 32×  | Full                   | 56.81 | 44.02    | 9.98    | 45.60 | 39.10 |
|      | −CFRS                  | 56.10 | 43.55    | 9.73    | 45.20 | 38.65 |
|      | Uniform $\rho = 0.625$ | 56.50 | 43.38    | 9.55    | 44.98 | 38.60 |
|      | −MTFRL                 | 56.52 | 43.12    | 9.78    | 45.30 | 38.68 |
|      | Forward-path-off       | 54.80 | 42.10    | 9.08    | 43.60 | 37.40 |
| 64×  | Full                   | 55.30 | 42.50    | 10.05   | 44.65 | 38.13 |
|      | −CFRS                  | 54.78 | 42.15    | 9.87    | 44.40 | 37.80 |
|      | Uniform $\rho = 0.625$ | 55.02 | 42.20    | 9.91    | 44.30 | 37.86 |
|      | −MTFRL                 | 55.10 | 41.88    | 9.93    | 44.48 | 37.85 |
|      | Forward-path-off       | 53.90 | 41.20    | 9.34    | 43.30 | 36.94 |
| 128× | Full                   | 54.05 | 41.65    | 9.55    | 42.40 | 36.91 |
|      | −CFRS                  | 53.44 | 41.28    | 9.38    | 42.04 | 36.54 |
|      | Uniform $\rho = 0.625$ | 52.80 | 42.10    | 9.95    | 44.30 | 37.29 |
|      | −MTFRL                 | 53.83 | 41.35    | 9.30    | 42.22 | 36.68 |
|      | Forward-path-off       | 52.58 | 41.08    | 9.72    | 43.25 | 36.66 |

Table A29: Independently trained soft- and hard-gate checkpoints under Normal retrieval (five-seed mean). Deltas are hard minus soft.

| CR   | Avg CEM |       |          | Avg F1 |       |          |
|------|---------|-------|----------|--------|-------|----------|
|      | Soft    | Hard  | $\Delta$ | Soft   | Hard  | $\Delta$ |
| 4×   | 42.11   | 41.93 | −0.18    | 45.93  | 45.70 | −0.23    |
| 16×  | 41.24   | 41.01 | −0.23    | 44.50  | 44.19 | −0.31    |
| 32×  | 39.10   | 38.86 | −0.24    | 42.67  | 42.35 | −0.32    |
| 64×  | 38.13   | 38.08 | −0.05    | 41.66  | 41.71 | +0.05    |
| 128× | 36.91   | 36.96 | +0.05    | 38.85  | 38.76 | −0.09    |

Table A30: Quick reference for the operational configurations used across Appendices A.18, A.20, and A.34. “Pipeline” columns indicate which components are enabled. All systems use the same Mistral-7B backbone, 5 seeds, Normal retrieval, CR = 16×. Avg F1 is the five-seed mean under this matched experimental setting. The labels state whether the compressor is ARIA- or CLaRa-trained; the pipeline columns separately show the retrieval and compression configuration.

| Configuration                               | CFRS | ACR | MTFRL | ARIA Retr. | ARIA Comp. | Used in  | Avg F1 |
|---------------------------------------------|------|-----|-------|------------|------------|----------|--------|
| ARIA (Full)                                 | ✓    | ✓   | ✓     | ✓          | ✓          | Tab. 1   | 44.50  |
| Forward-path-off                            | ×    | ×   | ×     | ✓          | ✓          | Tab. 5   | 42.00  |
| ARIA-COMP (CFRS only; CLaRa-trained)        | ✓    | ×   | ×     | ✓          | CLaRa      | Tab. A31 | 43.86  |
| ARIA-COMP (forward-path off; CLaRa-trained) | ×    | ×   | ×     | ✓          | CLaRa      | Tab. A32 | 42.02  |
| ARIA-COMP (ACR+CFRS; CLaRa retr.)           | ✓    | ✓   | ×     | CLaRa      | ✓          | Tab. A31 | 39.58  |
| ARIA-COMP (forward-path off; ARIA-trained)  | ×    | ×   | ×     | ✓          | ✓          | §A.34    | 42.00  |
| CLaRa                                       | ×    | ×   | ×     | CLaRa      | CLaRa      | Tab. 1   | 38.94  |

ing partition (19,938 examples), holding all other listed hyperparameters and training data sources fixed. All other benchmark training splits (NQ, HotpotQA, 2WikiMultiHopQA) are unchanged.

*The protocol-controlled ARIA-CLaRa MuSiQue gap changes by 0.5 pp.* Under the standard partition, ARIA records 18.9% MuSiQue F1, 2.2 points below the augmented-partition value of 21.1%; CLaRa records 16.0%, 1.7 points below its augmented-partition value of 17.7%. The ARIA-CLaRa gap narrows from 3.4 pp (augmented) to 2.9 pp (standard), a 0.5 pp change. The 2.9 pp standard-partition difference is an endpoint comparison between the two complete systems. On NQ, HotpotQA, and 2Wiki, both models change by at most 0.4 pp F1 in either direction; the

Table A31: Configured-system endpoints (F1%, Normal, CR=16 $\times$ , mean over 5 seeds). ARIA-COMP (CFRS only; CLaRa-trained): full ARIA retrieval pipeline plus CLaRa uniform compression and CFRS re-ranking. ARIA-COMP (ACR+CFRS; CLaRa retrieval): CLaRa single-pass retrieval followed by the ARIA-trained ACR compression and CFRS reranking operations. \*CLaRa values are 5-seed means.

| System                                | NQ           | HotpotQA     | MuSiQue      | 2Wiki        | Avg F1       |
|---------------------------------------|--------------|--------------|--------------|--------------|--------------|
| CLaRa (baseline)                      | 47.95        | 45.84        | 17.68        | 44.30        | 38.94*       |
| ARIA-COMP (ACR+CFRS; CLaRa retrieval) | 48.95        | 46.38        | 18.20        | 44.80        | 39.58        |
| ARIA-COMP (CFRS only; CLaRa-trained)  | 54.28        | 51.95        | 20.58        | 48.62        | 43.86        |
| <b>ARIA (full)</b>                    | <b>55.12</b> | <b>52.64</b> | <b>21.07</b> | <b>49.19</b> | <b>44.50</b> |

Table A32: CFRS-off and CFRS-on configured-system endpoints (F1%, Normal, CR=16 $\times$ , mean over 5 seeds). The CLaRa-trained forward-path-off variant uses ARIA’s five-stage retrieval pipeline without CFRS re-ranking, followed by CLaRa uniform compression.

| System                                      | NQ    | HotpotQA | MuSiQue | 2Wiki | Avg F1 |
|---------------------------------------------|-------|----------|---------|-------|--------|
| CLaRa (baseline)                            | 47.95 | 45.84    | 17.68   | 44.30 | 38.94  |
| ARIA-COMP (forward-path off; CLaRa-trained) | 52.66 | 50.03    | 18.60   | 46.79 | 42.02  |
| ARIA-COMP (CFRS only; CLaRa-trained)        | 54.28 | 51.95    | 20.58   | 48.62 | 43.86  |

Table A33: ARIA retrieval pipeline without compression (ARIA-NoCOMP) vs. main baselines (F1%/EM%/CEM%, Normal, CR=1 $\times$  for uncompressed systems, 5-seed mean). ARIA-NoCOMP: ARIA’s five-stage retrieval with direct top-5 context concatenation and no memory-token compression. DRO reports source-reported F1 and EM; its CEM cell is shown as –. \*CLaRa 16 $\times$  row: 5-seed means. The final row reports ARIA 16 $\times$  minus ARIA-NoCOMP.

| System                         | CR          | NQ    |       | HotpotQA |       | MuSiQue |       | 2Wiki |       | Avg F1 | Avg CEM |
|--------------------------------|-------------|-------|-------|----------|-------|---------|-------|-------|-------|--------|---------|
|                                |             | F1    | EM    | F1       | EM    | F1      | EM    | F1    | EM    |        |         |
| DRO                            | 1 $\times$  | 51.01 | 42.41 | 47.87    | 40.37 | 25.32   | 21.36 | 43.65 | 42.12 | 41.96  | –       |
| CLaRa                          | 16 $\times$ | 47.95 | 37.84 | 45.84    | 34.95 | 17.68   | 8.20  | 44.30 | 40.08 | 38.94  | 38.83*  |
| ARIA                           | 16 $\times$ | 55.12 | 44.83 | 52.64    | 41.17 | 21.07   | 11.34 | 49.19 | 44.73 | 44.50  | 41.24   |
| <b>ARIA-NoComp</b>             | 1 $\times$  | 54.31 | 44.12 | 51.84    | 41.03 | 27.41   | 22.80 | 48.14 | 44.87 | 45.43  | 42.40   |
| ARIA 16 $\times$ – ARIA-NoComp |             | +0.81 |       | +0.80    |       | –6.34   |       | +1.05 |       | –0.93  |         |

only larger observed change in this control is on MuSiQue.

*Standard-partition comparison.* Under the standard-partition control, ARIA records a +2.9 pp MuSiQue F1 difference from the protocol-controlled CLaRa reimplementation, 0.5 pp below the +3.4 pp reported in Table 1. The Avg F1 difference between ARIA and CLaRa changes from +5.56 pp (augmented, using 5-seed CLaRa means) to +5.50 pp (standard), a 0.06 pp change in the protocol-controlled ARIA–CLaRa gap.

### A.23 Per-Benchmark Pointwise Bootstrap Summaries (Mistral-7B)

Table A35 reports per-benchmark pointwise paired query-bootstrap  $p$ -value thresholds from 10,000 resamples after averaging the five seed-pair differences for ARIA vs. CLaRa on Mistral-7B, Normal retrieval. All values use the main-table configuration  $K_{\text{igfr}} = 2$  at 4 $\times$ –64 $\times$  and  $K_{\text{igfr}} = 1$  at

128 $\times$  (Appendix A.24). The symbols preserve cell-level resolution and are not multiplicity-adjusted. For the defined primary family of 50 Mistral ARIA–CLaRa CEM/F1 comparisons, Bonferroni correction at family-wise  $\alpha = 0.01$  gives  $p < 2.0 \times 10^{-4}$ ; the Avg-CEM and Avg-F1 gains at 4 $\times$ , 16 $\times$ , and 32 $\times$  meet that threshold. Claims for other cells follow the pointwise labels and are not presented as family-wise significant. All displayed CEM/F1 deltas have  $p < 0.01$  from 4 $\times$  through 64 $\times$  except MuSiQue CEM and F1 at 64 $\times$ , which both have  $p < 0.05$ . At 64 $\times$ , HotpotQA CEM reverses to –0.22 pp while the Avg CEM difference remains positive. At 128 $\times$ , NQ and HotpotQA retain  $p < 0.05$ ; the mean Avg-CEM difference of +0.16 pp has  $p \geq 0.05$ ; per-benchmark MuSiQue CEM reverses to –0.06 pp ( $p \geq 0.05$ ) and 2Wiki CEM reverses to –0.79 pp (\*\* $\dagger$ ). The F1 differences at 128 $\times$  have  $p < 0.05$  on NQ, HotpotQA, and MuSiQue.

Table A34: MuSiQue training set size ablation (F1%, Normal, CR = 16 $\times$ , mean over 5 seeds). Standard: 19,938 examples (the standard answerable partition). Augmented: 168,745 examples (main paper default). NQ, HotpotQA, and 2Wiki change by  $\leq 0.4$  pp in both conditions and are shown for reference only. \*CLaRa values are 5-seed means.

| Model                  | MuSiQue train    | NQ           | HotpotQA     | MuSiQue      | 2Wiki        | Avg F1       |
|------------------------|------------------|--------------|--------------|--------------|--------------|--------------|
| CLaRa                  | Augmented (168k) | 47.95        | 45.84        | 17.68        | 44.30        | 38.94*       |
| CLaRa                  | Standard (20k)   | 47.60        | 45.50        | 16.00        | 44.10        | 38.30        |
| <b>ARIA</b>            | Augmented (168k) | <b>55.12</b> | <b>52.64</b> | <b>21.07</b> | <b>49.19</b> | <b>44.50</b> |
| <b>ARIA</b>            | Standard (20k)   | <b>54.90</b> | <b>52.40</b> | <b>18.90</b> | <b>49.00</b> | <b>43.80</b> |
| ARIA-CLaRa (Augmented) |                  | +7.17        | +6.80        | +3.39        | +4.89        | +5.56        |
| ARIA-CLaRa (Standard)  |                  | +7.30        | +6.90        | +2.90        | +4.90        | +5.50        |

Table A35: Per-benchmark pointwise paired query-bootstrap  $p$ -value thresholds (10,000 resamples after averaging five seed-pair differences) for ARIA vs. CLaRa on Mistral-7B, Normal retrieval. All values use the main-table configuration ( $K_{\text{igfr}} = 2$  at 4 $\times$ -64 $\times$ ,  $K_{\text{igfr}} = 1$  at 128 $\times$ ). Markers denote pointwise thresholds:  $p < 0.001$  (\* \* \*),  $p < 0.01$  (\*\*),  $p < 0.05$  (\*), or  $p \geq 0.05$ . Markers are pointwise summaries. Dagger ( $\dagger$ ) indicates ARIA trails CLaRa on that benchmark.

| CR             | NQ    | Hotpot       | MuSiQue            | 2Wiki              | Avg        |
|----------------|-------|--------------|--------------------|--------------------|------------|
| <i>CEM (%)</i> |       |              |                    |                    |            |
| 4 $\times$     | * * * | * * *        | * * *              | * * *              | * * *      |
| 16 $\times$    | * * * | * * *        | * * *              | * * *              | * * *      |
| 32 $\times$    | * * * | * * *        | **                 | * * *              | * * *      |
| 64 $\times$    | * * * | ** $\dagger$ | *                  | **                 | * * *      |
| 128 $\times$   | * * * | **           | $\geq .05^\dagger$ | ** $\dagger$       | $\geq .05$ |
| <i>F1 (%)</i>  |       |              |                    |                    |            |
| 4 $\times$     | * * * | * * *        | * * *              | * * *              | * * *      |
| 16 $\times$    | * * * | * * *        | * * *              | * * *              | * * *      |
| 32 $\times$    | * * * | * * *        | **                 | * * *              | * * *      |
| 64 $\times$    | * * * | * * *        | *                  | * * *              | * * *      |
| 128 $\times$   | * * * | * * *        | *                  | $\geq .05^\dagger$ | * * *      |

## A.24 Hyperparameter Sensitivity Across Compression Ratios

*Selection protocol.* As with the main hyperparameter sweeps (Appendix A.15), the per-CR  $K_{\text{igfr}}$  setting was selected on the 500-query held-out dev partition, disjoint from every evaluation split. The tables below report evaluation sensitivity around those dev-selected settings.

The main sensitivity analyses are conducted at 16 $\times$ . Tables A36 and A37 additionally report the measured grids at 4 $\times$  and 128 $\times$ .

*CCEF threshold  $\tau$ .*  $\tau = 0.30$  has the highest measured aggregate value in the displayed 4 $\times$  and 128 $\times$  grids, matching the 16 $\times$  selected setting. The measured NQ CEM worst-to-best range is 0.72 pp at 4 $\times$  and 1.55 pp at 128 $\times$ , a 2.15 $\times$  range ratio. At 128 $\times$ ,  $\tau = 0.40$  records 53.60 NQ CEM but is 0.70 pp lower on HotpotQA and 0.55 pp lower on 2Wiki than  $\tau = 0.30$ .

*IGFR iteration count  $K$ .* At 4 $\times$ ,  $K = 2$  has the highest measured Avg CEM in the displayed grid,

Table A36: CCEF threshold  $\tau$  sensitivity at 4 $\times$  and 128 $\times$  (CEM%, Normal).  $\tau = 0.30$  has the highest measured aggregate value in both displayed grids. The NQ CEM worst-to-best range is 0.72 pp at 4 $\times$  vs. 1.55 pp at 128 $\times$ . **Note:**  $K_{\text{igfr}} = 2$  is held fixed across the displayed 128 $\times$   $\tau$  grid; main results use  $K = 1$  with Avg CEM 36.91%.

| CR                      | $\tau$      | NQ           | Hotpot       | MuSiQue      | 2Wiki        |
|-------------------------|-------------|--------------|--------------|--------------|--------------|
| 4 $\times$              | 0.10        | 59.10        | 46.85        | 11.80        | 48.72        |
|                         | 0.20        | 59.55        | 47.10        | 12.00        | 49.05        |
|                         | <b>0.30</b> | <b>59.82</b> | <b>47.31</b> | <b>12.15</b> | <b>49.17</b> |
|                         | 0.40        | 59.30        | 47.05        | 11.90        | 48.85        |
|                         | 0.50        | 59.45        | 46.50        | 11.65        | 48.55        |
| 128 $\times$ $^\dagger$ | 0.10        | 52.55        | 40.80        | 9.10         | 41.65        |
|                         | 0.20        | 53.40        | 41.20        | 9.30         | 42.00        |
|                         | <b>0.30</b> | <b>53.80</b> | <b>41.50</b> | <b>9.45</b>  | <b>42.20</b> |
|                         | 0.40        | 53.60        | 40.80        | 9.20         | 41.65        |
|                         | 0.50        | 54.10        | 40.30        | 8.95         | 41.40        |

$^\dagger$  128 $\times$  sensitivity rows use  $K_{\text{igfr}} = 2$  to hold  $K$  fixed across all  $\tau$  values; main results use  $K = 1$ . **Bold** marks the selected  $\tau = 0.30$  row. Note that  $\tau = 0.50$  achieves the highest individual NQ CEM (54.10) but is lower on HotpotQA and 2Wiki;  $\tau = 0.30$  has the highest observed aggregate result in the tested grid.

though the gap over  $K = 1$  is smaller (+0.30 pp Avg CEM at 4 $\times$  vs. +0.38 pp at 16 $\times$ ; computed from Table A15:  $K = 2$  avg 41.24 minus  $K = 1$  avg 40.86). At 128 $\times$ , the tested  $K = 1$  configuration records a 0.17 pp higher displayed Avg CEM than  $K = 2$ .

Table A37: IGFR iteration count  $K$  sensitivity at 4 $\times$ , 16 $\times$ , and 128 $\times$  (CEM%, Normal). 16 $\times$  values are reproduced from Table A15 for ease of comparison. At 128 $\times$ ,  $K = 1$  records a 0.17 pp higher displayed Avg CEM than  $K = 2$ .

| CR           | $K$ | NQ           | Hotpot       | MuSiQue      | 2Wiki        | Avg          |
|--------------|-----|--------------|--------------|--------------|--------------|--------------|
| 4 $\times$   | 0   | 58.21        | 45.72        | 11.20        | 47.92        | 40.76        |
|              | 1   | 59.50        | 47.00        | 11.85        | 48.90        | 41.81        |
|              | 2   | <b>59.82</b> | <b>47.31</b> | <b>12.15</b> | <b>49.17</b> | <b>42.11</b> |
|              | 3   | 59.55        | 47.15        | 11.95        | 48.80        | 41.86        |
| 16 $\times$  | 0   | 56.83        | 44.52        | 10.73        | 47.14        | 39.81        |
|              | 1   | 58.12        | 45.80        | 11.30        | 48.20        | 40.86        |
|              | 2   | <b>58.64</b> | <b>46.10</b> | <b>11.68</b> | <b>48.53</b> | <b>41.24</b> |
|              | 3   | 58.40        | 45.95        | 11.40        | 48.10        | 40.96        |
| 128 $\times$ | 0   | 52.80        | 40.60        | 8.90         | 41.50        | 35.95        |
|              | 1   | <b>54.05</b> | <b>41.65</b> | <b>9.55</b>  | <b>42.40</b> | <b>36.91</b> |
|              | 2   | 53.80        | 41.50        | 9.45         | 42.20        | 36.74        |
|              | 3   | 53.40        | 41.20        | 9.20         | 41.85        | 36.41        |

*Cross-CR configuration selection.* The dev-selected setting is  $K = 2$  at 4 $\times$ –64 $\times$  and  $K = 1$  at 128 $\times$ . In the displayed evaluation grids, the selected rows also record the highest measured aggregate values at 4 $\times$ , 16 $\times$ , and 128 $\times$ . **all 128 $\times$  main-table CEM results (Tables A6 and A7) report  $K_{\text{igfr}} = 1$  values** (NQ 54.05, HotpotQA 41.65, MuSiQue 9.55, 2Wiki 42.40, avg 36.91%), where the  $K_{\text{igfr}} = 1$  vs.  $K_{\text{igfr}} = 2$  switch was made before evaluation on the test set, based on the dev-set finding. On the evaluation set this setting records a 0.17 pp higher displayed Avg CEM than the  $K = 2$  configuration. ARIA records 36.91% versus CLaRa at 36.75%, a +0.16 pp mean difference ( $p \geq 0.05$  in the pointwise bootstrap summary; Appendix A.23). The coupling-mechanism ablation at 128 $\times$  (Appendix A.18, Table A28) is reported at the same  $K_{\text{igfr}} = 1$  as the main table.

#### A.25 CLaRa Mistral-7B: 5-Seed Reimplementation (All Compression Ratios)

We retrain CLaRa on Mistral-7B with 5 seeds  $\times$  5 CRs (25 runs) under matched settings (same data, lr  $2 \times 10^{-4}$ , warmup 500, 5 epochs, batch 32, LoRA rank  $r_{\text{LoRA}} = 16$ ; seeds {42, 123, 456, 789, 2024}). 5-seed CLaRa means are used in all main-text comparisons. For each

ratio, the query-reasoner and generator adapters are initialized from the corresponding Phase-I compressor checkpoint, while the compressor and offline document representations remain fixed. Phase II minimizes answer next-token cross-entropy and updates the query reasoner and generator. CLaRa’s straight-through selector uses hard top- $k$  document identities in the forward pass and the soft-selection surrogate in the backward pass, with a uniform memory-token count per selected document, following He et al. (2026). The largest per-cell deviation from CLaRa’s original single-run is 0.51 pp CEM (MuSiQue at 32 $\times$ ; on average  $\leq 0.25$  pp across all cells, with both positive and negative signs). Tables A38/A39 report CEM%/F1%.

*EM reimplementation results.* Table A40 reports 5-seed EM% under Normal retrieval. Per-benchmark 5-seed std values are: NQ EM  $\leq 0.58$  pp, HotpotQA EM  $\leq 1.14$  pp, MuSiQue EM  $\leq 1.73$  pp (the highest observed value; evaluation size 2,417), and 2Wiki EM  $\leq 0.82$  pp. 5-seed means shift by at most 0.18 pp in absolute value from original single-run values. All main-text comparison rows use the five-seed CLaRa means.

#### A.26 Supporting-Backbone Intervention Results Across Compression Ratios

Table A41 reports the five-stage supporting-backbone intervention analysis at 16 $\times$ . Tables A42 and A43 repeat the analysis at 4 $\times$  and 32 $\times$ . All rows are inference-time interventions on the same ARIA-trained checkpoints. Each row is a configured-system intervention: disabling an upstream stage changes the inputs received by downstream stages. The –MADS row has the largest Full-minus-variant CEM difference at every reported ratio. The –QCA NQ difference varies across CRs (+0.38 pp at 4 $\times$ , +2.54 pp at 16 $\times$ , +1.33 pp at 32 $\times$ ), with the largest observed difference at moderate compression. At 4 $\times$ , the Full-minus–IGFR NQ difference is +0.79 pp vs. +1.81 pp at 16 $\times$ ; the 4 $\times$  setting has the larger token budget. At 32 $\times$ , the –CCEF row also records lower NQ CEM.

#### A.27 MADS Scorer Combination Study Under Normal Retrieval

Appendix A.14 evaluated MADS scorer combinations under Oracle retrieval at 32 $\times$ . Table A44 reports the study at the measured 16 $\times$  Normal-retrieval setting. The full three-scorer

Table A38: CLaRa Mistral-7B CEM% under Normal and Oracle retrieval, mean over 5 seeds, across all compression ratios. The  $\Delta$  block reports per-CR differences (5-seed mean – original single-run value) under Normal retrieval. Per-benchmark shifts range from  $-0.51$  pp (MuSiQue, 32 $\times$ ; the largest absolute shift) to  $+0.45$  pp (HotpotQA, 32 $\times$ ); the largest average-level shift is  $0.14$  pp. Per-cell paired query-bootstrap  $p$ -value thresholds for ARIA vs. this 5-seed CLaRa baseline at the same nominal CR are reported in Table A35.

| Retr.                                                           | CR           | NQ      | HotpotQA | MuSiQue | 2Wiki   | Avg     |
|-----------------------------------------------------------------|--------------|---------|----------|---------|---------|---------|
| Normal                                                          | 4 $\times$   | 57.18   | 45.24    | 10.40   | 47.08   | 39.98   |
|                                                                 | 16 $\times$  | 55.72   | 43.86    | 10.22   | 45.52   | 38.83   |
|                                                                 | 32 $\times$  | 54.79   | 42.55    | 9.72    | 44.53   | 37.90   |
|                                                                 | 64 $\times$  | 53.91   | 42.72    | 9.88    | 44.10   | 37.65   |
|                                                                 | 128 $\times$ | 53.06   | 41.14    | 9.61    | 43.19   | 36.75   |
| Oracle                                                          | 4 $\times$   | 76.38   | 73.96    | 46.14   | 70.64   | 66.78   |
|                                                                 | 16 $\times$  | 75.37   | 70.93    | 43.06   | 66.31   | 63.92   |
|                                                                 | 32 $\times$  | 73.06   | 68.35    | 38.54   | 64.17   | 61.03   |
|                                                                 | 64 $\times$  | 71.42   | 64.25    | 35.12   | 60.89   | 57.92   |
|                                                                 | 128 $\times$ | 68.84   | 61.93    | 31.12   | 58.59   | 55.12   |
| $\Delta$ (5-seed mean – original single-run, Normal retrieval): |              |         |          |         |         |         |
|                                                                 | 4 $\times$   | +0.22   | +0.41    | $-0.18$ | +0.09   | +0.14   |
|                                                                 | 16 $\times$  | +0.16   | $-0.28$  | +0.37   | +0.23   | +0.12   |
|                                                                 | 32 $\times$  | $-0.19$ | +0.45    | $-0.51$ | $-0.14$ | $-0.10$ |
|                                                                 | 64 $\times$  | +0.11   | $-0.33$  | +0.08   | $-0.19$ | $-0.08$ |
|                                                                 | 128 $\times$ | $-0.24$ | +0.17    | $-0.29$ | $-0.16$ | $-0.13$ |

Table A39: CLaRa Mistral-7B F1% under Normal and Oracle retrieval, mean over 5 seeds, across all compression ratios.

| Retr.  | CR           | NQ    | HotpotQA | MuSiQue | 2Wiki | Avg   |
|--------|--------------|-------|----------|---------|-------|-------|
| Normal | 4 $\times$   | 48.34 | 46.07    | 17.55   | 47.32 | 39.82 |
|        | 16 $\times$  | 47.95 | 45.84    | 17.68   | 44.30 | 38.94 |
|        | 32 $\times$  | 47.28 | 44.28    | 16.72   | 42.82 | 37.77 |
|        | 64 $\times$  | 46.41 | 44.65    | 16.08   | 40.57 | 36.93 |
|        | 128 $\times$ | 45.54 | 42.29    | 14.50   | 38.31 | 35.16 |
| Oracle | 4 $\times$   | 75.78 | 69.81    | 33.28   | 74.02 | 63.22 |
|        | 16 $\times$  | 71.68 | 68.57    | 30.86   | 67.98 | 59.77 |
|        | 32 $\times$  | 69.88 | 66.08    | 28.94   | 64.52 | 57.36 |
|        | 64 $\times$  | 67.57 | 63.61    | 27.16   | 61.61 | 54.99 |
|        | 128 $\times$ | 65.00 | 61.36    | 23.65   | 58.43 | 52.11 |

Table A40: CLaRa Mistral-7B EM% under Normal retrieval, mean over 5 seeds, across all compression ratios. The  $\Delta$  block reports per-CR differences (5-seed mean – original single-run value). The largest absolute per-benchmark shift is  $0.18$  pp; the largest average-level shift is  $0.09$  pp.

| Retr.                                                           | CR           | NQ      | HotpotQA | MuSiQue | 2Wiki   | Avg     |
|-----------------------------------------------------------------|--------------|---------|----------|---------|---------|---------|
| Normal                                                          | 4 $\times$   | 38.16   | 35.12    | 8.11    | 43.11   | 31.12   |
|                                                                 | 16 $\times$  | 37.84   | 34.95    | 8.20    | 40.08   | 30.27   |
|                                                                 | 32 $\times$  | 37.22   | 33.95    | 7.85    | 38.52   | 29.38   |
|                                                                 | 64 $\times$  | 36.51   | 34.18    | 7.52    | 36.34   | 28.64   |
|                                                                 | 128 $\times$ | 35.63   | 31.92    | 6.68    | 34.11   | 27.08   |
| $\Delta$ (5-seed mean – original single-run, Normal retrieval): |              |         |          |         |         |         |
|                                                                 | 4 $\times$   | +0.08   | +0.18    | $-0.05$ | +0.11   | +0.08   |
|                                                                 | 16 $\times$  | +0.14   | $-0.07$  | +0.12   | +0.17   | +0.09   |
|                                                                 | 32 $\times$  | $-0.11$ | +0.15    | $-0.13$ | +0.06   | $-0.01$ |
|                                                                 | 64 $\times$  | +0.09   | $-0.12$  | +0.04   | $-0.15$ | $-0.04$ |
|                                                                 | 128 $\times$ | $-0.13$ | +0.08    | $-0.09$ | $-0.04$ | $-0.04$ |

Table A41: Supporting ablation of the experimental retrieval backbone (CEM% / F1%, Normal, 16 $\times$ , 5-seed mean). Q/A/I/M/C denote QCA/AHR/IGFR/MADS/CCEF. Rows are fixed-checkpoint inference-time interventions on the same ARIA-trained checkpoints.

| Variant     | Q | A | I | M | C | CEM%         |              |              |              |              | F1%          |              |              |              |              |
|-------------|---|---|---|---|---|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|             |   |   |   |   |   | NQ           | Hotpot       | MuSi.        | 2Wiki        | Avg          | NQ           | Hotpot       | MuSi.        | 2Wiki        | Avg          |
| <b>Full</b> | ✓ | ✓ | ✓ | ✓ | ✓ | <b>58.64</b> | <b>46.10</b> | <b>11.68</b> | <b>48.53</b> | <b>41.24</b> | <b>55.12</b> | <b>52.64</b> | <b>21.07</b> | <b>49.19</b> | <b>44.50</b> |
| −IGFR       | ✓ | ✓ | — | ✓ | ✓ | 56.83        | 44.52        | 10.73        | 47.14        | 39.81        | 53.42        | 50.83        | 19.36        | 47.81        | 42.86        |
| −CCEF       | ✓ | ✓ | ✓ | ✓ | — | 56.95        | 45.81        | 10.54        | 48.12        | 40.36        | 53.53        | 52.29        | 19.01        | 48.79        | 43.41        |
| −MADS       | ✓ | ✓ | ✓ | — | ✓ | 55.80        | 44.30        | 11.10        | 46.40        | 39.40        | 52.45        | 50.55        | 20.02        | 47.03        | 42.51        |
| −AHR        | ✓ | — | ✓ | ✓ | ✓ | 57.50        | 44.95        | 10.85        | 46.90        | 40.05        | 54.06        | 51.29        | 19.57        | 47.54        | 43.12        |
| −QCA        | — | ✓ | ✓ | ✓ | ✓ | 56.10        | 45.10        | 10.30        | 47.30        | 39.70        | 52.74        | 51.47        | 18.58        | 47.96        | 42.69        |
| CLaRa       | — | — | — | — | — | 55.72        | 43.86        | 10.22        | 45.52        | 38.83        | 47.95        | 45.84        | 17.68        | 44.30        | 38.94        |

Table A42: Supporting-backbone interventions (CEM%, Normal, CR=4 $\times$ , mean over 5 seeds). Q=QCA, A=AHR, I=IGFR, M=MADS, C=CCEF. 5-seed per-cell std lies in {NQ : 0.48–0.71, HotpotQA : 0.92–1.48, MuSiQue : 1.41–2.05, 2Wiki : 0.73–1.15} pp.

| Variant     | Q | A | I | M | C | NQ           | HotpotQA     | MuSiQue      | 2Wiki        |
|-------------|---|---|---|---|---|--------------|--------------|--------------|--------------|
| <b>Full</b> | ✓ | ✓ | ✓ | ✓ | ✓ | <b>59.82</b> | <b>47.31</b> | <b>12.15</b> | <b>49.17</b> |
| −IGFR       | ✓ | ✓ | — | ✓ | ✓ | 59.03        | 46.40        | 11.56        | 48.55        |
| −CCEF       | ✓ | ✓ | ✓ | ✓ | — | 59.14        | 47.10        | 11.38        | 48.88        |
| −MADS       | ✓ | ✓ | ✓ | — | ✓ | 57.90        | 45.92        | 11.71        | 47.53        |
| −AHR        | ✓ | — | ✓ | ✓ | ✓ | 59.44        | 46.85        | 11.60        | 48.42        |
| −QCA        | — | ✓ | ✓ | ✓ | ✓ | 59.44        | 47.05        | 11.80        | 49.05        |
| CLaRa       | — | — | — | — | — | 57.18        | 45.24        | 10.40        | 47.08        |

Table A43: Supporting-backbone interventions (CEM%, Normal, CR=32 $\times$ , mean over 5 seeds). 5-seed per-cell std lies in {NQ : 0.52–0.78, HotpotQA : 1.04–1.62, MuSiQue : 1.58–2.18, 2Wiki : 0.86–1.28} pp.

| Variant     | Q | A | I | M | C | NQ           | HotpotQA     | MuSiQue     | 2Wiki        |
|-------------|---|---|---|---|---|--------------|--------------|-------------|--------------|
| <b>Full</b> | ✓ | ✓ | ✓ | ✓ | ✓ | <b>56.81</b> | <b>44.02</b> | <b>9.98</b> | <b>45.60</b> |
| −IGFR       | ✓ | ✓ | — | ✓ | ✓ | 55.71        | 42.92        | 9.34        | 44.55        |
| −CCEF       | ✓ | ✓ | ✓ | ✓ | — | 56.17        | 43.83        | 9.68        | 45.20        |
| −MADS       | ✓ | ✓ | ✓ | — | ✓ | 54.50        | 42.48        | 9.55        | 43.60        |
| −AHR        | ✓ | — | ✓ | ✓ | ✓ | 55.90        | 43.10        | 9.64        | 44.80        |
| −QCA        | — | ✓ | ✓ | ✓ | ✓ | 55.48        | 43.25        | 9.28        | 44.62        |
| CLaRa       | — | — | — | — | — | 54.79        | 42.55        | 9.72        | 44.53        |

(Lex+Sem+Ent) configuration records the highest measured Avg F1. Relative to Lex+Sem, the Lex+Sem+Ent row records larger measured differences on HotpotQA (+1.35 pp F1) and MuSiQue (+0.91 pp) than on NQ/2Wiki (+0.30 pp). Learned scorer weights record 0.28 pp higher mean NQ F1 but 0.43 pp and 0.52 pp lower means on HotpotQA and MuSiQue, respectively. Equal weights were selected on the dev partition before this reported evaluation. As in the Oracle study, all rows are inference-time scoring interventions on the same trained checkpoints.

## A.28 ACR Adaptive Budget Floor at 128 $\times$

The 128 $\times$  bridge ablation in Appendix A.18 records a +0.38 pp mean Avg-CEM difference for the uniform-rate control over ACR with  $\rho_{\min} = 0.25$  (with  $K = 1$ ). We evaluate a *complexity-*

*adaptive budget floor*: retain  $\rho_{\min} = 0.25$  for QCA-routed Simple queries and raise  $\rho_{\min}$  for Multi-Hop and Multi-Aspect queries.

*Experimental setup.* All experiments use 5 independent seeds at 128 $\times$  Normal retrieval with  $K_{\text{igfr}} = 1$ , matching the main-table configuration. Three adaptive-floor settings are evaluated:  $\rho_{\min}^{\text{MH}} \in \{0.35, 0.45, 0.55\}$  for Multi-Hop/Multi-Aspect queries, with Simple queries fixed at  $\rho_{\min}^{\text{S}} = 0.25$ . The floor is applied at inference time only (without retraining); all other listed hyperparameters are held fixed to the main experiments.

*Results.* Table A45 reports CEM% and 2Wiki F1.

*The adaptive floor at  $\rho_{\min}^{\text{MH}} = 0.45$  records the highest measured Avg CEM (37.35%) in the tested grid, a +0.44 pp difference from Full*

Table A44: MADS scorer combination study (F1%, Normal, CR=16×, 5 seeds). Bold numeric cells mark the highest measured value in each score column; the bold configuration label marks the dev-selected equal-weight setting. Equal-weight Lex+Sem+Ent records the highest Avg F1 in this grid.

| Config             | Lex | Sem | Ent | NQ           | HotpotQA     | MuSiQue      | 2Wiki        | Avg F1       |
|--------------------|-----|-----|-----|--------------|--------------|--------------|--------------|--------------|
| Lex only           | ✓   | ×   | ×   | 52.10        | 49.35        | 18.80        | 47.35        | 41.90        |
| Sem only           | ×   | ✓   | ×   | 53.20        | 50.10        | 19.50        | 48.20        | 42.75        |
| Lex+Sem            | ✓   | ✓   | ×   | 54.82        | 51.29        | 20.16        | 48.89        | 43.79        |
| <b>Lex+Sem+Ent</b> | ✓   | ✓   | ✓   | 55.12        | <b>52.64</b> | <b>21.07</b> | <b>49.19</b> | <b>44.50</b> |
| Learned weights    | ✓   | ✓   | ✓   | <b>55.40</b> | 52.21        | 20.55        | 49.03        | 44.30        |

ARIA (36.91%) and a +0.06 pp difference from the uniform-rate control (37.29%). The largest changes relative to Full ARIA occur on 2Wiki (+1.48 pp CEM, +1.77 pp F1). The remaining CEM differences are −0.50 pp on NQ, +0.45 pp on HotpotQA, and +0.33 pp on MuSiQue. The 2Wiki F1 under  $\rho_{\min}^{\text{MH}} = 0.45$  is 39.62% (vs. Full ARIA 37.85%), a +1.77 pp difference (§5).

At  $\rho_{\min}^{\text{MH}} = 0.55$ , NQ changes by −0.90 pp while 2Wiki records a +1.70 pp difference. The 0.45 setting has the highest Avg CEM in the evaluated grid: the  $\rho_{\min}^{\text{MH}} = 0.55$  row records a lower Avg CEM of 37.27% vs. 37.35%, a 0.08 pp difference.

### A.29 Budget-Matched ACR Allocation Controls

We compare the default ACR allocation with uniform and adaptive-floor allocations while matching the mean evidence-token total at each ratio. All rows are fixed-checkpoint allocation interventions with retrieval, document count, decoding, and the remaining forward path held fixed. At 16×, all variants retain 108 tokens per query; the adaptive floor tracks default ACR within 0.05 Avg-CEM and 0.06 Avg-F1 points. At 128×, all variants retain 13 tokens, and the adaptive allocation records the highest Avg CEM, Avg F1, and 2Wiki F1 in the matched comparison (Table A46).

At 16×, the per-rank mean allocations for default ACR are 36.1/27.2/19.1/14.0/11.4 tokens (total 107.8 before displayed-value rounding). Uniform matching assigns 21.6 tokens to each rank. The adaptive matched allocation assigns 34.8/26.5/19.8/14.8/12.1 tokens, which sums to 108.0 and preserves the default rank ordering while raising the lower-rank floor.

### A.30 Focused Cross-Backbone Check: Qwen-2.5-14B-Instruct

To obtain one additional-backbone comparison at a larger model scale, we evaluate Qwen-2.5-14B-

Instruct (Qwen Team, 2024) at 16× Normal retrieval. Training follows the same Phase I/II protocol as the main experiments—including the same augmented MuSiQue training partition (Appendix A.16)—with these adjustments for the larger backbone: (i) global effective batch 16 per optimizer update; (ii) MTFRL projection head resized to 5120 → 2560 → 1024; (iii) LoRA rank  $r_{\text{LoRA}} = 16$  retained; (iv) learning rate reduced by 0.8× to  $1.6 \times 10^{-4}$ . All other reported hyperparameters (warmup and epochs) are unchanged. Both ARIA and CLaRa are trained on this augmented training data for 5 seeds.

ARIA records 42.23% Avg CEM (+1.74 pp over CLaRa) and 45.51% Avg F1 (+4.41 pp; Table A47). These mean differences are smaller than on Mistral-7B (+2.41 pp CEM; +5.56 pp F1) and Llama-3-8B (+2.85 pp CEM; +6.14 pp F1). Qwen-2.5-14B’s CLaRa baseline is also +1.66 pp higher in CEM than the Mistral-7B CLaRa baseline (40.49% vs. 38.83%). The per-benchmark F1-difference ordering (NQ > HotpotQA > 2Wiki > MuSiQue) matches the Mistral-7B pattern. The positive differences on this additional backbone provide a focused check at the evaluated 16× setting.

### A.31 QCA Reference-Label Routing Analysis

We use the 1,000-query annotated sample described in Appendix A.6 (NQ 225, HotpotQA 257, MuSiQue 84, 2Wiki 434). The two annotators’ disagreements are arbitrated after agreement measurement, yielding 279 Simple, 157 Multi-Aspect, and 564 Multi-Hop consensus labels. To measure the routing endpoint, we replace Stage 1 with these consensus labels while holding all other stages, hyperparameters, checkpoints, and five seeds fixed. The rule-based QCA confusion matrix in Table A48 gives 86.9% accuracy and 83.0% macro-F1 against the consensus reference.

Class-wise Simple/Multi-Aspect/Multi-Hop

Table A45: ACR adaptive budget floor at 128 $\times$  ( $K=1$ ) (CEM%, Normal retrieval, mean over 5 seeds); rightmost column shows 2Wiki F1% (the benchmark with the largest measured change in this tested grid).  $\rho_{\min}^S$  = Simple floor (fixed at 0.25);  $\rho_{\min}^{MH}$  = Multi-Hop/Multi-Aspect floor.  $\rho_{\min}^{MH} = 0.45$  records the highest measured Avg CEM and 2Wiki F1 in the tested grid; the differences from Full ARIA and the uniform-rate control are reported in the text. Bold metric entries mark the highest value in each column.

| Variant                                       | $\rho_{\min}^S$     | $\rho_{\min}^{MH}$ | NQ           | HotpotQA     | MuSiQue     | 2Wiki        | Avg CEM      | 2Wiki F1     |
|-----------------------------------------------|---------------------|--------------------|--------------|--------------|-------------|--------------|--------------|--------------|
| Full ARIA                                     | 0.25                | 0.25               | <b>54.05</b> | 41.65        | 9.55        | 42.40        | 36.91        | 37.85        |
| Uniform-rate control                          | $\rho \equiv 0.625$ |                    | 52.80        | <b>42.10</b> | <b>9.95</b> | <b>44.30</b> | 37.29        | 39.41        |
| Adaptive ( $\rho_{\min}^{MH} = 0.35$ )        | 0.25                | 0.35               | 53.80        | 41.95        | 9.78        | 43.30        | 37.21        | 38.74        |
| <b>Adaptive</b> ( $\rho_{\min}^{MH} = 0.45$ ) | 0.25                | <b>0.45</b>        | 53.55        | <b>42.10</b> | 9.88        | 43.88        | <b>37.35</b> | <b>39.62</b> |
| Adaptive ( $\rho_{\min}^{MH} = 0.55$ )        | 0.25                | 0.55               | 53.15        | 42.00        | 9.82        | 44.10        | 37.27        | 39.50        |

Table A46: Budget-matched ACR allocation controls under Normal retrieval (five-seed means). Every row within a ratio has the same mean evidence-token total.

| CR           | Allocation                    | Tokens/query | Avg CEM      | Avg F1       | 2Wiki F1     |
|--------------|-------------------------------|--------------|--------------|--------------|--------------|
| 16 $\times$  | Default ACR                   | 108          | 41.24        | 44.50        | —            |
|              | Uniform matched               | 108          | 40.70        | 43.86        | —            |
|              | Adaptive floor matched        | 108          | 41.19        | 44.44        | —            |
| 128 $\times$ | Default ACR                   | 13           | 36.91        | 38.85        | 37.85        |
|              | Uniform matched               | 13           | 37.21        | 39.20        | 39.05        |
|              | <b>Adaptive floor matched</b> | <b>13</b>    | <b>37.28</b> | <b>39.32</b> | <b>39.28</b> |

Table A47: Focused cross-backbone check on Qwen-2.5-14B-Instruct (CEM%/F1%, Normal, CR = 16 $\times$ , mean over 5 seeds). Positive ARIA–CLaRa mean differences are observed at 16 $\times$ .

| Model    | CEM%         |              |              |              |              | F1%          |              |              |              |              |
|----------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|          | NQ           | Hotpot       | MuSiQue      | 2Wiki        | Avg          | NQ           | Hotpot       | MuSiQue      | 2Wiki        | Avg          |
| CLaRa    | 57.90        | 45.50        | 11.35        | 47.20        | 40.49        | 50.10        | 48.00        | 19.80        | 46.50        | 41.10        |
| ARIA     | <b>59.72</b> | <b>47.25</b> | <b>12.54</b> | <b>49.40</b> | <b>42.23</b> | <b>55.77</b> | <b>53.29</b> | <b>22.32</b> | <b>50.66</b> | <b>45.51</b> |
| $\Delta$ | +1.82        | +1.75        | +1.19        | +2.20        | +1.74        | +5.67        | +5.29        | +2.52        | +4.16        | +4.41        |

*Mistral-7B reference:  $\Delta$  CEM = +2.41 pp,  $\Delta$  F1 = +5.56 pp (Table 1).*

Table A48: Rule-based QCA against the adjudicated consensus reference ( $n = 1,000$ ). Rows are reference labels and columns are predictions.

| Reference       | Simple | Multi-Aspect | Multi-Hop | Total |
|-----------------|--------|--------------|-----------|-------|
| Simple          | 245    | 23           | 11        | 279   |
| Multi-Aspect    | 21     | 118          | 18        | 157   |
| Multi-Hop       | 17     | 41           | 506       | 564   |
| Predicted total | 283    | 182          | 535       | 1000  |

precision is 86.6/64.8/94.6%, recall is 87.8/75.2/89.7%, and F1 is 87.2/69.6/92.1%. On the same annotated subset, consensus-reference routing records 41.74% Avg CEM and 44.91% Avg F1, compared with 41.14% and 44.38% for rule-based QCA: differences of +0.60 and +0.53 points, respectively (Table A49).

Table A49: Consensus-reference QCA endpoint on the same 1,000-query subset (Normal, 16 $\times$ , five-seed mean).

| Router                         | Avg CEM      | Avg F1       |
|--------------------------------|--------------|--------------|
| Rule-based QCA                 | 41.14        | 44.38        |
| <b>Consensus-reference QCA</b> | <b>41.74</b> | <b>44.91</b> |
| $\Delta$                       | +0.60        | +0.53        |

### A.32 First-Pass Retrieval and Phase-II Adaptation

**Phase-II retrieval adaptation.** Table A50 compares the operating point after  $W_{\text{BGE}}$  pre-fitting but before Phase-II QR adaptation with the corresponding Phase-II-trained checkpoint at nominal  $16\times$ . The frozen BGE encoder/index, corpus, retrieval backbone, candidate construction, seed set, and evaluation queries are shared. Phase II raises the four-benchmark mean by  $2.20/2.88/2.80$  points at Recall@1/3/5. Across five seeds, mean Recall@5 changes from  $58.5 \pm 0.4$  to  $61.3 \pm 0.3$ .

For Recall@5, the pre/post seed means and SDs are respectively  $71.7 \pm 0.5/74.3 \pm 0.4$  (NQ),  $60.8 \pm 0.8/64.2 \pm 0.7$  (HotpotQA),  $35.7 \pm 0.9/38.2 \pm 0.8$  (MuSiQue), and  $65.8 \pm 0.6/68.5 \pm 0.5$  (2Wiki). Paired query-bootstrap 95% CIs for the post-minus-pre difference are  $[1.9, 3.3]$ ,  $[2.7, 4.1]$ ,  $[1.3, 3.7]$ , and  $[2.2, 3.2]$  points, respectively ( $p < .01$ ,  $p < .001$ ,  $p < .05$ , and  $p < .001$ ; pointwise tests).

Table A51 reports the realized first-pass operating points for all five nominal ratios. The QR LoRA adapter is updated in Phase II, and each nominal compression ratio has a separately trained checkpoint. First-pass Recall@5 is measured before MTFRL and CFRS. The displayed average ARIA-CLaRa difference ranges from  $+4.1$  to  $+4.6$  percentage points across the five ratios.

### A.33 QCA Sensitivity Analysis: Rule-Based vs. Zero-Shot LLM Routing

We replace only ARIA’s rule-based QCA with a zero-shot Mistral-7B classifier using the same backbone as the compressor/generator. It returns {SIMPLE, MULTI-ASPECT, MULTI-HOP} with a brief textual rationale; no fine-tuning or in-context examples are used, and all downstream stages and 5 seeds remain fixed. QA metrics use the full benchmark evaluations at  $16\times$  Normal. AdaptiveRAG instead trains a query-complexity classifier on automatically constructed labels and uses different routing strategies and supervision (Jeong et al., 2024). Table A52 reports the results.

**Interpretation.** Zero-shot LLM routing records 41.43% Avg CEM and 44.69% Avg F1, versus 41.24% and 44.50% for rule-based QCA. Both rows otherwise use the same ARIA configuration and seed set.

### A.34 Forward-Path-Off Endpoint Comparison

ARIA-COMP (forward-path off; ARIA-trained) runs ARIA’s full five-stage retrieval pipeline with all three operations disabled (uniform memory-token allocation with  $\rho_i = 1$ , CCEF-only ranking, no MTFRL second round). At nominal  $16\times$ , this gives a measured mean of 184 evidence-memory tokens per query, versus 108 for Full ARIA. The compressor was trained under ARIA’s objective, but CFRS, ACR, and MTFRL are disabled at inference. Results use Mistral-7B at  $16\times$  under Normal retrieval and report five-seed means. This is the “Forward-path-off” configuration in Table 5. Table A53 reports the raw endpoints for CLaRa, the forward-path-off configuration, and Full ARIA.

### A.35 Realized Evidence-Token and Per-Document Allocation

On the measured 1,000-query sample, the five uncompressed source passages contain a mean of 2,950 tokens per query after rounding (mean passage length 590 tokens). CLaRa allocates  $K_i^{(0)}(r) = \max\{1, \lfloor L_i/r \rfloor\}$  memory tokens to each of the 5 retrieved documents, giving  $\sum_i K_i^{(0)}(r)$  retained evidence-memory tokens. Its empirically averaged totals are 738/184/92/46/23 tokens at nominal 4/16/32/64/128 $\times$ .<sup>1</sup> ARIA’s ACR further scales each document’s allocation by  $\rho_i \in [\rho_{\min}, \rho_{\max}]$ : with the measured profile (Table A54;  $\sum_i \rho_i = 2.93$ , retaining 58.7% of the CLaRa-equivalent allocation at  $16\times$ ), ARIA’s empirically averaged totals are 432/108/54/27/13 tokens at nominal 4/16/32/64/128 $\times$ .

*Per-document allocation.* Table A54 reports the empirical memory-token distribution under ACR at nominal  $16\times$ , averaged over 1,000 queries (250/benchmark). Every query in this audit retained five CCEF survivors, giving  $N = 5,000$  document slots in this audit sample. The top-ranked document receives  $\rho_1 = 0.98$  after rounding (near  $\rho_{\max} = 1.0$ ), the 5th-ranked receives  $\rho_5 = 0.31$  after rounding (near  $\rho_{\min} = 0.25$ ); the per-document effective CR ranges from  $16.3\times$  at rank 1 to  $51.8\times$  at rank 5 (harmonic mean  $27.3\times$ , arithmetic mean  $32.6\times$ ). Because ACR is a sparsifying mask rather

<sup>1</sup>The empirical totals average  $\sum_i K_i^{(0)}(r)$  over the per-document context-length distribution, with flooring applied before averaging. The per-rank  $T_i$  values in Table A54 are averaged in the same way.

Table A50: Pre- versus post-Phase-II first-pass retrieval at nominal 16 $\times$  (Recall%, five-seed means). Each  $\Delta$  is post minus pre. Retrieval is measured before compression, MTFRL, and CFRS.

| Dataset    | Recall@1     |              |              | Recall@3     |              |              | Recall@5     |              |              |
|------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|            | Pre          | Post         | $\Delta$     | Pre          | Post         | $\Delta$     | Pre          | Post         | $\Delta$     |
| NQ         | 37.6         | 40.2         | +2.6         | 62.6         | 65.5         | +2.9         | 71.7         | 74.3         | +2.6         |
| HotpotQA   | 29.0         | 31.4         | +2.4         | 51.2         | 54.7         | +3.5         | 60.8         | 64.2         | +3.4         |
| MuSiQue    | 15.7         | 17.2         | +1.5         | 28.1         | 30.4         | +2.3         | 35.7         | 38.2         | +2.5         |
| 2Wiki      | 31.5         | 33.8         | +2.3         | 56.5         | 59.3         | +2.8         | 65.8         | 68.5         | +2.7         |
| <b>Avg</b> | <b>28.45</b> | <b>30.65</b> | <b>+2.20</b> | <b>49.60</b> | <b>52.48</b> | <b>+2.88</b> | <b>58.50</b> | <b>61.30</b> | <b>+2.80</b> |

Table A51: First-pass Recall@5 (%) under Normal retrieval across nominal compression ratios, measured after AHR-IGFR-MADS-CCEF and before compression, MTFRL, and CFRS. Each ratio uses separately trained checkpoints.

| Model                              | Nominal CR   | NQ          | HotpotQA    | MuSiQue     | 2Wiki       | Avg         |
|------------------------------------|--------------|-------------|-------------|-------------|-------------|-------------|
| CLaRa                              | 4 $\times$   | 72.5        | 60.1        | 35.8        | 66.4        | 58.7        |
| CLaRa                              | 16 $\times$  | 71.2        | 58.4        | 34.1        | 64.7        | 57.1        |
| CLaRa                              | 32 $\times$  | 70.0        | 56.9        | 33.0        | 63.5        | 55.9        |
| CLaRa                              | 64 $\times$  | 68.5        | 55.2        | 31.5        | 62.0        | 54.3        |
| CLaRa                              | 128 $\times$ | 66.8        | 53.5        | 30.0        | 60.2        | 52.6        |
| <b>ARIA</b>                        | 4 $\times$   | <b>76.1</b> | <b>66.5</b> | <b>40.3</b> | <b>70.2</b> | <b>63.3</b> |
| <b>ARIA</b>                        | 16 $\times$  | <b>74.3</b> | <b>64.2</b> | <b>38.2</b> | <b>68.5</b> | <b>61.3</b> |
| <b>ARIA</b>                        | 32 $\times$  | <b>73.0</b> | <b>62.8</b> | <b>37.0</b> | <b>67.2</b> | <b>60.0</b> |
| <b>ARIA</b>                        | 64 $\times$  | <b>71.5</b> | <b>61.0</b> | <b>35.5</b> | <b>65.8</b> | <b>58.5</b> |
| <b>ARIA</b>                        | 128 $\times$ | <b>69.8</b> | <b>59.2</b> | <b>33.8</b> | <b>64.0</b> | <b>56.7</b> |
| $\Delta(\text{ARIA}-\text{CLaRa})$ | 4 $\times$   | +3.6        | +6.4        | +4.5        | +3.8        | +4.6        |
|                                    | 16 $\times$  | +3.1        | +5.8        | +4.1        | +3.8        | +4.2        |
|                                    | 32 $\times$  | +3.0        | +5.9        | +4.0        | +3.7        | +4.2        |
|                                    | 64 $\times$  | +3.0        | +5.8        | +4.0        | +3.8        | +4.2        |
|                                    | 128 $\times$ | +3.0        | +5.7        | +3.8        | +3.8        | +4.1        |

Table A52: Rule-based QCA vs. zero-shot LLM routing (Mistral-7B backbone). QA metrics (CEM%, F1%) are Normal retrieval at 16 $\times$ , mean over 5 seeds.

| Router           | CEM      | F1       |
|------------------|----------|----------|
| Rule-based (QCA) | 41.24    | 44.50    |
| Zero-shot LLM    | 41.43    | 44.69    |
| $\Delta$         | +0.19 pp | +0.19 pp |

Table A53: CLaRa, forward-path-off, and Full-ARIA endpoints (CEM%/F1%, Normal, 16 $\times$ , 5 seeds). \*CLaRa values are 5-seed means.

| System                                     | CEM%         |              |              |              |              | F1%          |              |              |              |              |
|--------------------------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
|                                            | NQ           | Hotpot       | MuSiQue      | 2Wiki        | Avg          | NQ           | Hotpot       | MuSiQue      | 2Wiki        | Avg          |
| CLaRa*                                     | 55.72        | 43.86        | 10.22        | 45.52        | 38.83        | 47.95        | 45.84        | 17.68        | 44.30        | 38.94        |
| ARIA-COMP (forward-path off; ARIA-trained) | 55.90        | 43.80        | 10.30        | 46.08        | 39.02        | 52.72        | 49.98        | 18.58        | 46.72        | 42.00        |
| <b>Full ARIA</b>                           | <b>58.64</b> | <b>46.10</b> | <b>11.68</b> | <b>48.53</b> | <b>41.24</b> | <b>55.12</b> | <b>52.64</b> | <b>21.07</b> | <b>49.19</b> | <b>44.50</b> |

than a budget-conserving reallocation, the total  $\sum_i T_i = 108$  tokens is 58.7% of CLaRa’s uniform 184. Figure A4 reports the measured per-rank soft-cutoff means from the corresponding 16 $\times$ –128 $\times$  evaluation runs.

*MTFRL inference-time intervention.* The MTFRL-off row in Table 5 records 0.63 pp lower Avg

CEM at 16 $\times$ , with the largest per-benchmark difference on HotpotQA (−1.15 pp). This fixed-checkpoint inference intervention uses the same ARIA-trained checkpoint as Full ARIA.

Table A54: Per-document memory-token allocation under ACR at nominal CR=16 $\times$  (averaged over 1,000 queries; measured mean document length rounded to  $L=590$  tokens).  $\rho_i$ : ACR scale factor;  $T_i$ : memory tokens allocated; Effective CR =  $L/T_i$  per document. The Mean  $\rho_i$  row averages the displayed per-rank means.

| Rank | Mean $\rho_i$ | Std $\rho_i$ | Mean $T_i$ | Effective CR        |
|------|---------------|--------------|------------|---------------------|
| 1    | 0.98          | 0.04         | 36.1       | 16.3 $\times$       |
| 2    | 0.74          | 0.09         | 27.2       | 21.7 $\times$       |
| 3    | 0.52          | 0.11         | 19.1       | 30.9 $\times$       |
| 4    | 0.38          | 0.08         | 14.0       | 42.1 $\times$       |
| 5    | 0.31          | 0.06         | 11.4       | 51.8 $\times$       |
| Mean | 0.586         | —            | 21.6       | 27.3 $\times$ harm. |

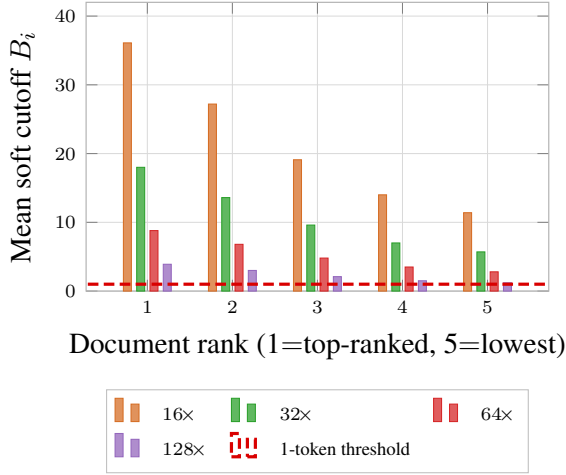


Figure A4: Mean per-rank soft cutoff  $B_i = \rho_i K_i^{(0)}(r)$  under ACR for nominal compression ratios 16 $\times$ –128 $\times$ , measured over the 1,000-query evaluation sample at each ratio. CR = 4 $\times$  is omitted because all five measured rank means lie above this  $y$ -scale and far above the threshold. The dashed red line marks the 1-token-per-document boundary. At 128 $\times$ , rank-4 and rank-5 soft cutoffs approach this boundary ( $B_4 = 1.5$ ,  $B_5 = 1.2$  after rounding); the inference realization  $T_i = \max\{1, \lfloor B_i \rfloor\}$  retains at least one position for each document. Appendix A.28 reports the adaptive-floor grid at this ratio. Soft-cutoff means and integer evidence-token totals apply flooring, the one-token clamp, averaging, and display rounding in different orders, so the sums need not match exactly (e.g., 53.9 vs. 54 at 32 $\times$  and 11.7 vs. 13 at 128 $\times$ ).

### A.36 MuSiQue Training-Set Augmentation Procedure

The Phase-II experiments use a 168,745-example MuSiQue training partition constructed from the 19,938-example answerable training split (Appendix A.16) with three transformation families.

(i) *Sub-question decomposition*. This transformation extracts bridge entities and intermediate questions from MuSiQue’s annotated decomposition and forms sub-question targets using a deterministic template (“Q: [hop-1 sub-question] → A: [hop-1 answer]”), while retaining the source labels and annotated chain structure.

(ii) *Partial-chain completion*. This transformation constructs variants from prefixes of the annotated evidence chain and pairs each prefix with the original final question and target. Evidence selection follows MuSiQue’s annotated hop structure; no text generation is involved. The input contains the selected evidence prefix.

(iii) *Entity-variant augmentation*. GPT-5.5 is prompted to generate surface-form variants of the question using prompts such as “Rephrase the following question keeping all named entities unchanged: [question]”. Generated paraphrases are filtered by (a) exact named-entity preservation (spaCy NER check); (b) ROUGE-L  $\geq 0.4$  with the original question. Generated questions are paired with the original decomposition and answer unchanged.

### A.37 Error Analysis Case Studies

We present structurally redacted versions of three observed error cases, one from each annotated failure mode in Table 4 (ARIA at CR = 16 $\times$ , Normal retrieval, Mistral-7B).

*Case 1: Retrieval failure (MuSiQue, 3-hop).*

**Query:** “Who directed the movie that [ACTOR-A] starred in, which was filmed in the city where [DIRECTOR-B]’s debut was set?”

**Ground truth:** [DIRECTOR-C]

**ARIA output:** [DIRECTOR-B] (incorrect)

**Retrieved evidence:** Top-5 contained biographical passages for [ACTOR-A] and [DIRECTOR-B], but *no passage* mentioning [DIRECTOR-C] or the specific film. IGFR’s second-hop query recovered additional passages about [DIRECTOR-B]’s debut city, but failed to issue a third-hop query linking city→movie→director. ACR allocated  $\rho_1 = 0.96$  to the [ACTOR-A] biography; mid-rank passages received lower allocations. This is

a retrieval-failure case: the answer entity is absent from the retrieved set, which contains no explicit support for that answer. It is an example of the MuSiQue missing-evidence annotation category, which accounts for 56.5% of the sampled errors.

*Case 2: Evidence-present answer failure (NQ, single-hop).*

**Query:** “What year was [HISTORICAL-EVENT] officially ratified?”

**Ground truth:** 1871

**ARIA output:** 1870

**Retrieved evidence:** Top-1 passage explicitly states “the ratification took effect the following year, in 1871”. The source passage contains both “1870” (the signing year) and the answer-bearing phrase “took effect ... in 1871”, while the model outputs the former. This case belongs to the NQ evidence-present answer-failure category, which constitutes 52.5% of the sampled NQ errors. The category pools compression-side and generator-side errors at the observed evidence-output level.

*Case 3: Joint failure (HotpotQA, 2-hop).*

**Query:** “[POLITICIAN-A] was governor of which state whose capital was designed by the architect who also designed [LANDMARK-B]?”

**Ground truth:** [STATE-C]

**ARIA output:** [STATE-D] (incorrect)

**Retrieved evidence:** Top-5 contained partial evidence for both hops ([POLITICIAN-A]→[STATE-C] was correctly retrieved; but the architect link for [LANDMARK-B] pointed to a different state [STATE-D]). ACR assigns the largest scale factor to the [POLITICIAN-A] biography ( $\rho_1 = 0.94$ ), and the trace records  $\rho_3 = 0.43$  for the architect passage. The retrieved source set contains one correct link and a partial, ambiguous chain in the compressed input, so the case is assigned to the joint category. This example belongs to the joint-failure category, which constitutes 14.5% of the annotated HotpotQA error sample.

### A.38 Evaluation-Metric Definitions and Normalization

For ARIA, the protocol-controlled CLaRa reimplementation, and the evaluated Self-RAG/CRAG raw outputs, CEM applies one normalization pipeline before substring matching. Other contextual CEM values in the complete table are source-native. Given a model output  $\hat{a}$  and a set of benchmark-provided gold aliases  $\mathcal{A}^*$ , we apply the following sequence to the output and

every alias: (1) Unicode NFKC normalization; (2) lowercasing; (3) removal of English articles a, an, the when they appear as standalone tokens; (4) stripping of ASCII punctuation other than apostrophes and hyphens; (5) whitespace collapse to single spaces. CEM is then defined as  $\max_{a \in \mathcal{A}^*} \mathbf{1}[a_{\text{norm}} \subseteq \hat{a}_{\text{norm}}]$  (substring match); EM uses equality in place of substring inclusion. Token-level F1 uses the same normalization, whitespace-token multisets, and the maximum F1 over gold aliases; if both output and alias are empty F1 is one, and if only one is empty it is zero. For a query  $q$  with annotated supporting corpus passages  $\mathcal{P}(q)$ , let  $D_q^{(k)}$  be the first  $k$  ranked documents after retaining the highest-ranked occurrence of each page ID, and  $P_q = \text{pid}(\mathcal{P}(q))$ . We report gold-document recall as

$$\text{Recall@}k = \frac{1}{|\mathcal{Q}_{\text{sup}}|} \sum_{q \in \mathcal{Q}_{\text{sup}}} \frac{|\text{pid}(D_q^{(k)}) \cap P_q|}{|P_q|}, \quad (19)$$

where  $\mathcal{Q}_{\text{sup}}$  contains evaluation queries with at least one annotated supporting page. Benchmark-level Recall@ $k$  is this query mean, and the reported cross-benchmark average is unweighted. All tabulated Recall@ $k$  values are the resulting proportions multiplied by 100. CEM is reported as a secondary character-substring inclusion metric, while token-level F1 is the primary metric. We apply the explicitly defined normalization and metrics identically to: (i) ARIA’s outputs across the evaluated CR/backbone configurations; (ii) the 5-seed CLaRa reimplementation (Appendix A.25); (iii) the evaluated Self-RAG and CRAG outputs. We use the exact sequence above throughout.

### A.39 Reproducibility and Responsible-Research Details

Table A55 consolidates ARIA’s objective weights, adapter placement, initialization, optimization, decoding, and pool sizes; Appendix A.25 specifies CLaRa’s distinct objective, initialization, and selection path. Model bases,  $W_{\text{BGE}}$ , and the BGE encoder/index are excluded from the optimizer. Mistral/Llama use three Phase-I epochs (batch 128) and five Phase-II epochs (batch 32); Qwen uses batch 16.

CEM, EM, token-level F1, and Recall@ $k$  follow the definitions above. Per-query differences are averaged across five matched seed pairs before 10,000 two-sided bootstrap resamples; the primary-family correction and pointwise appendix markers follow §4 and Appendix A.23. Figure 3 reports Avg-CEM seed SD. Public QA data and KILT retain their source licenses; annotation and model-assisted operations are specified in §6 and Appendix A.2.

Table A55: Reported ARIA hyperparameters; the explicitly labeled CLaRa row gives its distinct all-linear adapter placement. *Search range* reports the grid swept on the 500-query held-out dev partition; the chosen value is reported in *Value*. Hyperparameters without a search range were fixed by design or matched to CLaRa for the controlled comparison. Dataset/corpus licenses are NQ (CC BY-SA 3.0), HotpotQA (CC BY-SA 4.0), MuSiQue (CC BY 4.0), 2WikiMultiHopQA (Apache 2.0), and KILT Wikipedia text (CC BY-SA 3.0).

| Symbol                                                   | Value                                      | Search range                         |
|----------------------------------------------------------|--------------------------------------------|--------------------------------------|
| <i>Pipeline / coupling</i>                               |                                            |                                      |
| $\tau$ (CCEF threshold)                                  | 0.30                                       | {0.10, 0.20, 0.30, 0.40, 0.50}       |
| $\alpha$ (CCEF floor)                                    | 0.5                                        | {0.3, 0.5, 0.7}                      |
| $\beta$ (ACR sharpness)                                  | 10                                         | {5, 10, 20}                          |
| $\rho_{\min}$                                            | 0.25                                       | {0.10, 0.25, 0.40}                   |
| $\rho_{\max}$                                            | 1.0                                        | fixed                                |
| $w_{\text{cfrs}}$                                        | 0.30                                       | {0.00, 0.10, 0.20, 0.30, 0.40, 0.50} |
| $\gamma$ (IGFR coverage threshold)                       | 0.5                                        | {0.3, 0.5, 0.7}                      |
| $K_{\text{igfr}}$ (4–64 $\times$ )                       | 2                                          | {0, 1, 2, 3, 4}                      |
| $K_{\text{igfr}}$ (128 $\times$ )                        | 1                                          | {0, 1, 2, 3}                         |
| $\lambda$ (MSE loss weight)                              | 0.1                                        | {0, 0.01, 0.05, 0.1, 0.2, 0.5}       |
| $\mu$ (CFRS auxiliary weight)                            | 0.1                                        | fixed by design                      |
| $\nu$ (QR auxiliary weight)                              | 0.05                                       | fixed by design                      |
| $\xi$ (MTFRL auxiliary weight)                           | 0.05                                       | fixed by design                      |
| $\tau_{\text{fb}}$ (MTFRL temperature)                   | 0.05                                       | fixed by design                      |
| $\epsilon$ (numerical floor)                             | $10^{-6}$                                  | fixed                                |
| <i>Training</i>                                          |                                            |                                      |
| ARIA LoRA target / rank / $\alpha_{\text{LoRA}}$         | q_proj / 16 / 32                           | fixed                                |
| CLaRa LoRA target / rank / $\alpha_{\text{LoRA}}$        | all linear layers / 16 / 32                | method-specific                      |
| LoRA dropout / bias                                      | 0.10 / none                                | fixed                                |
| Optimizer / schedule                                     | AdamW / cosine decay                       | fixed                                |
| AdamW ( $\beta_1, \beta_2$ ) / $\epsilon$ / weight decay | (0.9, 0.95) / $10^{-8}$ / 0                | fixed                                |
| Phase-I learning rate / warmup                           | $1 \times 10^{-4}$ / 3% of steps           | fixed                                |
| Phase-II learning rate (Mistral/Llama)                   | $2 \times 10^{-4}$                         | $\{1, 2, 3\} \times 10^{-4}$         |
| Phase-II learning rate (Qwen-14B)                        | $1.6 \times 10^{-4}$                       | 0.8 $\times$ Mistral lr              |
| Phase-II warmup steps                                    | 500                                        | {250, 500, 1000}                     |
| Phase-I epochs                                           | 3                                          | fixed                                |
| Phase-II epochs                                          | 5                                          | {3, 5, 7}                            |
| Phase-I effective batch (Mistral/Llama)                  | 128                                        | fixed                                |
| Phase-II effective batch (Mistral/Llama)                 | 32                                         | fixed                                |
| Effective batch (Qwen-14B; both phases)                  | 16                                         | fixed                                |
| Precision / maximum gradient norm                        | bfloat16 / 1.0                             | fixed                                |
| Passage / query maximum length                           | 768 / 256 tokens                           | fixed                                |
| Phase-I / Phase-II input maximum                         | 2,048 / 1,024 tokens                       | fixed                                |
| Phase-I target maximum                                   | 512 tokens                                 | fixed                                |
| Phase-II target / evaluation maximum                     | 128 / 64 tokens                            | fixed                                |
| Evaluation decoding                                      | greedy (one beam); EOS or length stop      | fixed                                |
| Phase-I LoRA initialization                              | Kaiming-uniform $A$ ; zero $B$             | fixed                                |
| Phase-II compressor-adapter initialization               | corresponding Phase-I CR checkpoint        | fixed                                |
| Phase-II QR-adapter initialization                       | corresponding Phase-I compressor adapter   | fixed                                |
| Phase-II generator-adapter initialization                | fresh LoRA ( $A$ Kaiming; $B$ zero)        | fixed                                |
| Phase-II $P_{\text{fb}}$ initialization                  | Xavier-uniform weights; zero biases        | fixed                                |
| <i>Pre-fitting <math>W_{\text{BGE}}</math></i>           |                                            |                                      |
| Optimizer / learning rate                                | AdamW / $5 \times 10^{-4}$                 | fixed                                |
| Pairs / batch / epochs                                   | 50,000 / 128 / 2                           | fixed                                |
| <i>Pipeline pool sizes</i>                               |                                            |                                      |
| AHR pool $ D_{\text{AHR}} $                              | 4,000                                      | fixed                                |
| IGFR additions per iteration                             | at most 200                                | fixed                                |
| MADS $\rightarrow$ CCEF                                  | 100 $\rightarrow$ at most 5                | fixed                                |
| <i>MTFRL projection-head dimensions</i>                  |                                            |                                      |
| Mistral-7B/Llama-3-8B                                    | 4096 $\rightarrow$ 2048 $\rightarrow$ 1024 | —                                    |
| Qwen-2.5-14B                                             | 5120 $\rightarrow$ 2560 $\rightarrow$ 1024 | —                                    |

Table A56: Declarative QCA rule inventory, Part 1: Multi-Hop and Multi-Aspect categories (26 rules).  $w_\ell$ : rule weight. “Entity count” refers to the number of distinct named entities (PERSON, ORG, GPE, LOC, FAC, EVENT, PRODUCT, WORK\_OF\_ART types) detected by spaCy in the query string. All phrase patterns use case-insensitive substring match. Simple-category rules are in Table A57.

| Category     | Rule ID | $w_\ell$ | Trigger pattern / condition                                                                                      |
|--------------|---------|----------|------------------------------------------------------------------------------------------------------------------|
| Multi-Hop    | H01     | 1.0      | entity count $\geq 3$                                                                                            |
|              | H02     | 1.0      | contains: “in which country”, “in which city”, “in which state”, “in which year”                                 |
|              | H03     | 0.9      | contains: “born in the same”, “headquartered in the”, “located in the same”                                      |
|              | H04     | 0.9      | entity count $\geq 2$ AND contains: “who invented”, “who founded”, “who directed”, “who wrote”, “who discovered” |
|              | H05     | 0.9      | contains: “where was ... born”, “where did ... die”, “where is ... headquartered”                                |
|              | H06     | 0.85     | entity count = 2 AND contains bridge connective: “that”, “which”, “whose”, “where”                               |
|              | H07     | 0.85     | contains: “the {ENTITY} of {ENTITY}” (nested possessive pattern)                                                 |
|              | H08     | 0.8      | contains: “same {noun} as”, “the same ... as”                                                                    |
|              | H09     | 0.8      | contains: “also known as”, “formerly known as”, “previously called”                                              |
|              | H10     | 0.75     | sub-question trigger: starts with “what is the ... of the {ENTITY} that”                                         |
|              | H11     | 0.75     | contains: “after {ENTITY} {verb}”, “before {ENTITY} {verb}”                                                      |
|              | H12     | 0.7      | contains: “the {superlative} ... in the {ENTITY}” (e.g., “tallest mountain in the country where”)                |
|              | H13     | 0.65     | entity count = 2 AND question type = “what/who/when/where” AND verb is transitive                                |
|              | H14     | 0.6      | contains: “through which”, “via which”, “by means of which”                                                      |
| Multi-Aspect | A01     | 1.0      | starts with: “compare”, “what are the differences between”, “contrast”                                           |
|              | A02     | 1.0      | contains: “advantages and disadvantages”, “pros and cons”, “benefits and drawbacks”                              |
|              | A03     | 0.9      | contains: “both ... and ...”, “either ... or ...” (with two distinct entities)                                   |
|              | A04     | 0.9      | contains: “difference between”, “similarities between”, “relationship between”                                   |
|              | A05     | 0.85     | contains enumeration trigger: “list”, “enumerate”, “name all”, “give examples of”                                |
|              | A06     | 0.85     | contains: “how does ... compare to”, “how is ... different from”                                                 |
|              | A07     | 0.8      | contains: “what are the main”, “what are the key”, “what are the primary” (plural aspect noun)                   |
|              | A08     | 0.75     | contains: “in terms of”, “with respect to”, “regarding ... and ...”                                              |
|              | A09     | 0.75     | contains: “as well as”, “in addition to” (with entity count $\geq 2$ )                                           |
|              | A10     | 0.7      | contains: “multiple {noun}”, “various {noun}”, “several {noun}” (plural aspect)                                  |
|              | A11     | 0.65     | contains: “what factors”, “what aspects”, “what characteristics”                                                 |
|              | A12     | 0.6      | contains: “overview of”, “describe the ... of”, “explain the {noun} and the {noun}”                              |

Table A57: Declarative QCA rule inventory, Part 2: Simple category (12 rules). Continues Table A56; same notation. Together, Tables A56–A57 list all 38 rule families.

| Category | Rule ID | $w_\ell$ | Trigger pattern / condition                                                         |
|----------|---------|----------|-------------------------------------------------------------------------------------|
| Simple   | S01     | 1.0      | entity count = 1 AND starts with: “what is”, “who is”, “when is”                    |
|          | S02     | 1.0      | entity count = 1 AND starts with: “when was”, “where was”, “how many”               |
|          | S03     | 0.9      | entity count = 1 AND starts with: “who wrote”, “who invented” (single known entity) |
|          | S04     | 0.9      | starts with: “what year”, “what date”, “in what year”, “how old is”                 |
|          | S05     | 0.85     | contains: “the capital of”, “the population of”, “the president of” (simple fact)   |
|          | S06     | 0.85     | entity count = 0 AND factoid verb: “is”, “was”, “are”, “were” (definition query)    |
|          | S07     | 0.8      | starts with: “what is the name of the”, “who is the current”                        |
|          | S08     | 0.8      | contains: “the birthplace of”, “the nationality of”, “the height of”                |
|          | S09     | 0.75     | starts with: “how tall”, “how long”, “how far”, “how deep”, “how wide”              |
|          | S10     | 0.7      | starts with: “what language”, “what currency”, “what color”, “what type of”         |
|          | S11     | 0.65     | entity count = 1 AND no bridge connective, no comparative/superlative               |
|          | S12     | 0.6      | fallback (default): no hop or aspect indicators triggered                           |

*Note:* Rules H02–H05 and A01–A06 are matched before NER-count rules; when conflicting, explicit phrase patterns override count-only rules. In particular, rules H04 (“who invented/wrote” with multi-entity context) and S03 (“who invented/wrote” with single entity) are disambiguated by entity count: if entity count  $\geq 2$ , H04 fires and Multi-Hop is assigned; if entity count = 1, S03 fires and Simple is assigned. Brace-delimited placeholders and ellipses denote declarative pattern families. The tables specify the declarative rule families, and the precedence rule above resolves conflicts among rule families. The  $w_\ell$  values were set manually from development-set error analysis and held fixed throughout evaluation.